MA3D1 – Fluid Dynamics

Lecture notes

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Overview

These lecture notes are for the MA3D1 Fluid Dynamics course at the University of Warwick. They have evolved through a number of iterations, with key contributions from Sergey Nazarenko, James Sprittles, Shreyas Mandre, and Ferran Brosa Planella. The notes are not intended to be a comprehensive reference, and the reader is encouraged to consult the references given in the notes for further reading. If you spot any errors or typos, please let me know by emailing me at J.E.Sprittles@warwick.ac.uk. Any other comments about the material or the lectures are also very welcome.

Reading list

The lecture notes are sufficient to cover the course material, but if you want to learn more there are a huge array of books on the topic of fluid dynamics and you should feel free to engage with any of them. My recommendation is:

- **D.J. Acheson**, Elementary Fluid Dynamics, OUP. (Well-written accessible exposition of fluid dynamics with a useful set of questions and solutions.).

However other famous possibilities include:

- **G. K. Batchelor** An Introduction to Fluid Dynamics, CUP (A classic which goes into greater detail than the present course).
- **L.D. Landau and E.M. Lifshitz**, Fluid Mechanics, OUP. (A classic for those with a deep interest in fluid dynamics in modern physics.)
- **A.R. Paterson**, A First Course in Fluid Dynamics, CUP. (Easier than Acheson.)
- **D.J. Tritton**, Physical Fluid Dynamics, Oxford Science Publs. (Emphasis is on the physical phenomena and less on the mathematics.)

Videos

In the 1960's a fantastic set of videos were filmed by the National Committee for Fluid Mechanics Films (see `https://web.mit.edu/hml/ncfmf.html`). These can now be found on YouTube (see `https://www.youtube.com/playlist?list=PL0EC6527BE871ABA3` though you'll need to put up with the sound being out of sync with the presenter's voice!). Two other good resources for fluid videos and pictures are `https://www.efluids.com/efluids/pages/gallery.htm` and `https://gfm.aps.org/`. If you find any other interesting videos/pictures, please let me know.

Chapter 1

Preliminaries

The theory of fluid dynamics rests on the developments in some greater disciplines such as mathematics and physics, especially mechanics. In this chapter, let us summarize and revise some of these topics.

1.1 Newtonian mechanics

While it is not possible to do justice to Newtonian mechanics here, we present the basic tenets. [1] Newtonian mechanics, at its core, describes physical laws governing the motion of point material particles. Consider a point material object of mass m moving through empty space. Let us say that the objects location is given by $\boldsymbol{x}(t)$ as a function of time t. The velocity of the body is $\boldsymbol{v}(t) = \mathrm{d}\boldsymbol{x}/\mathrm{d}t$. The three laws of motion according to Newtonian mechanics are:

1. **First law:** In an inertial reference frame, every body continues in its state of rest or uniform motion until and unless acted upon by an external unbalanced force. Essentially, this law defines the notion of an inertial reference frame, i.e. one in which a free body moves uniformly.
2. **Second law:** When an external unbalanced force, $\boldsymbol{F}$, acts on a body it must equal the rate of change of momentum of this body. The momentum (also known as the linear momentum) of this point mass is defined as $\boldsymbol{p} = m\boldsymbol{v}$. Therefore, the mathematical statement of the second law is:
$$\boldsymbol{F} = \frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}t}. \tag{1.1}$$
Noting that the acceleration of the particle, $\boldsymbol{a} = \mathrm{d}\boldsymbol{v}/\mathrm{d}t$, if the mass of the body does not change with time, then this law is also written as $\boldsymbol{F} = m\boldsymbol{a}$. This law describes the motion of the particle.
3. **Third law:** Every action has an equal and opposite reaction. In other words, if a body A exerts a force $\boldsymbol{F}$ on a body B, then the body B must exert a force of $-\boldsymbol{F}$ on body A. This law describes the interaction between particles.

As a consequence of Newton's laws of motion for point particles, it is possible to derive the following "conservation laws" of momentum and angular momentum for extended deformable bodies. A common terminology used in physics is the moment of a quantity, which is obtained by making a cross product of the position vector $\boldsymbol{x}$ of a point particle with that quantity. Suppose that an extended material object at time t occupies a region of space given by Ω. Then its total momentum is obtained by breaking it up into infinitesimal pieces of mass $\mathrm{d}m$, determining the momentum of these individual pieces, and summing the momentum. Mathematically, this amounts to integrating as

$$\boldsymbol{p} = \int_\Omega \boldsymbol{v}(\boldsymbol{x}, t)\ \mathrm{d}m, \tag{1.2}$$

where $\boldsymbol{v}$ is the velocity of the piece at position $\boldsymbol{x} \in \Omega$. The total mass m of the body is simply $m = \int_\Omega \mathrm{d}m$. The moment of momentum , also known as the angular momentum $\boldsymbol{l}$, is then

$$\boldsymbol{l} = \int_\Omega \boldsymbol{x} \times \boldsymbol{v}\ \mathrm{d}m. \tag{1.3}$$

[1] For more detailed treatments, refer to a book on undergraduate mechanics, e.g., Halliday and Resnick, Physics - I, or, if you prefer the economy of Soviet authors, I. E. Irodov, Fundamental Laws of Mechanics.

These conservation laws invoke the "centre of mass" $\boldsymbol{x}_c$ of this object, defined as

$$\boldsymbol{x}_c = \frac{\int_\Omega \boldsymbol{x}\ \mathrm{d}m}{\int_\Omega \mathrm{d}m} = \frac{1}{m}\int_\Omega \boldsymbol{x}\ \mathrm{d}m. \tag{1.4}$$

Suppose that the external forces $\boldsymbol{F}_i$, $i = 1, 2, \ldots, n$ act on this object at locations $\boldsymbol{x}_i$. The net torque $\boldsymbol{\tau}$ on this body is the moment of the external forces,

$$\boldsymbol{\tau} = \sum_{i=1}^{n} \boldsymbol{x}_i \times \boldsymbol{F}_i. \tag{1.5}$$

So long as the object does not exchange material with anything outside it, the laws of conservation state that:

1. **Conservation of (linear) momentum:** The rate of change of total momentum must equal the total external unbalanced force on the object, i.e.

   $$\frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}t} = \sum_{i=1}^{n} \boldsymbol{F}_i. \tag{1.6}$$

   Noting the relation between $\boldsymbol{p}$ and $\boldsymbol{x}_c$, this law is also stated as

   $$m\frac{\mathrm{d}^2\boldsymbol{x}_c}{\mathrm{d}t^2} = \sum_{i=1}^{n} \boldsymbol{F}_i, \tag{1.7}$$

   which means that the mass times the acceleration of the centre of mass must equal the external unbalanced force.

2. **Conservation of angular momentum:** The rate of change of angular momentum must equal the net torque on the body. Mathematically,

   $$\frac{\mathrm{d}\boldsymbol{l}}{\mathrm{d}t} = \boldsymbol{\tau}. \tag{1.8}$$

While the momentum and angular momentum are not constants of motion, here the term "conservation law" is used in the sense that it is possible to keep an account of these quantities in relation to their respective sources.

1.2 Units and dimensions

Units are used to describe the values of physical quantities. For example, Prof. James Sprittles is 182 cm tall, weighs 11 stones and consumes about 100 Watts when resting (about 2000kcal a day). The choice of the unit is arbitrary, as reflected in the plethora of units available for the same quantity, e.g. 182 cm $\approx$ 72 inches $\approx$ 1.9 $\times 10^{-16}$ light-years. The SI system is popular for the purpose of standardisation (and that purpose alone).

The fundamental nature of a quantity attributes it with a unique dimension. Example of dimensions are length L, mass M, time T, force F, etc. Many different units are associated with each dimension, as described above. The basic rules for analysis based on dimensions are as follows. The dimension of product of physical quantities is the product of the dimensions of the individual multiplicands. It is an error to add quantities of different dimensions, i.e. lengths may be added with lengths but not mass. For an equation in physics to be dimensionally consistent, the dimensions on both sides of the equation must be identical.

Dimensions of quantities are not necessarily independent of each other. For example, the dimensions of volume L^3 depend on the dimensions of length. This dependence is best expressed in systems of dimensions. In the MLT system, the dimensions M, L and T are considered basic, from which all other dimensions are derived, e.g. $F = ML/T^2$. There is some arbitrariness in constructing systems of dimensions, for example, equally valid is the FLT system where $M = FT^2/L$ is a derived dimension.

Exercise: Construct a ρVL system, where ρ is the dimension of density and V of velocity. Express the dimensions of mass and time in terms of ρ, V and L.

Exercise: What are the dimensions of p, μ and ν?

This arbitrariness in the nature of physical quantities underlies some of the most powerful theoretical tools in physics. It was on the basis of such considerations that Sir G. I. Taylor estimated the yield of the atomic bomb, a heavily guarded American secret at the time, from publicly available images. Ignore dimensions at your own peril!

1.3 Cartesian tensors

Vectors and tensors offer a compact representation to tackle the large number of variables whilst accounting for the relationship between them. Vectors are quantities that, in the colloquial sense, have a magnitude and a direction. Mathematically, vectors are elements of $\mathbb{R}^3$, where the components commonly represent Cartesian components. Tensors are generalization of vectors which have more than three components in 3-dimensions. An nth rank tensor has 3^n components in 3 dimensions. In this way, a vector is a first-rank tensor and a scalar is a rank zero tensor. Examples of second rank tensors include moment of inertia of a solid body. Here the nine Cartesian components of the moment-of-inertia tensor are

$$I_{xx} = \int_V x^2 \, \mathrm{d}m, \qquad I_{yy} = \int_V y^2 \, \mathrm{d}m, \qquad I_{zz} = \int_V z^2 \, \mathrm{d}m,$$
$$I_{yx} = I_{xy} = \int_V xy \, \mathrm{d}m, \qquad I_{zy} = I_{yz} = \int_V yz \, \mathrm{d}m, \qquad I_{zx} = I_{xz} = \int_V xz \, \mathrm{d}m,$$

where $\mathrm{d}m$ represents an infinitesimal mass element spanning the extent of the body V.

Two different notations are available to represent vector and tensor expressions, they are called the vector notation and the index notation. Depending on the circumstance, one notation may be preferred to the other either for clarity or compactness.

We will use boldface symbols to represent vectors and tensors in vector notation. In this notation, it is crucial to declare the rank of the tensor a symbol represents, e.g., let $\boldsymbol{u}$ represent first-rank velocity tensor, and let $\boldsymbol{T}$ represent the second rank stress tensor. The operations on these symbols are understood as a matter of convention, e.g., $\boldsymbol{a} + \boldsymbol{b}$ is the sum of two tensors $\boldsymbol{a}$ and $\boldsymbol{b}$ of equal rank. Vector notation is not equipped to unambiguously represent dot products and cross products, but it is denoted as $\boldsymbol{a} \cdot \boldsymbol{b}$ and $\boldsymbol{a} \times \boldsymbol{b}$, respectively, when no ambiguity exists. Examples will be abundant in the rest of the notes. The advantage of this notation is that it treats tensors in their abstraction without reference to any specific coordinate system.

The index notation explicitly invokes (for this module Cartesian) coordinate system. The Cartesian coordinates are themselves written as $x_1 = x$, $x_2 = y$ and $x_3 = z$. Let $\boldsymbol{e}_1$, $\boldsymbol{e}_2$ and $\boldsymbol{e}_3$ be the unit vectors along x_1, x_2 and x_3 coordinates, respectively. Then the components u_1, u_2 and u_3 of the tensor a first rank tensor (i.e., a vector) $\boldsymbol{u}$ are written as u_i, where the index i ranges from 1 to 3. The components of a second rank tensor are represented using two indices, e.g., the components of the moment-of-inertia tensor are represented as I_{ij}, where

$$I_{ij} = \int_V x_i x_j \, \mathrm{d}m, \quad \text{and } i \text{ and } j \text{ both independently range from 1 to 3.}$$

It is customary to omit the range of the indices in this notation. In general, n^{th} rank tensors have n indices.

The summation convention: Expressions, such as the dot product between two vectors are written as a sum in index notation, e.g.

$$\boldsymbol{a} \cdot \boldsymbol{b} = a_1 b_1 + a_2 b_2 + a_3 b_3 = \sum_{i=1}^{3} a_i b_i. \tag{1.9}$$

As a matter of convention due to Einstein, it is customary to also omit the summation sign over an index which appears twice in any product. Hence, $\boldsymbol{a} \cdot \boldsymbol{b} = a_i b_i$, with an implied sum over i from 1 to 3. When a sum on an index appearing twice is not implied, this must be stated explicitly. Similarly, whether a sum is implied on an index that repeats more than two times must be explicitly stated.

Two special tensors: Let us introduce two special and basic tensors. The first is called the Kronecker delta, defined as

$$\delta_{ij} = \begin{cases} 0 & \text{for } i \neq j \\ 1 & \text{for } i = j \end{cases}. \tag{1.10}$$

The second is the third-rank alternating tensor, also known as the Levi-Civita, which is defined as

$$\epsilon_{ijk} = \begin{cases} 0 & \text{if } \; i = j, j = k \;\; \text{or} \;\; i = k, \\ 1 & \text{for } \; (i,j,k) = (1,2,3), (2,3,1) \;\; \text{or} \;\; (3,1,2), \\ -1 & \text{for } \; (i,j,k) = (1,3,2), (2,1,3) \;\; \text{or} \;\; (3,2,1). \end{cases} \tag{1.11}$$

Tensor algebra: Using these rules, the notation for tensor algebra is as follows. Here w is a scalar, $\boldsymbol{a}$, $\boldsymbol{b}$ and $\boldsymbol{c}$ are vectors, $\boldsymbol{T}$, $\boldsymbol{S}$, $\boldsymbol{A}$ and $\boldsymbol{B}$ are second rank tensors.

	Vector notation	**Index notation**	**Tensorial rank**
Sum of vectors:	$\boldsymbol{c} = \boldsymbol{a} + \boldsymbol{b}$	$c_i = a_i + b_i$	1
Dot product of vectors:	$w = \boldsymbol{a} \cdot \boldsymbol{b}$	$w = a_i b_i$	0
Cross product of vectors :	$\boldsymbol{c} = \boldsymbol{a} \times \boldsymbol{b}$	$c_i = \epsilon_{ijk} a_j b_k$	1
Multiplication of tensors:	$\boldsymbol{T} = \boldsymbol{A}\,\boldsymbol{B}$	$T_{ij} = A_{ik} B_{kj}$	2
Tensor-vector product:	$\boldsymbol{c} = \boldsymbol{A}\boldsymbol{b}$	$c_i = A_{ij} b_j$	1
Outer product of vectors:	$\boldsymbol{A} = \boldsymbol{a} \otimes \boldsymbol{b} = \boldsymbol{a}\boldsymbol{b}^{\mathrm{T}}$	$A_{ij} = a_i b_j$	2
Tensor product:	$\boldsymbol{A} \otimes \boldsymbol{B}$	$A_{ij} B_{km}$	4
Trace:	$\mathrm{tr}(\boldsymbol{A})$	A_{ii}	0
Contraction:	$\boldsymbol{A} : \boldsymbol{B} = \mathrm{tr}(\boldsymbol{A}\boldsymbol{B})$	$A_{ik} B_{ki}$	0
Scalar triple product:	$\boldsymbol{a} \cdot (\boldsymbol{b} \times \boldsymbol{c})$	$\epsilon_{ijk} a_i b_j c_k$	0
Vector triple product:	$\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c})$	$\epsilon_{ijk} \epsilon_{kmn} a_j b_m c_n$	1

We can define the transposition of a tensor as $[\boldsymbol{A}^{\mathrm{T}}]_{ij} = [\boldsymbol{A}]_{ji}$. A symmetric second-rank tensor is on that satisfies $\boldsymbol{S}^{\mathrm{T}} = \boldsymbol{S}$ (i.e. $S_{ij} = S_{ji}$ for all combinations of i and j). An anti-symmetric one is that satisfies $\boldsymbol{A}^{\mathrm{T}} = -\boldsymbol{A}$ (i.e. $A_{ij} = -A_{ji}$).

Components of vectors transform in a predictable way when represented in a different coordinate system. In other words, if a vector $\boldsymbol{u}$ has components u_i, i.e. $\boldsymbol{u} = u_i \boldsymbol{e}_i$ (note: sum implied), then the components of $\boldsymbol{u}$ change when the basis vectors change to $\boldsymbol{e}'_i$. Similarly, tensor components also change in a predictable way under coordinate transformations. This can be seen using the representation of tensors in terms of the Cartesian basis vectors as $\boldsymbol{T} = T_{ij} \boldsymbol{e}_i \otimes \boldsymbol{e}_j$ (sum implied). Note that $\boldsymbol{e}_i \otimes \boldsymbol{e}_j$ is the outer product between $\boldsymbol{e}_i$ and $\boldsymbol{e}_j$, and just like $\boldsymbol{e}_i$ is a unit vector, the term $\boldsymbol{e}_i \otimes \boldsymbol{e}_j$ is one of the nine unit second-rank tensors.

The reason to introduce the Kronecker delta and the alternating tensor are that they are special second and third rank tensors. They are the so-called "unit isotropic" tensors, i.e. the Cartesian components of these tensors do not change under coordinate transformations. (We shall take it for granted here without proof.) By extension, any algebraic expressions constructed using these unit isotropic tensors are also isotropic. In fact, the most general second, third, and fourth rank isotropic tensors are

$$I^2_{ij} = \lambda \delta_{ij} \tag{1.12a}$$

$$I^3_{ijk} = \lambda \epsilon_{ijk} \tag{1.12b}$$

$$I^4_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \alpha \delta_{ik} \delta_{jl} + \beta \delta_{il} \delta_{jk}, \tag{1.12c}$$

where λ, α and β are arbitrary scalars.

The so-called $\epsilon - \delta$ identity is useful in simplifying the inner products of two alternating tensors as

$$\epsilon_{ijk} \epsilon_{kmn} = \delta_{im} \delta_{jn} - \delta_{in} \delta_{jm}. \tag{1.13}$$

Here note that the left hand side is an isotropic fourth-rank tensor, which is expressed on the right-hand side as a special case of the most general isotropic fourth-rank tensor.

1.4 Tensor calculus

Tensorial notation also aids in representing multivariate calculus of fields. Here are the basic elements in terms of scalar field $\phi(\boldsymbol{x}, t)$, vector field $\boldsymbol{u}(\boldsymbol{x}, t)$ and tensor fields $\boldsymbol{T}(\boldsymbol{x}, t)$. A special notation called the comma notation is used the denote derivatives with respect to Cartesian coordinates, $\partial \phi / \partial x_i = \phi_{,i}$, i.e. an index following a comma in the subscript of a symbol implies differentiation with respect to the corresponding Cartesian coordinate.

	Vector notation	**Index notation**	**Tensorial rank**
Gradient of a scalar:	$\nabla\phi$	$\dfrac{\partial\phi}{\partial x_i} = \phi_{,i}$	1
Divergence of a vector:	$\nabla \cdot \boldsymbol{u}$	$\dfrac{\partial u_i}{\partial x_i} = u_{i,i}$	0
Gradient of a vector:	$\nabla \boldsymbol{u}$	$\dfrac{\partial u_i}{\partial x_j} = u_{i,j}$	2
Divergence of a tensor:	$\nabla \cdot \boldsymbol{T}$	$\dfrac{\partial T_{ij}}{\partial x_j} = T_{ij,j}$	1
Curl of a vector:	$\nabla \times \boldsymbol{u}$	$\epsilon_{ijk}\dfrac{\partial u_k}{\partial x_j} = \epsilon_{ijk}u_{k,j}$	1

Results in integral calculus may also be condensed using the index notation. Consider a volume Ω in space , with its unit normal oriented outwards denoted by $\boldsymbol{n}$. The divergence theorem expressed in vector and index notation is

$$\int_{\partial\Omega} \boldsymbol{u}\cdot\boldsymbol{n}\ \mathrm{d}A = \int_{\Omega} \nabla\cdot\boldsymbol{u}\ \mathrm{d}V \quad \text{or} \quad \int_{\partial\Omega} u_i n_i\ \mathrm{d}A = \int_{\Omega} u_{i,i}\ \mathrm{d}V, \tag{1.14}$$

where $\partial\Omega$ represents the bounding surface of Ω and $\mathrm{d}A$ is the infinitesimal area element on it. A generalization of this theorem for second-rank tensors is

$$\int_{\partial\Omega} \boldsymbol{T}\cdot\boldsymbol{n}\ \mathrm{d}A = \int_{\Omega} \nabla\cdot\boldsymbol{T}\ \mathrm{d}V \quad \text{or} \quad \int_{\partial\Omega} T_{ij} n_j\ \mathrm{d}A = \int_{\Omega} T_{ij,j}\ \mathrm{d}V. \tag{1.15}$$

Stokes theorem applies to closed curves C, the unit tangent to it $\boldsymbol{t}$, the surface it encloses S and the unit normal to the surface $\boldsymbol{n}$. For vectors, it reads

$$\int_{C} \boldsymbol{u}\cdot\boldsymbol{t}\ \mathrm{d}s = \int_{S} (\nabla\times\boldsymbol{u})\cdot\boldsymbol{n}\ \mathrm{d}A \quad \text{or} \quad \int_{C} u_i t_i\ \mathrm{d}s = \int_{S} \epsilon_{ijk}u_{kj}n_i\ \mathrm{d}A, \tag{1.16}$$

where $\mathrm{d}s$ is the infinitesimal arc-length element.

Chapter 2

Mathematical modelling of fluid flow

In this chapter, we will derive the mathematical model governing the motion of fluids. In summary, these will be the Navier-Stokes equations, but on the way we will encounter the Eulerian and the Lagrangian descriptions of fluid motion, the deformation of infinitesimal fluid particles, Cauchy's notion of stress and his equation for motion of continuum, Euler's equations for motion of an ideal fluid, etc.

2.1 Fluid dynamics

Before we delve into the mathematics, we need to briefly introduce the subject of fluid dynamics. As the name suggests, fluid dynamics (sometimes also called fluid mechanics) is concerned with the motion of fluids.

Fluids, unlike solids, are characterised by their ability to flow and change shape. This category encompasses not only liquids (e.g. water and honey), but also gases (e.g. air) and even plasma. In fact, the line between a fluid and a solid is quite blurry: as ice might look to us like a solid, but over long timescales it can flow like a fluid (e.g. glacier motion). The study of fluids spans an incredible range of scales (as shown in Figure 2.1), from the microscopic, such as the behaviour of microdroplets in a 3D printer, to the macroscopic, like the dynamics of tornadoes. Furthermore, the flow of fluids can also occur at vastly different speeds, from the slow movement of glaciers to the rapid motion in aeronautics.

These fluid phenomena are not only observable but also frequently counter-intuitive in their behaviour (Figure 2.2). And quite astonishingly, the same set of equations (the Navier-Stokes equations) can be used to describe the flow of both gases and liquids in all of the aforementioned situations.

2.2 What is a fluid? The continuum hypothesis

We have given a quite intuitive definition for a fluid. However, for the purpose of this module, the following definition will prove more useful.

Definition: Fluid

A fluid is a continuum substance that cannot withstand shear stress without continually deforming.

While this definition addresses some of the shortcomings of the more elementary definitions, it has left some terms undefined. For instance, what is a continuum substance? What is a shear stress? How does one quantify continual deformation? Also note that the definition does not constrain the magnitudes of the shear stress or that of the deformation, nor does it specify any relation between the two.

Based on this guiding definition, we will derive mathematical representations of the laws of conservation of mass, momentum and angular momentum later in this chapter. These laws, jointly with some constitutive relations for the fluid behaviour will provide the governing equations for fluids. In the process, the concepts of deformation rate and stress will emerge. A fluid can then be defined as any substance that obeys the assumptions that go into the derivations of these governing. On the one hand, such an exercise rigorously constrains the materials to which the governing equations could be applied. And on the other hand it broadens the scope to include materials that are not commonly classified as fluids but whose mechanics are governed by the equations



Figure 2.1: Top: Small (inkjet microdroplets) → Large (tornado). Bottom: Slow (glacial flow) → Fast (aeronautics)

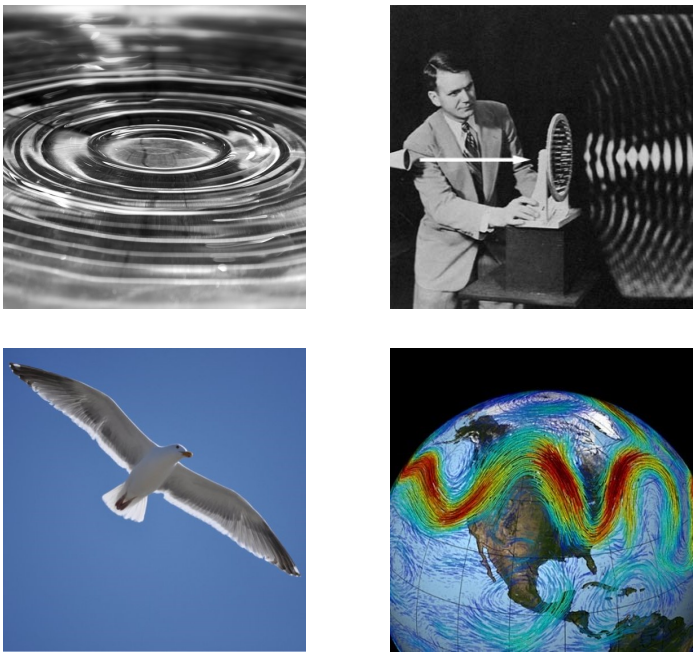
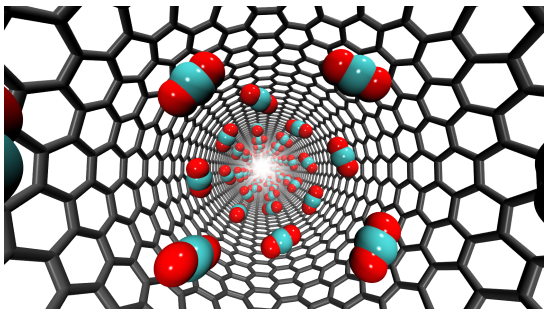


Figure 2.2: Everyday fluid phenomena - water waves, sound, flight and weather



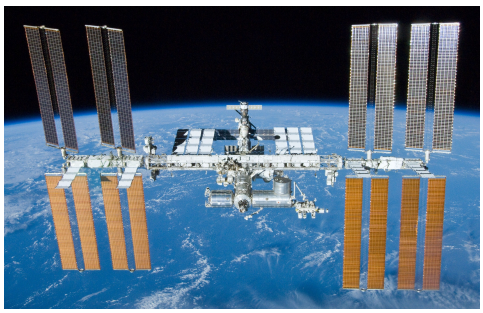


Figure 2.3: Left: An example where continuum mechanics is unlikely to work - a molecular dynamics simulation of a desalination membrane, where water molecules flow through a carbon nanotube (see `www.micronanoflows.ac.uk`. Right: The International Space Station (source: `www.wikipedia.org/wiki/International_Space_Station`.

we derive because they satisfy the underlying assumptions. Understanding the significance of the assumptions that form the basis of the derivation of the governing equations is the objective of this chapter.

Of all the assumptions used in the definition of a fluid, one of them is the assumption of a continuum material. This assumption is so fundamental that it basically defines the mathematical approach to take. In fact, it is so fundamental, that it will almost never be mentioned explicitly in the derivations of the governing equations as it underpins all the tools we will use.

Matter on earth is made up of atoms and molecules, each obeying classical or quantum laws of physics. When dealing with a large collection of molecules, the granularity of molecular composition somehow is lost and/or becomes irrelevant. Properties of matter can then be considered to vary smoothly with location within the material. The emergence of thermodynamics is an example of this. Similarly, when describing motion of macroscopic matter on length scales much larger than molecular scales, it becomes possible to consider the material as a infinitely divisible continuum. The theory of mechanics that arises out of such a consideration is called continuum mechanics.

The continuum approach is accurate when the mean free path ℓ, which is the length a molecule of the medium travels between collisions with other molecules, is much smaller than the dimensions of the flow L, so that $\ell/L \ll 1$ (the continuum limit takes $\ell/L \to 0$). Note that ℓ/L is a 'dimensionless number or parameter', whose value does not depend on the system of units used. When $\ell/L \approx 1$ or $\ell \gg L$, microscopic methods (molecular dynamics for liquids and kinetic theory for gases) are required.

It is worthwhile to ponder briefly about cases where continuum hypothesis fails. Here are two situations:

1. **Flow of water through a carbon nanotube:** Nanotubes are modern materials consisting of a tube of diameter as small as 0.4 nm. Figure 2.3 shows the use of nanotubes made out of carbon for desalinating water. The basic idea is that while water can flow through the tube, the impurities do not. In this case, the water near the centre of the tube flows faster compared to that near the tube walls, where friction impedes the flow. Thus, the water speed changes over a distance compared to the diameter of the tube, so $L \approx 0.4$ nm. The molecular mean-free path is about 0.25 nm. In this case, $\ell/L \approx 0.25/0.4 = 0.625$, which is not small. Therefore, the continuum hypothesis is likely to fail.

2. **Motion of the International Space Station (ISS):** The ISS is about 79 m $\times$ 109 m, say approximately $L = 100$ m across and orbits the earth at an altitude of 400 km. The atmosphere there is so rarified that the mean free path of gas molecules is $\ell \approx 20$ km. The ratio $\ell/L = 20 \times 10^3/100 = 200$ is clearly not small, so the surrounding gas cannot be treated as a continuum.

In summary, the continuum approximation of the fluid may fail because (i) the length-scale over which the flow varies is too small, or (ii) the mean free path of the fluid molecules is too large. Both these cases are adequately encapsulated by the ratio ℓ/L.

In day-to-day situation, e.g. flow of water through ordinary pipes, continuum approximation does apply. In that case, every point in space is assigned physical properties such as density, pressure, velocity, stress, etc. Continuous functions of space and time, called fields, are used to represent these physical variables.

2.3 Material properties of fluids

There are five main properties that will be important for us in this module.

1. **Density (ρ):** The mass of the fluid per unit volume. Air has a density of about 1.2 kg/m^3 and water of 1000 kg/m^3. Physically, the density of a fluid is determined by its thermodynamic state, and could thus change with pressure and temperature, for instance. If the conditions are such that the density of the fluid does not change appreciably in the flow then such a flow is termed "incompressible". Under ordinary circumstances, flows of air and water are incompressible. For flow of air around a jet plane, the flow is compressible. And it is certainly so for flow around a supersonic plane.

2. **Velocity ($\boldsymbol{u}$ or u_i):** The combination of speed and direction of motion of a fluid particle. The velocity is a vector quantity, which may depend on space and time, and its magnitude is called the speed. We can also define the velocity field is a vector field, i.e. a function that assigns a vector to each point in space and time.

3. **Pressure (p):** The force a fluid exerts on a wall per unit area. At sea level, air exerts a pressure of about 10^5 N/m^2. In SI system, N/m^2 = Pa, a Pascal. Other common units of pressure, especially in atmospheric sciences, are the bar, 1 bar = 10^5 Pa, and the atmosphere, 1 atm = 101325 Pa. Pressure is also measured in terms of a height of a fluid column, where the fluid is commonly but not necessarily either water or mercury. Normal human blood pressure is purported to range from 80 to 120 mm of mercury.

4. **Dynamic Viscosity (μ):** Also known as Newtonian viscosity. It represents the resistance of a fluid to shear. The viscosity is the ratio of the traction applied to the shear rate generated in steady state. Dynamic viscosity has the SI units of Ns/m^2 or, equivalently, Pa s. An "ideal" fluid or an "inviscid" fluid is one with zero viscosity.

5. **Kinematic viscosity (ν):** The kinematic viscosity is the dynamic viscosity divided by fluid density, i.e. $\nu = \mu/\rho$. In the SI system, it has units of m^2/s (note the absence of kg in the units, hence the epithet kinematic).

The conditions termed as incompressible and inviscid or ideal are idealizations of real conditions. They are never exactly realized.

2.4 The Navier-Stokes equations

The equations that govern the motion of fluids are called the Navier-Stokes equations, named after the French engineer and physicist Claude-Louis Navier (1785 – 1836) and the Irish physicist and mathematician George Gabriel Stokes (1819 – 1903).

They are a set of partial differential equations that describe how the velocity field of a fluid and its pressure evolve in time, and they are given by

$$\frac{\partial u_i}{\partial x_i} = 0, \qquad \text{and} \qquad \rho\left(\frac{\partial u_i}{\partial t} + u_j\frac{\partial u_i}{\partial x_j}\right) = -\frac{\partial p}{\partial x_i} + \mu\frac{\partial^2 u_i}{\partial x_j^2} + \rho g_i, \tag{2.1}$$

or, in vector notation,

$$\nabla\cdot\boldsymbol{u} = 0, \qquad \text{and} \qquad \rho\left(\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u}\cdot\nabla\boldsymbol{u}\right) = -\nabla p + \mu\nabla^2\boldsymbol{u} + \rho\boldsymbol{g}. \tag{2.2}$$

Here, in addition to the properties introduced in the previous section, we have $\boldsymbol{g}$ which is the acceleration due to gravity.

This version in particular is the incompressible Navier-Stokes equation, which is the form we will use in this module. The first equation represents conservation of mass, which means that the fluid cannot be created or destroyed. The second equation represents conservation of momentum, which means that the fluid cannot gain or lose momentum (it is analogous to Newton's second law, but for a fluid). Conservation of angular momentum is already baked into the second equation. The remainder of this chapter will be dedicated to the derivation of these equations, and in the following chapters we will explore various tools to analyse and solve them.

2.5 Lagrangian and Eulerian descriptions

Watch the video on 'Lagrangian and Eulerian descriptions of fluid flow' narrated by Prof. John Lumley of Cornell University from the National Committee for Fluid Mechanics. YouTube link: `www.youtube.com/watch?v=mdN8OOkx2ko`, duration: 27 mins.

The objective in developing the continuum theory of fluids is to decompose the fluid into an infinite number of infinitesimal fluid elements and impose Newton's laws of motion on each one of them. To do so, it is necessary to label each infinitesimal element of a fluid. There are two common approaches historically employed for doing so. These approaches are called the Eulerian and the Lagrangian descriptions of fluid flow.

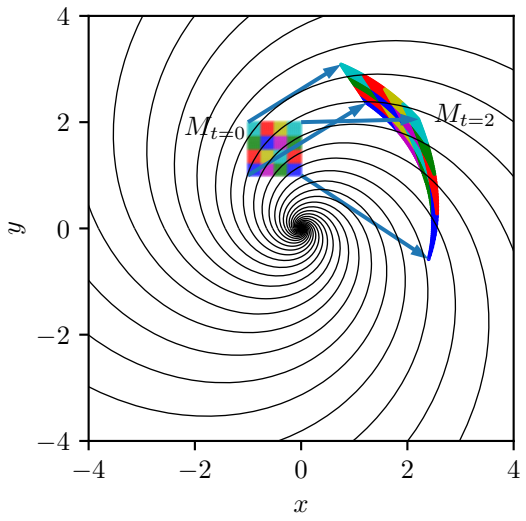


Figure 2.4: The motion and deformation of a Lagrangian fluid volume in a flow. The black curves denote streamlines of the flow spiralling outwards. The square coloured patch labelled $M_{t=0}$ shows the Lagrangian fluid volume at $t = 0$. The Lagrangian volume at $t = 2$ is shown as the deformed coloured patch labelled $M_{t=2}$. Blue arrows show the displacement of Lagrangian fluid particles forming the corners of the $M_{t=0}$. The colour of the sub-patches is preserved between $M_{t=0}$ and $M_{t=2}$.

2.5.1 Lagrangian description

The Lagrangian description seeks to label and track infinitesimal material volumes of fluid. Their boundaries move with the average speed of the molecules composing the boundaries (only possible in the continuum limit, when there are a large number of molecules on the boundary). Consequently, these volumes always contains the same amount of mass within them. Intrinsic[1] material properties are attributed to the volume by virtue of the material contained in the volume. For example, the velocity, the acceleration, the density, the temperature, the pressure, the concentration of any dissolved species could be attributed to the material volumes. These infinitesimal fluid volumes are also called (Lagrangian) fluid particles or fluid parcels.

A convenient label is the position of an infinitesimal fluid volume, X_j, at some reference time t_0. In the Lagrangian description, the fluid velocity is then given by the fields $U_i(X_j, t)$. This expression stands for the velocity of the Lagrangian fluid particle at time t, which was at position X_j at time t_0. Similar interpretation is applied to the density $R(X_j, t)$, acceleration $A_i(X_j, t)$, pressure $P(X_j, t)$, etc. As an informal convention, we will used capital letters to denote Lagrangian fields.

Caution: It is a common misunderstanding that the fluid inside the material volume is composed of the same material throughout the flow. This is not true. Instead, the correct statement, as mentioned earlier, is that the fluid inside the volume always contains the same amount of mass. Molecules composing the fluid move in and out of the volume such that no net mass leaves or enters the volume.

[1]Here intrinsic quantities are those which do not depend on the total amount of material in the volume. Examples of intrinsic quantities are pressure, temperature, density, as opposed to *extrinsic* quantities such as mass, volume, etc, which are proportional to the mass of material in the volume.

2.5.2 Eulerian description

The Eulerian description seeks to label the fluid particles by their current location in space. A convenient and common label are the coordinates of the points in space. All physical properties of the fluid are then described as fields over these labels, i.e., the Cartesian coordinates, x_j. For example, the velocity of the fluid is described as a field $u_i(x_j, t)$, which is interpreted as the velocity of the Lagrangian fluid particle that is at location x_j at time t. Note that even in the Eulerian description, physical quantities are attributed to the fluid elements and not to the space. Similarly, the density is denoted $\rho(x_j, t)$, the pressure as $p(x_j, t)$, the acceleration as $a_i(x_j, t)$, the stress as $T_{ij}(x_k, t)$, etc.

Definition: A Steady Eulerian Field

A steady field is one for which $\partial/\partial t = 0$. One of the advantages of the Eulerian description of fluid flow is that in many cases and in properly chosen reference frames, the resulting fields of velocity and pressure are steady. This significantly simplifies the analysis of such flows.

2.5.3 Eulerian versus Lagrangian descriptions

The two descriptions are invoked in fluid dynamics because they simplify different stages of the analysis. Newton's laws of motion are formulated in the Lagrangian description. Later, we will write the acceleration of Lagrangian particles in terms of the forces they experience. In this formulation, we will need to represent the force exerted by the neighbouring fluid particle as it pushes against the one under consideration, or slides past it. This will require us to track the neighbouring fluid particles as well. Now, the nature of fluid flows is such that the Lagrangian fluid volumes deform significantly with time and so do their neighbours, see Figure 2.4.

The Eulerian description does not suffer from the deformation of volumes. The volume remains fixed because we have defined it to be so. Neighbouring particles also remain in place without deforming. However, the laws of mechanics have been formulated in Lagrangian description. Whilst in the Eulerian description one sits at a fixed position and watches the flow pass (like standing at the bank of a river), in the Lagrangian description we 'follow the fluid' (as in a boat flowing with the river or on a weather balloon).

To bridge this gap, we use an Eulerian-Lagrangian map $\boldsymbol{x} = \boldsymbol{F}(\boldsymbol{X}, t)$, which tells us where the particle which was at $\boldsymbol{X}$ at time t_0 (i.e. $\boldsymbol{X} = \boldsymbol{F}(\boldsymbol{X}, t_0)$) is now to be found, in the Eulerian frame, at time t. I.e. it gives us the particle paths.

The inverse $\boldsymbol{X} = \boldsymbol{F}^{-1}(\boldsymbol{x}, t)$ can also be constructed, if paths don't cross, and tells us which particle (i.e. the one labelled by $\boldsymbol{X}$) is currently at position $\boldsymbol{x}$.

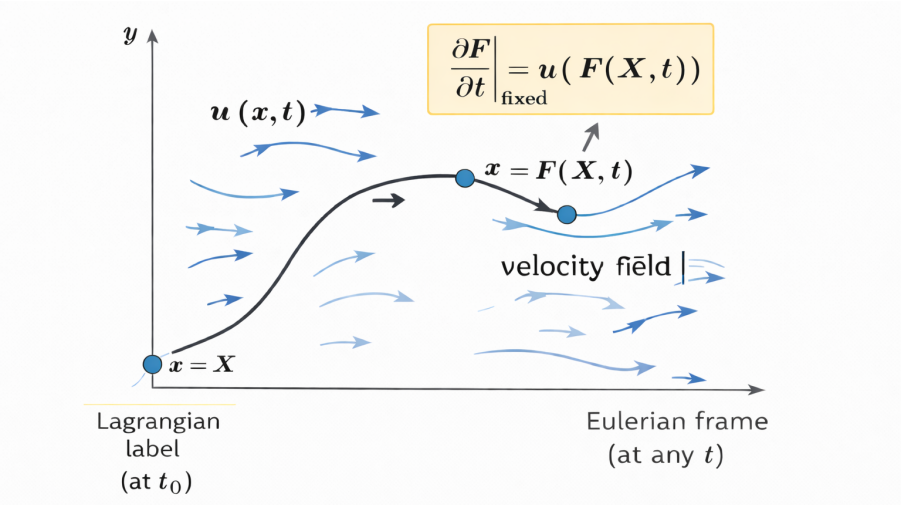


Figure 2.5: Creation of map between Lagrangian and Eulerian descriptions.

What we would like to know if how to transform between the descriptions and we do this by knowing the velocity of a fluid particle at a particular point in both descriptions. In the Lagrangian description the velocity of a particle labelled by $\boldsymbol{X}$ $(= \boldsymbol{F}(\boldsymbol{X}, t_0))$ at time t is given by $\partial \boldsymbol{F}/\partial t|_{\text{fixed } \boldsymbol{X}}$, where the 'fixed $\boldsymbol{X}$' isolates a particular Lagrangian particle/label, and we know this must be equal to the Eulerian velocity at that same

point ($\boldsymbol{x} = \boldsymbol{F}(\boldsymbol{X}, t)$). Therefore, we must solve

$$\left.\frac{\partial \boldsymbol{F}}{\partial t}\right|_{\text{fixed } \boldsymbol{X}} = \boldsymbol{u}\left(\boldsymbol{F}(\boldsymbol{X}, t), t\right), \tag{2.3}$$

with $\boldsymbol{F}(\boldsymbol{X}, t_0) = \boldsymbol{X}$. In other words, the rate of change of position of a Lagrangian fluid particle is simply the Eulerian velocity at the current location of the fluid particle.[2]

2.5.4 The material derivative and acceleration

There is a section on 'Material Derivative applied to the concentration of a pollutant in a river' starting at 12:00 mins in the video on 'Lagrangian and Eulerian descriptions of fluid flow' by Prof. John Lumley of Cornell University from the National Committee for Fluid Mechanics. YouTube link: `www.youtube.com/watch?v=mdN8OOkx2ko`, starting 12:00 mins to the end.

We now use this relation to develop the most common Eulerian to Lagrangian conversion of an operation, differentiation with time. For a generic intrinsic fluid property $c(\boldsymbol{x}, t)$ of arbitrary tensorial rank, the partial derivative with time $\partial c/\partial t$ merely represents the rate of change of c at the Eulerian location $\boldsymbol{x}$. It does not account for the fact that at an infinitesimally later time, a different fluid particle now occupies the location $\boldsymbol{x}$. Since most physical laws are formulated in terms of the properties of the Lagrangian fluid particle, it is useful to instead derive an expression for the rate of change of c following the fluid particle.

To do so, note that $c\left(F_i(\boldsymbol{X}, t), t\right)$ is the property c evaluated at the position F_i of the Lagrangian particle labelled by $\boldsymbol{X}$[3]. The location of the particle itself changes with time. Therefore, the rate of change of c following the fluid particle is

$$\begin{aligned}
\left.\frac{\partial c\left(F_i(\boldsymbol{X}, t), t\right)}{\partial t}\right|_{\text{fixed } \boldsymbol{X}} &= \left.\frac{\partial c}{\partial t}\right|_{\text{fixed } \boldsymbol{x}} + \left.\frac{\partial F_i}{\partial t}\,\frac{\partial c}{\partial x_i}\right|_{\text{fixed } t} \qquad \ldots \quad \text{recalling } x_i = F_i(\boldsymbol{X}, t) \\
&= \left.\frac{\partial c}{\partial t}\right|_{\text{fixed } \boldsymbol{x}} + u_i\left.\frac{\partial c}{\partial x_i}\right|_{\text{fixed } t} \qquad \ldots \quad \text{using } \frac{\partial F_i}{\partial t} = u_i \text{ from (2.3)} \\
&= \left.\frac{\partial c}{\partial t}\right|_{\text{fixed } \boldsymbol{x}} + \boldsymbol{u} \cdot \nabla c \qquad \ldots \quad \text{in vector notation.}
\end{aligned} \tag{2.4}$$

Definition: The Material Derivative a.k.a. The Lagrangian Derivative a.k.a. The Total Derivative a.k.a. The Substantial Derivative

The material derivative is the rate of change of an intrinsic property of a Lagrangian fluid particle expressed in terms of Eulerian quantities. It is symbolically denoted as $\dfrac{D}{Dt}$ and is given by the expression

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + (\boldsymbol{u} \cdot \nabla). \tag{2.5}$$

As written earlier, when this operator is applied to a scalar field c it gives

$$\frac{Dc}{Dt} = \frac{\partial c}{\partial t} + \boldsymbol{u} \cdot \nabla c = \frac{\partial c}{\partial t} + u_i \frac{\partial c}{\partial x_i}. \tag{2.6}$$

Interpretation of the terms in the Material derivative:

1. The first term is the local rate of change at a fixed Eulerian position $\boldsymbol{x}$ due to temporal changes.
2. The second is the 'convective rate of change' caused by the velocity $\boldsymbol{u}$ driving fluid elements through spatial gradients in the quantity of interest. The next example illustrates how a steady (i.e. time independent) Eulerian field could imply a non-zero time-rate-of-change for Lagrangian properties because of the convective rate-of-change term.

[2] Note that there is a slight abuse of notation here. For the Lagrangian coordinates and the Eulerian-Lagrangian map we use bold upper case letters, which we usually reserve for tensors. However, given that we use specifically $\boldsymbol{X}$ and $\boldsymbol{F}$ (which we typically don't use with tensors) and that they only appear in this section, confusion should be minimal.

[3] Here we mix index notation for $\boldsymbol{F}$ with vector notation for $\boldsymbol{X}$.

Example: If there is a concentration of pollutant $c = c(x)$ in a river that flows steadily with constant speed $\boldsymbol{u} = (u_0, 0, 0)$ (see Figure 2.6), how does the concentration change in a fluid element that 'follows the fluid'?

Note that (a) the concentration at a fixed point does not change with time $\partial_t c = 0$ but (b) the concentration inside a fluid element that 'follows the fluid' will change in time, at a rate given by the material derivative

$$\frac{Dc}{Dt} = \frac{\partial c}{\partial t} + u\frac{\partial c}{\partial x} + v\frac{\partial c}{\partial y} + w\frac{\partial c}{\partial z} = u_0\frac{\partial c}{\partial x}, \tag{2.7}$$

which is non-zero if and only if there is flow $u_0 \neq 0$ and spatial gradients in $\partial_x c \neq 0$.

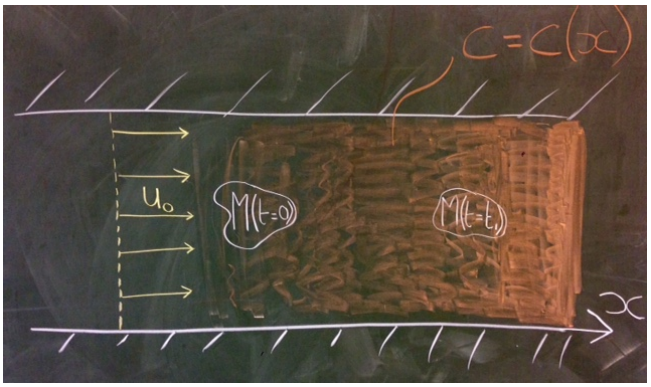


Figure 2.6: Evolution of a fluid element through a spatial gradient in concentration $c = c(x)$ showing that the concentration inside the fluid element will change from $t = 0$ to $t = t_1$ even though the flow is steady $\partial_t \boldsymbol{u} = \boldsymbol{0}$.

It is the material derivative which gives us the rate of change of velocity, i.e. the acceleration $\boldsymbol{a}$ or a_i, of the fluid element

$$a_i = \frac{Du_i}{Dt} = \frac{\partial u_i}{\partial t} + u_j\frac{\partial u_i}{\partial x_j} \qquad \text{or} \qquad \boldsymbol{a} = \frac{D\boldsymbol{u}}{Dt} = \frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}. \tag{2.8}$$

Notably, even when the flow is *steady* in the Eulerian frame, so that $\partial_t \boldsymbol{u} = 0$, the acceleration still has a component $(\boldsymbol{u}\cdot\nabla)\boldsymbol{u}$ due to the spatial variation of the velocity field—this is the convective acceleration.

Example: If we take a one-dimensional steady flow $\boldsymbol{u} = (u(x), 0, 0)$ (see Figure 2.7), then what is the acceleration of fluid particles?

$$a_1 = \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} = u\frac{\partial u}{\partial x}, \quad a_2 = a_3 = 0, \tag{2.9}$$

so that where there is a spatial gradient in u there is an acceleration.

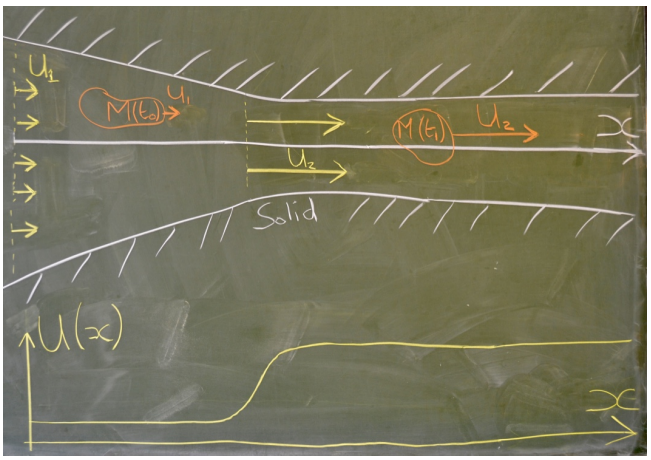


Figure 2.7: The acceleration of a fluid element M as it travels from an area of low speed (left) at time t_0 to a region of high speed (right) at $t = t_1$. Therefore, despite the flow being steady, so that at a fixed point $\partial_t u = 0$, the fluid inside M feels an acceleration due to its motion.

2.5.5 Reynolds' transport theorem

To describe laws of physics in terms of continuum quantities, we also need the time-rate-of-change of a total quantity enclosed within a finite Lagrangian volume Ω. If the volumetric density of a property B is b, i.e., if the total amount of B in any volume Ω may written as

$$B = \int_{\Omega} b \, \mathrm{d}V, \tag{2.10}$$

then

$$\frac{D}{Dt}\left(\int_{\Omega(t)} b \, \mathrm{d}V\right) = \int_{\Omega} \frac{\partial b}{\partial t} \, \mathrm{d}V + \int_{\partial\Omega} b(\boldsymbol{u}\cdot\boldsymbol{n}) \, \mathrm{d}A, \tag{2.11}$$

where $\partial\Omega$ is the bounding surface of Ω and $\boldsymbol{u}$ is the Eulerian velocity. The first term on the right hand side is simply the contribution that arises from the rate of change of b within the volume, even when the Lagrangian volume stays fixed. The second term arises from the motion of the Lagrangian volume under the Eulerian velocity $\boldsymbol{u}$. Here $(\boldsymbol{u}\cdot\boldsymbol{n})\,\mathrm{d}A$ is the rate at which volume of fluid enters the surface $\partial\Omega$ through an infinitesimal area $\mathrm{d}A$ (as shown in Figure 2.8), and its product with b gives the amount of B entering the volume per unit time.

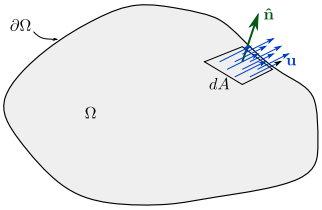


Figure 2.8: Schematic showing a motion of a small area element $\mathrm{d}A$ of $\partial\Omega$ of the surface bounding the volume Ω. Here $\boldsymbol{u}$ is the Eulerian velocity field on the surface and $\boldsymbol{n}$ is the unit normal. The quantity $(\boldsymbol{u}\cdot\boldsymbol{n})\,\mathrm{d}A$ is the rate at which volume of fluid is entrained into Ω through $\mathrm{d}A$.

2.6 Flow visualisation

Watch the video on 'Flow Visualisation' narrated by Prof. Stephen J. Kline from Stanford University. YouTube link: `www.youtube.com/watch?v=nuQyKGuXJOs`, duration: 31 mins.

The most common way to visualise a flow, theoretically (once we have $\boldsymbol{u}$) or experimentally, is by drawing a set of curves. Four types of curves are commonly used in fluid dynamics. Some are purely theoretical concepts, while others are used for experimental convenience.

To visualise a flow field experimentally, one usually places illuminated passive tracers (e.g. dye, bubbles or beads that don't alter the flow) into the flow (e.g. Figure 2.9). The various curves are then realized by the choice of the initial release of the tracer particles and the visualization technique.

1. **Particle paths or pathlines:** The paths of fluid elements ('*particle paths*') $F_i(t)$, which are simply the locus of points traced by a Lagrangian fluid element. Equation (2.3) may be used to theoretically determine these from an Eulerian description of the velocity field. Typically a small amount of illuminated passive tracers are introduced in the flow, and the resulting images superimposed in post-processing to reveal the pathlines for a flow. Long exposures, if possible, also yield pathlines.

2. **Streaklines:** The instantaneous locus of all the Lagrangian fluid elements that have passed through or will pass through a fixed point in space is a 'streakline'. It is easier to visualize streaklines experimentally, which can be done by introducing a steady streak of a passive tracer in the fluid (hence the name). Theoretically, streakline at time t passing through a point $\boldsymbol{X}_0$ may be obtained by solving the equation

$$\boldsymbol{F}(\boldsymbol{X}, s) = \boldsymbol{X}_0, \quad t_i \le s \le t_f, \tag{2.12}$$

for $\boldsymbol{X}(s)$, where t_i and t_f are the initial and final times of interest, and then plotting $\boldsymbol{F}(\boldsymbol{X}(s), t)$ for fixed t while varying s as a parameter.

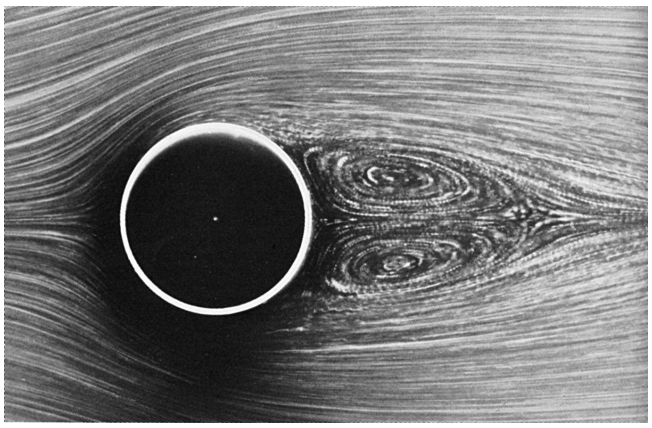


Figure 2.9: Steady laminar flow (from left to right) around a solid cylinder with streamlines/particle paths showing the trajectories of fluid elements

3. **Timelines:** A 'time line' is the subsequent shape of a curve made of Lagrangian fluid elements. It is more convenient to visualize a timeline experimentally than it is to calculate it theoretically. For the experimental visualization, a passive tracer is released at a time instance in the shape of a curve. The subsequent evolution of the curve naturally visualizes the timeline. To construct the timeline from an Eulerian velocity field, we start with a curve parameterized as $\boldsymbol{X}_0(s)$ by parameter s. The subsequent shape of the curve at time t is $\boldsymbol{F}(\boldsymbol{X}_0(s), t)$, which is the timeline at time t.

4. **Streamlines:** A 'streamline' is a curve which is everywhere tangent to the velocity vector of an Eulerian flow field. Streamlines are mathematical constructions, and, therefore, are easier to construct theoretically. Parameterized by an independent parameter s, the streamlines $(x, y, z) = (x(s), y(s), z(s))$ at time t are obtained by solving

$$\left.\frac{\partial x_i(s)}{\partial s}\right|_{\text{fixed } t} = u_i. \tag{2.13}$$

   To visualize streamlines in an experiment, it is necessary to measure the Eulerian velocity field. This measurement can be accomplished using techniques such as particle image velocimetry (PIV).

   Streamlines, although theoretical, have deep physical significance, especially for incompressible flows. Because the volume of a Lagrangian fluid element is conserved for an incompressible fluid, the only way a fluid element can move is by displacing the one ahead of it. At the same instance, the one ahead in turn displaces the one ahead of it, and so on. An incompressible fluid can flow only by forming such chains of instantaneous displacement of fluid elements. A streamline is nothing but one such chain.

   Because of the aforementioned interpretation, streamlines cannot terminate in an incompressible fluid. They must extend all the way to infinity or form closed curves.

In general, the particle paths differ from the streamlines, which differ from streaklines. The following example illustrates that.

Example (from Acheson Ex 1.8): Consider the two-dimensional flow given by

$$\boldsymbol{u} = (u_0, kt, 0), \qquad \text{with} \qquad u_0, k > 0.$$

What are the pathlines, streamlines and streaklines?

The pathlines are obtained from solving

$$\frac{\partial F_1}{\partial t} = u_0, \qquad \frac{\partial F_2}{\partial t} = kt \qquad \frac{\partial F_3}{\partial t} = 0$$

for fixed initial position $(F_1, F_2, F_3) = (X, Y, Z)$ at $t = 0$. Therefore,

$$(F_1, F_2, F_3) = \left(u_0 t + X, \frac{1}{2}kt^2 + Y, Z\right)$$

which, notably, could be used to transform from Eulerian to Lagrangian coordinates (by expressing (x, y, z) in terms of (X, Y, Z)). We can eliminate t to find the particle paths

$$y = \frac{k}{2}\left(\frac{x - X}{u_0}\right)^2 + Y$$

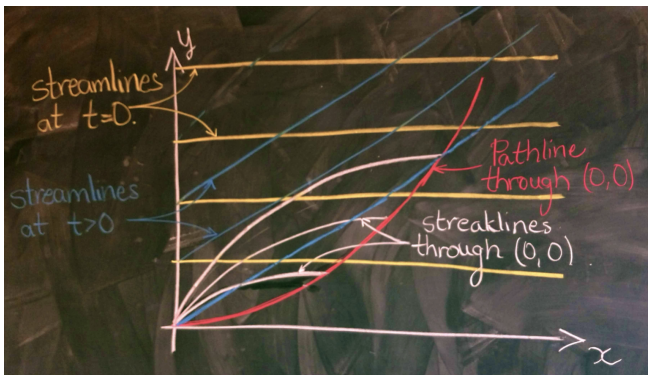


Figure 2.10: Sketch showing how streamlines change in time, differing at $t = 0$ and $t = t_1$, the particle path $y = \frac{k}{2}(x/u_0)^2$ of a fluid element starting from $X = Y = 0$ at $t = 0$, and streaklines for three different times passing through $X = Y = 0$.

which are parabolas with their minimum at (X, Y) and $k/2u_0^2$ as the coefficient of the quadratic term (see figure 2.10 for the pathline through the origin).

The streamlines come from

$$\frac{\partial x}{\partial s} = u_0, \qquad \frac{\partial y}{\partial s} = kt, \qquad \frac{\partial z}{\partial s} = 0$$

for a *fixed* t. Therefore

$$(x, y, z) = (u_0 s + a, kts + b, c)$$

for a, b, c constant. Eliminating s we find that

$$y = \frac{kt}{u_0} x + (b - kta/u_0)$$

showing that the streamlines are straight lines whose slope will increase in time. The result is figure 2.10.

The streaklines are also easy to calculate. Upon solving equation (2.12), we get for $\boldsymbol{X}(s)$

$$X(s) = X_0 - us, \quad Y(s) = Y_0 - \frac{1}{2}ks^2, \quad Z(s) = Z_0.$$

The equation for the curve for the streakline is $\boldsymbol{F}(\boldsymbol{X}(s), t)$, which is

$$x = X_0 - us + ut, \quad y = Y_0 - \frac{1}{2}ks^2 + \frac{1}{2}kt^2, \quad z = Z_0.$$

These are to be plotted for fixed t while varying s as a parameter, which means that they are inverted parabolas through $(X_0 + ut, Y_0 + kt^2/2, Z_0)$. The one through the origin is plotted in figure 2.10.

2.7 Deformation of infinitesimal fluid elements

The study of the deformation of continuous media (including solid and fluid dynamics) is a course in its own right, so we will only consider the concepts we will require to formulate the equations of fluid mechanics. In particular, we will attempt to understand how flow deforms fluid elements; as it is this (rate of) deformation that is related to the internal stress (and hence forces) required for formulating Newtons second law for the fluid.

The velocity can be decomposed into fundamental components which specify the kinematics of the flow (the geometry of the motion), i.e. how small fluid elements are deformed by the flow. To do so, consider the flow at an infinitesimal distance δx_j from a reference point x_j, see figure 2.11.

The Taylor expansion of $u_i(x_j + \delta x_j)$, keeping only the terms up to the first power in δx_j, gives

$$u_i(x_j + \delta x_j, t) = u_i(x_j, t) + \frac{\partial u_i}{\partial x_j}\,\delta x_j = \underbrace{\underbrace{u_i(x_j)}_{\text{Translation}} + \underbrace{r_{ij}\,\delta x_j}_{\text{Rotation}}}_{\text{Rigid Body Motions}} + \underbrace{e_{ij}\,\delta x_j}_{\text{Shearing and Extension}} \tag{2.14}$$

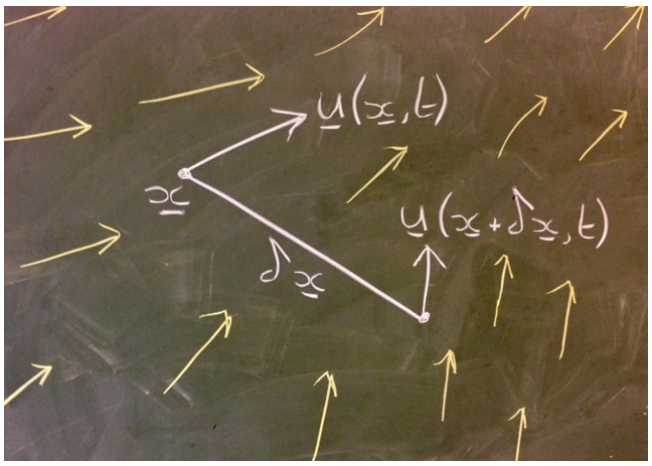


Figure 2.11: Flow kinematics - how the flow velocity varies in the vicinity of (a distance $\delta\mathbf{x}$ from) a point $\mathbf{x}$.

where we have decomposed $\partial_{x_j} u_i$ into an anti-symmetric ($r_{ij} = -r_{ji}$) *rate of rotation tensor*

$$r_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i}\right) \quad \text{or in vector notation} \quad \boldsymbol{r} = \frac{1}{2}\left(\nabla \boldsymbol{u} - \nabla \boldsymbol{u}^T\right),$$

and the symmetric ($e_{ij} = e_{ji}$) *rate of strain tensor*

$$e_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) \text{or in vector notation} \quad \boldsymbol{e} = \frac{1}{2}\left(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^T\right).$$

Note that $\nabla \boldsymbol{u} = \boldsymbol{e} + \boldsymbol{r}$.

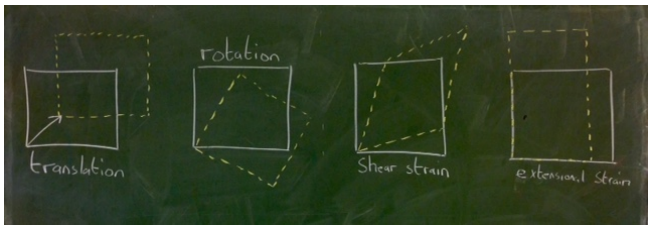


Figure 2.12: Different modes of deformation to a fluid element (in 2D).

Each of the terms contributing to the velocity

$$u_i(x_j + \delta x_j, t) = u_i^T + u_i^R + u_i^S, \qquad \text{where} \qquad u_i^T = u_i(x_j), \quad u_i^R = r_{ij}\ \delta x_j, \quad u_i^S = e_{ij}\ \delta x_j \tag{2.15}$$

can now be analysed. The first term in (2.14) is due to rigid body translation $u_i^T = u_i(x_j, t)$. If $r_{ij} = e_{ij} = 0$, then all fluid elements have the same velocity.

Rate of rotation OR Vorticity

Due to the antisymmetry of r_{ij}, there are only 3 independent components (all diagonal elements zero) and we can rewrite the term $u_i^R = r_{ij}\ \delta x_j$ as a cross product $u_i^R = \epsilon_{ijk}\Omega_j \delta x_k$ or $\boldsymbol{u}^R = \boldsymbol{\Omega} \times \delta\mathbf{x}$ (see figure 2.13), where $\boldsymbol{\Omega} = (r_{32}, r_{13}, r_{21})$ is the rate of rotation *vector* (you can check this). Thus we see that this term is associated with rigid body rotation about the reference point with rotational rate $\boldsymbol{\Omega}$.

Notably, the rate of rotation vector $\boldsymbol{\Omega}$ is equal to half the *vorticity* vector

$$\boldsymbol{\omega} = \nabla \times \boldsymbol{u} = 2\boldsymbol{\Omega}, \qquad \text{so that} \qquad \boldsymbol{u}^R = \frac{1}{2}\boldsymbol{\omega} \times \delta\mathbf{x} \tag{2.16}$$

which is an important fluid mechanical quantity we will repeatedly encounter - now we can recognise it as a measure of the local rate of rotation in the fluid. In Cartesian coordinates it is given by

$$\boldsymbol{\omega} = \begin{vmatrix} \mathbf{e}_x & \mathbf{e}_y & \mathbf{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix} = \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}, \frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}, \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right).$$

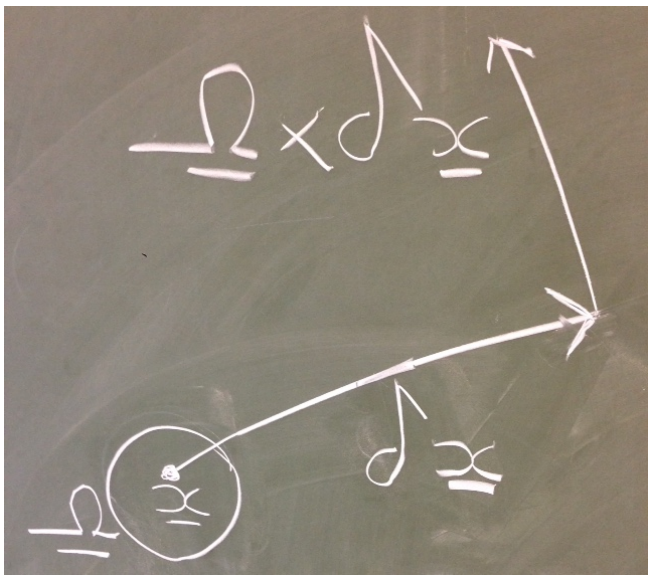


Figure 2.13: Sketch illustrating the rigid body rotation $\boldsymbol{u}^R = \boldsymbol{\Omega} \times \delta\mathbf{x}$ induced a distance $\delta\mathbf{x}$ away from a point $\mathbf{x}$ by the rotation rate $\boldsymbol{\Omega}$. Here, $\boldsymbol{\Omega}$ points out of the paper.

Example: How many components of vorticity does a 2D flow have?

For a two-dimensional flow $\boldsymbol{u} = (u(x, y, t), v(x, y, t), 0)$, the vorticity has only one component $(0, 0, \omega)$ where $\omega = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$, with a rotation axis in the z-direction.

Strain rate:

As the first two terms in (2.14) are associated with rigid body motion, it is the third term $u_i^S = e_{ij}\delta x_j$ which represents the straining motions which distinguish the flow through the relative motion of adjacent fluid elements. The rate of strain tensor e_{ij} can be further decomposed into (a) diagonal elements which are associated with extensional motion and (b) off-diagonal elements give the shear rate of strain.

Consider how the components of e_{ij} deform an infinitesimal rectangular fluid element of size $\delta x \times \delta y$ (figure 2.14) in a two-dimensional flow $\boldsymbol{u} = (u(x, y, t), v(x, y, t), 0)$. For simplicity, we will remove translation of the fluid element by considering flow relative to the point A, so that a Taylor expansion gives the flow components at adjacent points to A shown in figure 2.14.

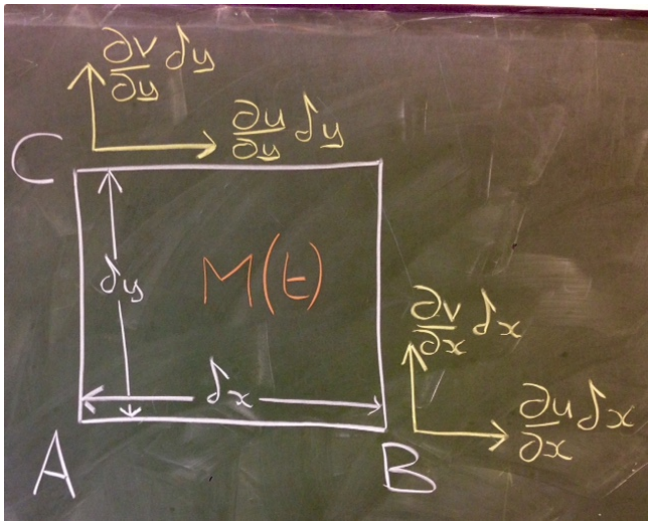


Figure 2.14: The velocity components relative to a point A in a material element M at time t.

From figure (2.15) we can see that *extensional rate of strain* acts to increase the lengths of the sides. For

example, taking the line AB, in a time δt the point B moves so that the new length of AB at time $t+\delta t$ is

$$\delta x(t+\delta t) = \delta x(t) + \underbrace{\frac{\partial u}{\partial x}\delta x}_{\text{speed}}\underbrace{\delta t}_{\text{time}}. \tag{2.17}$$

Therefore, the (infinitesimal) strain (extension/original length)

$$\delta S_x = (\delta x(t+\delta t) - \delta x(t))/\delta x(t) \overset{(2.17)}{=} \frac{\partial u}{\partial x}\delta t \quad \text{so the rate of strain is} \quad \frac{\partial S_x}{\partial t} = \frac{\partial u}{\partial x} = e_{11}$$

as expected.

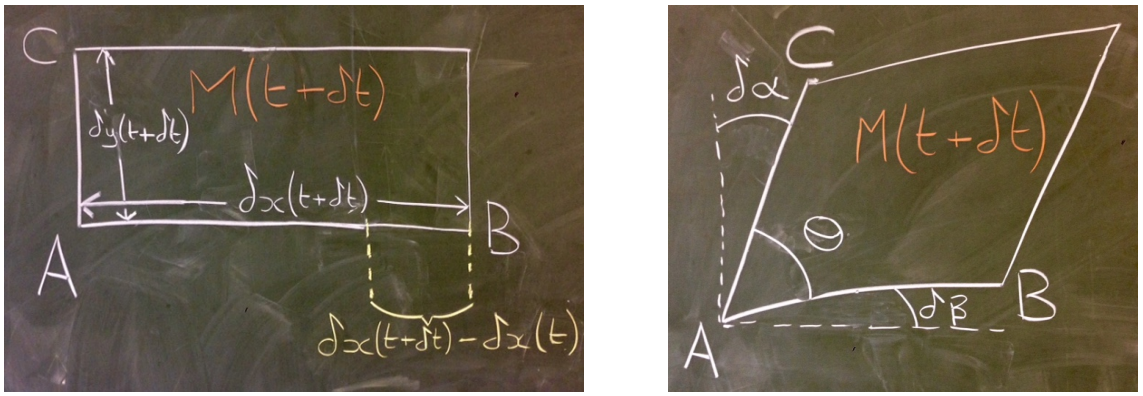


Figure 2.15: Left: deformation due to extensional strain on the material element at a time $t+\delta t$. Right: influence of shear strain on the material element at a time $t+\delta t$.

It can be shown that the extensional components change the volume V of the fluid element as

$$\frac{1}{V}\frac{DV}{Dt} = e_{kk} \qquad \text{or in vector notation} \qquad \frac{1}{V}\frac{DV}{Dt} = \nabla\cdot\boldsymbol{u} \tag{2.18}$$

so that for an incompressible fluid, where the volume of the fluid element does not change in time ($D_tV = 0$), we have $e_{kk} = e_{11} + e_{22} + e_{33} = 0$. In other words, extension of a fluid element in one direction must lead to contraction in one of the other directions.

The *shear* rate of strain acts to drive the element from its rectangular shape. This can be quantified by measuring the angle between the sides. It can be shown that (see Examples Sheet 2) the angle θ at CAB in figure 2.14 evolves according to

$$-\frac{\partial\theta}{\partial t} = 2e_{12}.$$

So we have seen that it is the vorticity and rate of strain that give us all of the information we require to understand how fluid elements are locally deformed, with the former telling us about the rotation of the flow and the latter about the relative motion of fluid elements (which is related to viscous stress).

Example (Adapted from Acheson §1.4): Consider the 'shear flow' $\boldsymbol{u} = (\beta y, 0, 0)$. How does this simple flow deform fluid elements? E.g., what is the contribution from deformation to (2.15)?

Let's consider all the different contributions to (2.15).

- The translational term is just $u_i^T = (\beta y, 0, 0)$.
- The vorticity will only have a component in the z-direction $\boldsymbol{\omega} = (0, 0, \omega)$ which is

$$\omega = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = -\beta.$$

So there is rotation in the flow, despite the streamlines and particle paths being straight. This is because the velocity at the point C is greater than that at A so the line CA rotates clockwise (figure 2.16). The rate of rotation will be $\omega/2 = -\beta/2$.

Therefore, the contribution to (2.15) is

$$u_i^R = \frac{1}{2}\epsilon_{ijk}\omega_j\delta x_k = \frac{1}{2}(\beta dy, -\beta dx, 0)$$

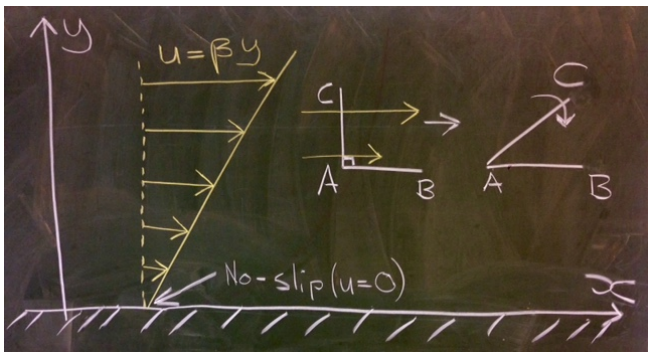


Figure 2.16: The deformation of two initially perpendicular fluid line element (i.e. they 'follow the fluid') in a shear flow showing how AC rotates - hence the flow contains vorticity.

- The only non-zero entries in the rate of strain tensor will be due to $\partial_y u = \beta$. Therefore, extensional strains are all zero (ensuring incompressibility) and the only non-zero entries are $e_{12} = e_{21} = \beta/2$. Consequently, the contribution to (2.15) is
$$u_i^S = \frac{1}{2}(\beta dy, \beta dx, 0).$$
- Therefore, even in this simple flow there are rigid body motions of translation and rotation as well as deformation.

2.8 Conservation laws in continuum mechanics

In continuum mechanics, it is customary to refer to laws of nature as conservation laws. An obvious example is the law of conservation of mass. As the fluid flows, mass can neither be created nor could it disappear. A less obvious one is Newton's second law of motion. While momentum is not "conserved", the law constrains the rate of change of momentum to be equal to force. Thus, by considering forces to be "sources" of momentum, one may still use the language of "conservation laws" to describe Newton's second law. Examples of quantities to develop conservation laws for include mass, (linear) momentum, angular momentum (or moment of momentum), internal energy, total energy, and entropy. We will limit ourselves to mass, momentum and angular momentum.

Three main concepts need to be introduced to develop the conservation law for an abstract quantity, which we will call B. These are the volumetric density of B, the volumetric rate of generation (also known as the source) of B, and the rate of exchange of B between neighbouring fluid volumes.

1. Volumetric density of B is the quantity b such that the total amount of B in any volume Ω could be written as
$$\int_\Omega b \, \mathrm{d}V.$$
Examples are presented in Table 2.1. The tensorial rank of b and B are identical. Furthermore, b is an Eulerian field, i.e., $b(\boldsymbol{x}, t)$.

B	b	Notes
Mass	ρ	ρ is the mass density
Momentum	$\rho\boldsymbol{u}$	$\boldsymbol{u}$ is the Eulerian fluid velocity
Angular momentum	$\boldsymbol{x} \times \rho\boldsymbol{u}$	

Table 2.1: Examples of physical quantities B and their volumetric density.

2. Volumetric source term also known as **body source term,** q^B, is the volumetric density of the rate of generation of B. The rate of increase of B in any volume Ω due to the volumetric source is
$$\int_\Omega q^B \, \mathrm{d}V.$$

Examples include gravitational force as a source for momentum and ohmic generation of heat as an example for energy. The source $q^B(\boldsymbol{x}, t)$ is a field with its tensorial rank being equal to that of b.

3. Surface source term, h^B, originates from the fact that the volume Ω may gain the quantity B from the neighbouring fluid. For example, the force exerted on Ω by the neighbouring fluid contributes to the rate of change of momentum in Ω. By Newton's third law, the volume Ω exerts an equal and opposite force on the surrounding fluid. In this sense, the surface source may be interpreted as an exchange of B between neighbouring fluid volumes.

A concrete example of such a term is the action of fluid pressure to generate force, or rate of change of momentum, on a volume Ω. Due to the pressure, $h^B = -p\boldsymbol{n}$ because the net force due to pressure is

$$\int_{\partial\Omega} -p\boldsymbol{n} \, \mathrm{d}A.$$

Generally, the strength of such a surface source term is proportional to area of the bounding surface, rather than volume as for the volumetric source term. In general, the contribution to the rate of change of B due to the surface source is written as

$$\int_{\partial\Omega} h^B \, \mathrm{d}A.$$

The tensorial rank of h^B and of b are equal. Clearly, h^B must also be a field, i.e. $h^B(\boldsymbol{x}, t)$. One important way h^B differs from both b and q^B is that in addition to being a field, h^B also has dependency on the orientation of the surface $\partial\Omega$. In other words, if a point $\boldsymbol{X}$ is on the boundary of two different volumes, h^B can have different values depending on the unit outward normal to their bounding surfaces. This is represented through an explicit dependence on the unit normal as $h^B(\boldsymbol{x}, t; \boldsymbol{n})$.

The conservation law for B may now be written as

$$\frac{D}{Dt}\int_\Omega b \, \mathrm{d}V = \int_\Omega q^B \, \mathrm{d}V + \int_{\partial\Omega} h^B \, \mathrm{d}A. \tag{2.19}$$

2.8.1 The dimensional inconsistency of the surface source term

Let us consider an arbitrary shape for Ω. Now consider the limit of (2.19) as Ω shrinks in size indefinitely, whilst retaining its shape. Let L be a length characterizing Ω. The volume of Ω scales a L^3 and the area as L^2. Thus, the first two terms of (2.19) scale as proportional to L^3, while the surface source term scales proportional to L^2, i.e.,

$$\underbrace{\frac{D}{Dt}\int_\Omega b \, \mathrm{d}V}_{\propto L^3} = \underbrace{\int_\Omega q^B \, \mathrm{d}V}_{\propto L^3} + \underbrace{\int_{\partial\Omega} h^B \, \mathrm{d}A}_{\propto L^2}. \tag{2.20}$$

As the volume shrinks, L becomes successively smaller, and relative to the surface source term, the first two terms become smaller. In the limit $L \to 0$, the conservation law for B (2.19) reduces to

$$\int_{\partial\Omega} h^B(\boldsymbol{x}, t; \boldsymbol{n}) \, \mathrm{d}A = 0. \tag{2.21}$$

In other words, for *any* infinitesimal volume, the contribution of the surface source term that is proportional to the area of the bounding surface must vanish. The consequence of this condition is the basis for all continuum mechanics.

2.8.2 Cauchy's tetrahedron argument

Augustine-Louis Cauchy, a french mathematician, engineer and physicist, reasoned the compelling consequence of (2.21) by taking the shape of the volume Ω to be a tetrahedron, as shown in Figure 2.17. The tetrahedron has four faces; the area of the inclined face is given by the (vector[4]) $A\boldsymbol{n}$, while the remaining three faces are $-A_1\boldsymbol{e}_1$, $-A_2\boldsymbol{e}_2$ and $-A_3\boldsymbol{e}_3$, respectively, so that $A\boldsymbol{n} = A_i\boldsymbol{e}_i$. We will now evaluate the terms of the integral in (2.21), but note because of the infinitesimal size of the tetrahedron, we will not need the distribution of h^B over the areas. The value of h^B may be considered a constant with $\boldsymbol{x}$, and we will drop the dependence on $\boldsymbol{x}$ in the

[4]Reminder: the direction of a planar area is normal to its face.

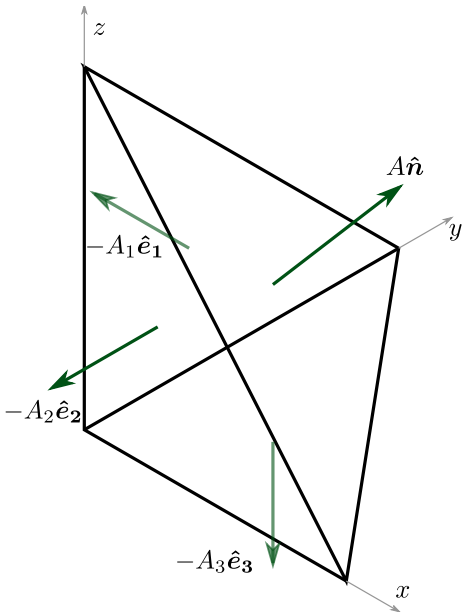


Figure 2.17: Cauchy's tetrahedron.

notation here. But because we could choose a tetrahedron of any proportions, the dependence on $\boldsymbol{n}$ remains, which we will explicitly write.

The contribution to the integral in (2.21) from the four face may now be written as

$$h^B(\boldsymbol{n})A + h^B(-\boldsymbol{e}_1)A_1 + h^B(-\boldsymbol{e}_2)A_2 + h^B(-\boldsymbol{e}_3)A_3 = 0. \tag{2.22}$$

By the principle of exchange of B, $h^B(-\boldsymbol{e}_1) = -h^B(\boldsymbol{e}_1)$, which arises from the fact that the amount of B gained by the tetrahedron is lost by the neighbouring fluid. Let's further define (in index notation) $T^B_{\ldots j} = h^B(\boldsymbol{e}_j)$, where the $\ldots$ on the indices of T^B accounts for the free indices of h^B, which may have a nonzero tensorial rank. In this way, T_B has a tensorial rank one higher than h^B. Furthermore, it follows from the geometry for tetrahedrons, $A_i = An_i$. With this notation, (2.22) becomes

$$h^B(\boldsymbol{n}) == T^B_{\ldots j} n_j. \tag{2.23}$$

This dependence of the surface source term on the surface normal helps satisfies the condition (2.21) not only for a tetrahedron but also for any Ω. This is so because

$$\int_{\partial\Omega} h^B(\boldsymbol{x}, t; \boldsymbol{n})\ \mathrm{d}A = \int_{\partial\Omega} T^B_{\ldots j} n_j\ \mathrm{d}A = \int_\Omega \frac{\partial T^B_{\ldots j}}{\partial x_j}\ \mathrm{d}V, \tag{2.24}$$

where the last step follows from the tensorial form of the divergence theorem. This step converts the surface integral to a volume integral, and in the limit of vanishing volume Ω, this term scales as L^3 and not as L^2.

The conservation law for B now reads

$$\frac{D}{Dt}\int_\Omega b\ \mathrm{d}V = \int_\Omega q^B\ \mathrm{d}V + \int_{\partial\Omega} T^B_{\ldots j} n_j\ \mathrm{d}A = \int_\Omega q^B\ \mathrm{d}V + \int_\Omega \frac{\partial T^B_{\ldots j}}{\partial x_j}\ \mathrm{d}V. \tag{2.25}$$

Applying the Reynolds transport theorem and the divergence to the left hand side yields two additional forms of this equation.

$$\int_\Omega \frac{\partial b}{\partial t}\ \mathrm{d}V + \int_{\partial\Omega} bu_j n_j\ \mathrm{d}A = \int_\Omega q^B\ \mathrm{d}V + \int_{\partial\Omega} T^B_{\ldots j} n_j\ \mathrm{d}A = \int_\Omega q^B\ \mathrm{d}V + \int_\Omega \frac{\partial T^B_{\ldots j}}{\partial x_j}\ \mathrm{d}V, \tag{2.26}$$

$$\int_\Omega \left(\frac{\partial b}{\partial t} + \frac{\partial (bu_j)}{\partial x_j}\right)\ \mathrm{d}V = \int_\Omega q^B\ \mathrm{d}V + \int_{\partial\Omega} T^B_{\ldots j} n_j\ \mathrm{d}A = \int_\Omega q^B\ \mathrm{d}V + \int_\Omega \frac{\partial T^B_{\ldots j}}{\partial x_j}\ \mathrm{d}V, \tag{2.27}$$

Equation (2.25-2.27) are integral representations of the law of conservation of B. The term $bu_j - T_{\ldots j}$ is the total flux of b and bu_j is the convective flux. (In general, the flux of b is the rate at which b crosses a surface

per unit area.) Conservation laws also have a differential form. To derive it, note that (2.27) holds irrespective of the volume Ω. This is only possible if the integrand itself vanishes everywhere, i.e.,

$$\frac{\partial b}{\partial t} + \frac{\partial (b u_j)}{\partial x_j} = q^B + \frac{\partial T^B_{\ldots j}}{\partial x_j}. \tag{2.28}$$

We will see that in the differential representation of general conservation laws, the rate of change of the density b is balanced by the divergence of the flux of b and the volumetric generation of b.

We now apply the result of this general analysis to specific quantities.

2.9 Conservation of mass (mass balance)

Now we use the (classical) physical law that mass can neither be created or destroyed. This implies, when $b = \rho$, then $q^B = T^B_j = 0$, and (2.25-2.27) becomes

$$\frac{D}{Dt} \int_\Omega \rho \, \mathrm{d}V = 0, \tag{2.29}$$

$$\int_\Omega \frac{\partial \rho}{\partial t} \, \mathrm{d}V + \int_{\partial\Omega} \rho u_j n_j \, \mathrm{d}A = 0, \tag{2.30}$$

$$\int_\Omega \left(\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u_j)}{\partial x_j} \right) \, \mathrm{d}V = 0. \tag{2.31}$$

Equations (2.29-2.31) are integral representations of the law of conservation of mass. The differential representation corresponding to (2.28)

$$\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u_j)}{\partial x_j} = 0 \quad \text{or in vector notation} \quad \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) = 0. \tag{2.32}$$

We will now interpret the terms in some of representations of the law of conservation of mass, starting with (2.30). The first term in this equation is the rate of change of mass in a fixed (i.e. Eulerian) volume Ω. The second term is the rate at which mass exits the volume due to the Eulerian velocity $\boldsymbol{u}$. Conservation of mass implies that the two rates must balance, which is the statement of (2.30).

Next we will discuss the terms in (2.32). The first term is the Eulerian rate-of-change of density the second is the divergence of the convective mass flux $\rho \boldsymbol{u}$. Another form of (2.32), written using the expression

$$\frac{\partial \rho}{\partial t} + \boldsymbol{u} \cdot \nabla \rho + \rho \nabla \cdot \boldsymbol{u} = 0 \tag{2.33}$$

as

$$\frac{1}{\rho} \frac{D\rho}{Dt} = -\nabla \cdot \boldsymbol{u} = -\frac{1}{V} \frac{DV}{Dt}, \tag{2.34}$$

can be interpreted as the relative Lagrangian rate of change of density is balanced by the negative of the relative Lagrangian rate of change of volume. In other words, if volume of an infinitesimal Lagrangian element increases, its density must decrease.

If either the fluid or the flow is "incompressible", that means that the density of infinitesimal Lagrangian fluid particles does not change with time, i.e., $D\rho/Dt = 0$. This implies

$$\nabla \cdot \boldsymbol{u} \equiv 0 \text{ everywhere.} \tag{2.35}$$

2.10 Conservation of momentum (momentum balance)

The density of momentum is $\rho \boldsymbol{u}$, and its rate of change is a force as per Newton's second law of motion. The volumetric source term is a force per unit volume that acts on the fluid and is termed a "body force", e.g. the force of gravity. We will denote it by $q^B = \rho \boldsymbol{g}$, where $\boldsymbol{g}$ is acceleration due to the body force. Here the density and the volumetric source of momentum are vectors. The second rank tensor resulting from the surface

force $T^B = \boldsymbol{T}$ is the stress tensor, which when acting on an area element $\mathrm{d}A$ with normal n_j exerts the force $\mathrm{d}F_i = T_{ij} n_j \, \mathrm{d}A$.

The three different integral forms representing conservation of momentum are

$$\frac{D}{Dt}\int_\Omega \rho u_i \, \mathrm{d}V = \int_\Omega \rho g_i \, \mathrm{d}V + \int_{\partial\Omega} T_{ij} n_j \, \mathrm{d}A = \int_\Omega \rho g_i \, \mathrm{d}V + \int_\Omega \frac{\partial T_{ij}}{\partial x_j} \, \mathrm{d}V. \tag{2.36}$$

$$\int_\Omega \frac{\partial(\rho u_i)}{\partial t} \, \mathrm{d}V + \int_{\partial\Omega} \rho u_i u_j n_j \, \mathrm{d}A = \int_\Omega \rho g_i \, \mathrm{d}V + \int_{\partial\Omega} T_{ij} n_j \, \mathrm{d}A = \int_\Omega \rho g_i \, \mathrm{d}V + \int_\Omega \frac{\partial T_{ij}}{\partial x_j} \, \mathrm{d}V, \tag{2.37}$$

$$\int_\Omega \left(\frac{\partial(\rho u_i)}{\partial t} + \frac{\partial(\rho u_i u_j)}{\partial x_j} \right) \mathrm{d}V = \int_\Omega \rho g_i \, \mathrm{d}V + \int_{\partial\Omega} T_{ij} n_j \, \mathrm{d}A = \int_\Omega \rho g_i \, \mathrm{d}V + \int_\Omega \frac{\partial T_{ij}}{\partial x_j} \, \mathrm{d}V. \tag{2.38}$$

The differential form of momentum balance is

$$\frac{\partial(\rho u_i)}{\partial t} + \frac{\partial(\rho u_i u_j)}{\partial x_j} = \rho g_i + \frac{\partial T_{ij}}{\partial x_j}. \tag{2.39}$$

The left-hand side of (2.39) is $D(\rho \boldsymbol{u})/Dt$ can be simplified further by using the product rule and mass balance (2.32) as

$$\frac{\partial(\rho u_i)}{\partial t} + \frac{\partial(\rho u_i u_j)}{\partial x_j} = u_i \underbrace{\left(\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u_j)}{\partial x_j} \right)}_{0, \text{ mass balance}} + \rho \left(\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} \right). \tag{2.40}$$

This simplifies the differential form of momentum balance (2.39) to the Cauchy momentum balance equation as

$$\rho \frac{Du_i}{Dt} \equiv \rho \left(\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} \right) = \rho g_i + \frac{\partial T_{ij}}{\partial x_j}, \quad \text{or} \quad \rho \frac{D\boldsymbol{u}}{Dt} = \rho \boldsymbol{g} + \nabla \cdot \boldsymbol{T}. \tag{2.41}$$

The left-hand side of this equation is mass times acceleration per unit volume of an infinitesimal Lagrangian element.

The right-hand side contains the net forces on this volume per unit volume. It is obvious that $\rho \boldsymbol{g}$ is the body force per unit volume. To see that $\nabla \cdot \boldsymbol{T}$ is the unbalanced surface force per unit volume, consider forces acting on a infinitesimal cubic fluid element with side d and bottom corner at $\mathbf{x} = (x_0, y_0, z_0)$; see figure 2.18. We will start with the x-component of this force which will involve us calculating $T_{1j} n_j \delta A$ (with $\delta A = d^2$) over all six sides of the cube.

$$\delta F_1 = d^2 \left(T_{11}(x_0 + d, y, z) - T_{11}(x_0, y, z) \right) \tag{2.42}$$

$$+ d^2 \left(T_{12}(x, y_0 + d, z) - T_{12}(x, y_0, z) \right) \tag{2.43}$$

$$+ d^2 \left(T_{13}(x, y, z_0 + d) - T_{13}(x, y, z_0) \right). \tag{2.44}$$

Taylor expanding this expression in small d and keeping the leading order only, we have:

$$\delta F_1 = d^3 \left[\frac{\partial T_{11}(x,y,z)}{\partial x} + \frac{\partial T_{12}(x,y,z)}{\partial y} + \frac{\partial T_{13}(x,y,z)}{\partial z} \right] = d^3 \, \frac{\partial T_{1j}(x,y,z)}{\partial x_j}$$

where we have used the summation convention (with j the repeated index).

Generalising to the other force components, we have:

$$\delta F_i = d^3 \, \frac{\partial T_{ij}(x,y,z)}{\partial x_j}.$$

symbolically this is the divergence of the stress tensor $\delta \boldsymbol{F} = d^3 \, \nabla \cdot \boldsymbol{T}$.

2.10.1 Force exerted by a fluid

We will often need to compute the force exerted by a fluid on a given surface, either at an internal surface within the fluid or at an external boundary (e.g. boundary between a fluid and an immersed body). As defined before, the force $\mathrm{d}\boldsymbol{F}$ acting on an infinitesimal surface element $\mathrm{d}A$ with normal $\boldsymbol{n}$ is

$$\mathrm{d}F_i = T_{ij} n_j \, \mathrm{d}A, \qquad \text{or in vector form} \qquad \mathrm{d}\boldsymbol{F} = \boldsymbol{T} \cdot \boldsymbol{n} \, \mathrm{d}A. \tag{2.45}$$

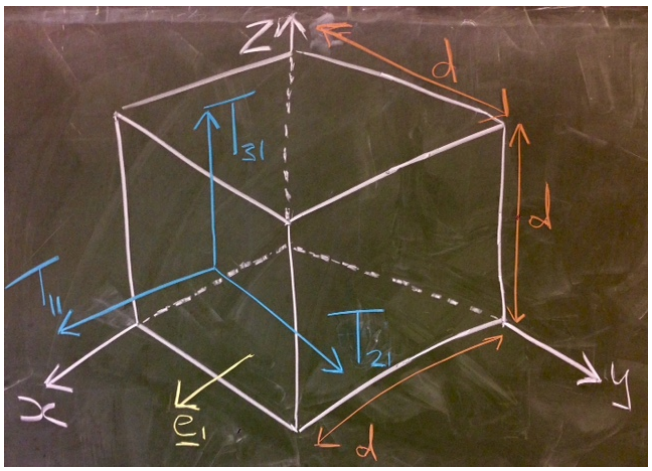


Figure 2.18: An infinitesimal cubic fluid element. Shown are the components of the stress tensor acting on one face whose normal aligns with the x-axis.

The total force over a surface $\partial\Omega$ is then obtained by integrating the stress tensor over the surface, i.e.

$$\boldsymbol{F} = \int_{\partial\Omega} \boldsymbol{T}\cdot\boldsymbol{n}\ \mathrm{d}A = \int_{\Omega} \nabla\cdot\boldsymbol{T}\ \mathrm{d}V, \tag{2.46}$$

where the latter expression is obtained applying the divergence theorem, which is usually more convenient for computation.

Naturally, the force is a vector, but if instead we only need to compute the force in a particular direction, we can take the dot product from the left of the force with a unit vector in that direction. For example, the force in the i-direction is

$$F_i = \int_{\partial\Omega} \boldsymbol{e}_i\cdot\boldsymbol{T}\cdot\boldsymbol{n}\ \mathrm{d}A. \tag{2.47}$$

2.11 Conservation of angular momentum (torque balance)

The angular momentum changes at a rate equal to the net external torque applied on a body. For angular momentum, $b = \boldsymbol{x}\times\rho\boldsymbol{u}$, $q^B = \boldsymbol{x}\times\rho\boldsymbol{g}$, and $h^B = \boldsymbol{x}\times(\boldsymbol{T}\cdot\boldsymbol{n})$. Note that the sources of angular momentum are related to the sources of linear momentum. We will go straight to the differential form of the angular momentum balance, which reads

$$\frac{\partial(\epsilon_{ijk}x_j\rho u_k)}{\partial t} + \frac{\partial(\epsilon_{ijk}x_j\rho u_k u_m)}{\partial x_m} = \epsilon_{ijk}x_j\rho g_k + \frac{\partial(\epsilon_{ijk}x_j T_{km})}{\partial x_m}. \tag{2.48}$$

Differentiating each term using the product rule yields

$$\epsilon_{ijk}x_j\left(\frac{\partial(\rho u_k)}{\partial t} + \frac{\partial(\rho u_k u_m)}{\partial x_m} - \rho g_k - \frac{\partial(T_{km})}{\partial x_m}\right) + \epsilon_{ijk}\rho u_j u_k = \epsilon_{ijk}T_{jk}. \tag{2.49}$$

The first term is the cross product of $\boldsymbol{x}$ with the linear momentum equation (2.39), and the second term is zero due to the anti-symmetry of ϵ_{ijk} in the indices j,k and the symmetry of $u_j u_k$. Thus the only remaining term is $\epsilon_{ijk}T_{jk} = 0$. A direct consequence of this is that $T_{jk} = T_{kj}$, or, in other words, $\boldsymbol{T}$ is a symmetric second-rank tensor.

2.11.1 Torque exerted by a fluid

Similarly to a force, we can compute the torque exerted by a fluid on a given surface. The torque $\mathrm{d}\boldsymbol{\Gamma}$ acting on an infinitesimal surface element $\mathrm{d}A$ with normal $\boldsymbol{n}$ is

$$\mathrm{d}\Gamma_i = \epsilon_{ijk}x_j T_{km}n_m\ \mathrm{d}A, \qquad \text{or in vector form} \qquad \mathrm{d}\boldsymbol{\Gamma} = \boldsymbol{x}\times(\boldsymbol{T}\cdot\boldsymbol{n})\ \mathrm{d}A, \tag{2.50}$$

We can integrate over a surface to obtain the total torque exerted by the fluid on that surface:

$$\boldsymbol{\Gamma} = \int_{\partial\Omega} \boldsymbol{x}\times(\boldsymbol{T}\cdot\boldsymbol{n})\ \mathrm{d}A. \tag{2.51}$$

2.12 Constitutive laws

The differential equations governing a continuum material in an Eulerian description that we have derived so far are mass balance (2.32) and momentum balance (2.41). These equations are partial differential equations for determining the density and velocity by integrating in time, i.e., if suitable initial conditions are available for ρ and $\boldsymbol{u}$, and the volumetric forces $\boldsymbol{g}$ and stress $\boldsymbol{T}$ were known, then one may use these equations to find the rate of change of ρ and $\boldsymbol{u}$, which one then uses to determine the subsequent evolution of ρ and $\boldsymbol{u}$. The situation is similar to the use of Newton's second law of motion for describing the motion of rigid bodies; knowing the external forces one uses it to integrate for velocity and position. Here we think about integrating for ρ and $\boldsymbol{u}$. Let's examine quantities we do not know yet; viz. $\boldsymbol{g}$ and $\boldsymbol{T}$.

The simplest one is $\boldsymbol{g}$: it must be prescribed as part of the problem. The stress $\boldsymbol{T}$ needs to determined based on the fluid and the flow, so that the system of equations (2.32) and (2.41) is closed. The closure of this system is made by the constitutive law for the continuum material. This constitutive law is what distinguishes between a solid and a fluid, and for that matter, between different kinds of "complex" fluids. Next we present some considerations for a constitutive law.

2.12.1 General considerations

An infinitesimal fluid element must continue to deform when a shear stress acts on it – this is the definition of a fluid. Therefore, we expect the components of the stress to be related to the components of the rate of shear tensor, i.e.,

$$T_{ij} = f_{ij}[e_{mn}; \rho]. \tag{2.52}$$

We will start with some simple models.

2.12.2 Ideal or inviscid fluid

In an ideal or inviscid fluid there is no friction between fluid elements, so that the stress is given by an isotropic pressure that acts normal to the surface and inward. Therefore, our particular form of the stress tensor (known as a constitutive relation) is

$$T_{ij} = -p\delta_{ij}. \tag{2.53}$$

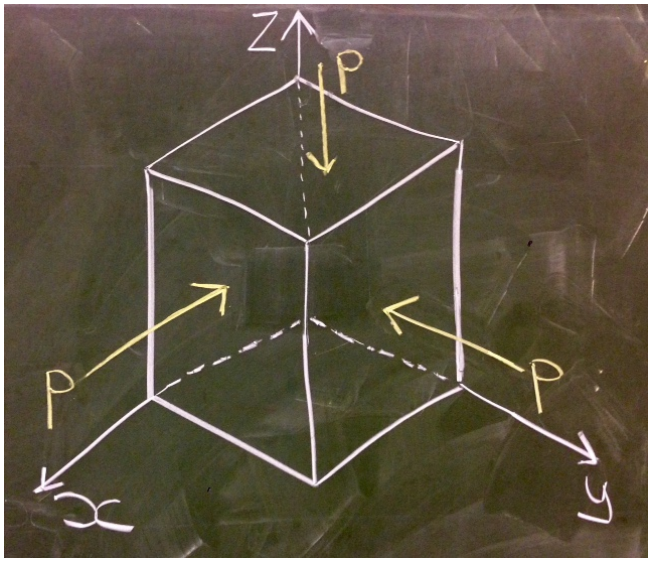


Figure 2.19: In an inviscid fluid the stress is generated entirely by the pressure, which acts with equal magnitude on all faces of our infinitesimal cubic fluid element. Clearly then, flow must be driven by this pressure changing from point to point, i.e. by spatial gradients in pressure.

Substituting into Cauchy's equation (2.41), we arrive at the so-called Euler equation which describes inviscid fluid flow

$$\rho\frac{Du_i}{Dt} = -\frac{\partial p}{\partial x_i} + \rho g_i \quad \text{or in vector notation} \quad \rho\frac{D\boldsymbol{u}}{Dt} = -\nabla p + \rho\boldsymbol{g}. \tag{2.54}$$

This equation is named after Leonhard Euler, the celebrated Swiss mathematician.

Here the pressure p is a thermodynamic quantity, so an equation of state that relates p to other thermodynamic quantities would serve this purpose. The problem is that thermodynamic relations also introduce some other thermodynamic variable independent of density (say temperature) into consideration, which in itself could be non-uniformly distributed throughout the fluid. So an equation incorporating the law of conservation of thermal energy must be derived and treated for this new variable with mass and momentum balance. In some limited but practically useful circumstances, it is possible to eliminate the temperature and simply write $p(\rho)$. Examples include the propagation of sound waves. Under those circumstances, p is well-defined. A very singular special case of this scenario is for an incompressible flow, for which ρ is a constant, and, therefore, there is no explicit form for $p(\rho)$ but p must be deduced such that no changes in ρ occur.

We will see later that an ideal fluid has a zero Newtonian viscosity.

In the absence of fluid motion $D_t\boldsymbol{u} = 0$ we recover a hydrostatic balance $p = p_0 - \rho g z$.

2.12.3 Newtonian fluid

To describe a viscous fluid, we separate the pressure contribution to the stress tensor from the remaining terms — the latter will be associated with an internal friction or viscosity as

$$T_{ij} = -p\delta_{ij} + \sigma_{ij}, \tag{2.55}$$

where σ_{ij} is called the *deviatoric stress tensor*. For a Newtonian fluid, the deviatoric stress arises from viscosity, and therefore, σ_{ij} is also called the *viscous stress tensor*. An ideal fluid is characterized by $\sigma_{ij} \equiv 0$. For a general fluid the dependence on the rate-of-strain tensor in (2.52) is passed on to σ_{ij} as $\sigma_{ij} = s_{ij}[e_{mn}; \rho]$ for some tensorial function s_{ij}.

The Newtonian constitutive law corresponds to the relation where the relation between σ_{ij} and e_{mn} is linear and isotropic. Linearity means that

$$\sigma_{ij} = A_{ijmn}e_{mn}, \tag{2.56}$$

where A_{ijmn} is a constant fourth rank tensor. For the relation to be isotropic, A_{ijmn} must be isotropic, i.e., it's components may not depend of the coordinate system used to represent the tensor. We saw in (1.12c) that the most general fourth rank isotropic tensor has the form $\lambda\delta_{ij}\delta_{kl} + \alpha\delta_{ik}\delta_{jl} + \beta\delta_{il}\delta_{jk}$, where λ, α and β are scalar constants. Substituting in (2.56) yields $\sigma_{ij} = (\alpha+\beta)e_{ij} + \lambda\delta_{ij}e_{mm}$. Here we learn that, in general, two constants are needed to specify the Newtonian constitutive law. However, for incompressible fluids (which Newton tacitly assumed), the term $e_{mm} = \nabla \cdot \boldsymbol{u}$ is identically zero, and setting $(\alpha + \beta)$ to 2μ, where μ is the coefficient of Newtonian viscosity, yields the tensorial form of the Newtonian constitutive law for an incompressible fluid[5]

$$\sigma_{ij} = 2\ \mu\ e_{ij} \quad \text{so that} \quad T_{ij} = -p\ \delta_{ij} + \mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right). \tag{2.58}$$

Substituting (2.58) into Cauchy's equation (2.41), we have the *incompressible Navier-Stokes equations*:

$$\rho\frac{Du_i}{Dt} = -\frac{\partial p}{\partial x_i} + \mu\frac{\partial^2 u_i}{\partial x_j^2} + \rho g_i \qquad \text{or in vector notation} \qquad \rho\frac{D\boldsymbol{u}}{Dt} = -\nabla p + \mu\nabla^2\boldsymbol{u} + \rho\boldsymbol{g}. \tag{2.59}$$

For zero viscosity, (2.59) reduces to the Euler equation (2.54) for ideal fluid.

[5]For a *compressible flow,* where σ_{ij} can depend both on e_{ij} and the identity tensor, the most general form of such a tensor is

$$\sigma_{ij} = \underbrace{2\ \mu\ \left(e_{ij} - \frac{1}{3}\delta_{ij}\, e_{kk}\right)}_{\text{Stress without volume change}} + \underbrace{\xi\, e_{kk}\delta_{ij}}_{\text{Stress due to volume change}}, \tag{2.57}$$

where μ is the viscosity ('shear viscosity' or 'first viscosity coefficient'), ξ is the "bulk viscosity" or the "second viscosity coefficient".

2.13 Boundary and initial conditions

As usual for PDEs, in order to find a unique solution one has to specify appropriate boundary conditions and initial conditions.

2.13.1 Initial conditions

The set of relevant fields must be specified at $t = 0$ at each point $\boldsymbol{x}$ in the domain occupied by the fluid. For incompressible fluids, one only has to specify the initial velocity field $\boldsymbol{u}_0$

$$\boldsymbol{u}(\boldsymbol{x}, t = 0) = \boldsymbol{u}_0. \tag{2.60}$$

2.13.2 Boundary conditions

The number of boundary conditions required depends on the bulk equations (Euler or Navier Stokes in our case) and their form is determined by the properties of the boundaries (i.e. the physics at the boundary). For this module we will mostly use no-slip and free slip boundary conditions, but the kinematic and dynamic boundary conditions are also introduced for completeness.

No-slip condition (viscous fluids)

At solid impenetrable boundaries, it has been found empirically that the fluid velocity at the boundary $\partial\Omega$ of the retaining volume Ω adjusts itself to the boundary's velocity $\mathbf{U}_b$: this is the famous *no-slip boundary condition*:

$$\boldsymbol{u}(\boldsymbol{x}, t) = \mathbf{U}_b(\boldsymbol{x}, t), \quad \boldsymbol{x} \in \partial\Omega. \tag{2.61}$$

Free slip condition (inviscid fluids)

For inviscid fluids, where one only has first derivatives of fluid velocity (no $\nabla^2\boldsymbol{u}$ term), one can only enforce inpenetrability of the boundary, i.e. that the normal to the boundary velocity component has to match to the one of the boundary, whereas the parallel velocity component remains arbitrary, since the fluid can slip freely along the boundary in the absence of internal friction. This is the so-called *free-slip boundary conditions* written as

$$\boldsymbol{u}_\perp(\boldsymbol{x}, t) = \mathbf{U}_{b\,\perp}(\boldsymbol{x}, t), \quad \boldsymbol{x} \in \partial\Omega. \tag{2.62}$$

In addition, at every point on the fluid where the fluid flows inward, the vorticity needs to be specified. This is an advanced condition, which we will not pursue in this course.

Kinematic boundary condition (moving material boundaries)

For cases like the free surface, the motion of the boundary itself occurs under the influence of the fluid flow. In other words, the reason the free surface changes shape is because the fluid particles at the free surface flow. Mathematically, the free surface is a *material boundary*, i.e.,

$$\frac{D\boldsymbol{X}}{Dt} = \boldsymbol{u}(\boldsymbol{X}, t), \quad \text{for the Lagrangian label } \boldsymbol{X} \in \partial\Omega. \tag{2.63}$$

Dynamic (or traction) boundary condition (moving material boundaries)

In lieu of specifying velocity, one may also specify the traction t_i, i.e. the force per unit area on the boundary. In general, the traction condition is written as

$$T_{ij}n_j = t_i, \quad \boldsymbol{x} \in \partial\Omega, \text{ where } n_i \text{ is the unit normal.} \tag{2.64}$$

This condition is especially useful for moving free surfaces, where two conditions are needed, viz., one for specifying the rate of motion of the interface and another as a boundary condition for the fluid partial differential equations. In that case, along with the kinematic boundary condition, the traction information furnishes the second condition.

2.14 Summary of the incompressible Navier-Stokes equations

In this chapter we have derived the incompressible version of the Navier-Stokes equation, which we will use for the rest of the module. In vector form, it reads

$$\nabla \cdot \boldsymbol{u} = 0, \tag{2.65}$$

$$\rho\left(\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u}\cdot\nabla\boldsymbol{u}\right) = -\nabla p + \mu\nabla^2\boldsymbol{u} + \rho\boldsymbol{g}. \tag{2.66}$$

The vector form is independent of the coordinate system used, but we will usually work with the component form of the equations. This form depends on the coordinate system used, so here we present the equations in Cartesian, cylindrical and spherical coordinates.

2.14.1 Cartesian coordinates

In Cartesian coordinates we define the velocity field (u, v, w) along the x, y and z directions, so conservation of mass becomes

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0, \tag{2.67}$$

and conservation of momentum becomes

$$\rho\left(\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z}\right) = -\frac{\partial p}{\partial x} + \rho g_x + \mu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right), \tag{2.68a}$$

$$\rho\left(\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z}\right) = -\frac{\partial p}{\partial y} + \rho g_y + \mu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2}\right), \tag{2.68b}$$

$$\rho\left(\frac{\partial w}{\partial t} + u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z}\right) = -\frac{\partial p}{\partial z} + \rho g_z + \mu\left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2}\right). \tag{2.68c}$$

Here g_x, g_y and g_z are the x, y and z components of gravitational acceleration (for generality we do not assume gravity is in the z direction here).

2.14.2 Cylindrical coordinates

In cylindrical coordinates we define the radial, azimuthal and axial components of velocity (u, v, w) in a cylindrical polar coordinate system (r, θ, z), so conservation of mass becomes

$$\frac{1}{r}\frac{\partial (ru)}{\partial r} + \frac{1}{r}\frac{\partial v}{\partial \theta} + \frac{\partial w}{\partial z} = 0, \tag{2.69}$$

and conservation of momentum becomes

$$\rho\left(\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial r} + \frac{v}{r}\frac{\partial u}{\partial \theta} + w\frac{\partial u}{\partial z} - \frac{v^2}{r}\right) = -\frac{\partial p}{\partial r} + \rho g_r + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial (ur)}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} - \frac{2}{r^2}\frac{\partial v}{\partial \theta} + \frac{\partial^2 u}{\partial z^2}\right] \tag{2.70a}$$

$$\rho\left(\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial r} + \frac{v}{r}\frac{\partial v}{\partial \theta} + w\frac{\partial v}{\partial z} + \frac{uv}{r}\right) = -\frac{1}{r}\frac{\partial p}{\partial \theta} + \rho g_\theta + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial (vr)}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 v}{\partial \theta^2} + \frac{2}{r^2}\frac{\partial u}{\partial \theta} + \frac{\partial^2 v}{\partial z^2}\right] \tag{2.70b}$$

$$\rho\left(\frac{\partial w}{\partial t} + u\frac{\partial w}{\partial r} + \frac{v}{r}\frac{\partial w}{\partial \theta} + w\frac{\partial w}{\partial z}\right) = -\frac{\partial p}{\partial z} + \rho g_z + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial w}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 w}{\partial \theta^2} + \frac{\partial^2 w}{\partial z^2}\right]. \tag{2.70c}$$

Here g_r, g_θ and g_z are the r, θ and z components of gravitational acceleration.

2.14.3 Spherical coordinates

In spherical coordinates we define the radial, polar and azimuthal components of velocity (u, v, w) in a spherical polar coordinate system (r, θ, ϕ), so conservation of mass becomes

$$\frac{1}{r^2}\frac{\partial (r^2 u)}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial (v\sin\theta)}{\partial \theta} + \frac{1}{r\sin\theta}\frac{\partial w}{\partial \phi} = 0, \tag{2.71}$$

and conservation of momentum becomes

$$\rho\left(\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial r}+\frac{v}{r}\frac{\partial u}{\partial \theta}+\frac{w}{r\sin\theta}\frac{\partial u}{\partial \phi}-\frac{v^2+w^2}{r}\right)$$
$$=-\frac{\partial p}{\partial r}+\rho g_r+\mu\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial u}{\partial r}\right)+\frac{1}{r^2\sin\theta}\left(\frac{1}{\sin\theta}\frac{\partial^2 u}{\partial \phi^2}+\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial u}{\partial\theta}\right)-2\frac{\partial w}{\partial\phi}\right)-\frac{2}{r^2}\left(u+\frac{\partial v}{\partial\theta}+v\cot\theta\right)\right], \tag{2.72a}$$

$$\rho\left(\frac{\partial v}{\partial t}+u\frac{\partial v}{\partial r}+\frac{v}{r}\frac{\partial v}{\partial \theta}+\frac{w}{r\sin\theta}\frac{\partial v}{\partial \phi}+\frac{uv-w^2\cot\theta}{r}\right)$$
$$=-\frac{1}{r}\frac{\partial p}{\partial\theta}+\rho g_\theta+\mu\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial v}{\partial r}\right)+\frac{1}{r^2\sin\theta}\left(\frac{1}{\sin\theta}\left(\frac{\partial^2 v}{\partial\phi^2}-v\right)+\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial v}{\partial\theta}\right)-2\cot\theta\frac{\partial w}{\partial\phi}\right)+\frac{2}{r^2}\frac{\partial u}{\partial\theta}\right], \tag{2.72b}$$

$$\rho\left(\frac{\partial w}{\partial t}+u\frac{\partial w}{\partial r}+\frac{v}{r}\frac{\partial w}{\partial \theta}+\frac{w}{r\sin\theta}\frac{\partial w}{\partial \phi}+\frac{uv+vw\cot\theta}{r}\right)$$
$$=-\frac{1}{r\sin\theta}\frac{\partial p}{\partial\phi}+\rho g_\phi+\mu\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial w}{\partial r}\right)+\frac{1}{r^2\sin\theta}\left(\frac{1}{\sin\theta}\left(\frac{\partial^2 w}{\partial\phi^2}-w\right)+\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial w}{\partial\theta}\right)+2\frac{\partial u}{\partial\phi}+2\cot\theta\frac{\partial v}{\partial\phi}\right)\right]. \tag{2.72c}$$

Here g_r, g_θ and g_ϕ are the r, θ and ϕ components of gravitational acceleration.

Chapter 3

One-dimensional flows

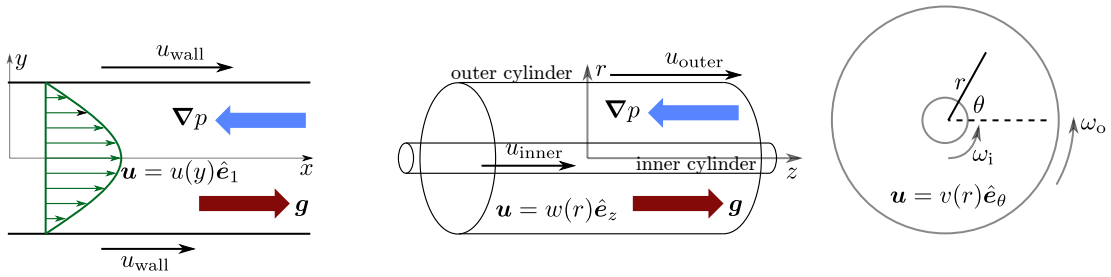
In this chapter, we will treat exact solutions of Navier-Stokes equations that result from the assumption of "one-dimensionality". Such flows do occur in nature and are highly relevant for many common applications.

3.1 Steady flows

Several key features are common amongst the one-dimensional solutions of Navier-Stokes equations, which are summarized here.

3.1.1 One-dimensionality

The assumption of one-dimensionality underlies all the flows presented in this chapter. One-dimensionality means that in either Cartesian or cylindrical-polar coordinates, the velocity has only one non-zero component of velocity, which varies in a direction perpendicular to the flow. Here are examples, which are illustrated in Figure 3.1:



(a) Plane Couette-Poiseuille flow (b) Cylindrical Couette-Poiseuille (c) Circular Couette flow

Figure 3.1: Schematic of three cases of one-dimensional flows.

1. **Plane Couette-Poiseuille flow:** This is a flow of a fluid sandwiched between two infinite planar walls, shown in Figure 3.1(a). In this case the flow is along a single direction x parallel to the wall and the flow velocity varies in the direction y perpendicular to the walls.
2. **Cylindrical Couette-Poiseuille flow**, some cases of which are also known as **Hagen-Poiseuille flow:** This is the flow in the axial (i.e., z) direction through the gap between two infinitely-long coaxial cylinders. The flow varies only with r, the radial distance from the axis.
3. **Circular Couette flow:** This is the flow of a fluid in the azimuthal (i.e., $\boldsymbol{e}_\theta$) direction between two coaxial cylinders. The flow velocity varies only with the radial distance from the axis, i.e. r.

In practice, the walls on plane Couette-Poiseuille flow and the cylinders in the cylindrical Couette Poiseuille flow are not infinitely long. This is only meant as an approximation for the case where they are much longer than the width of the gap through which the flow occurs.

In all these cases, the component of the acceleration of the fluid in the direction of the flow is zero. In fact, the acceleration is zero for the case of plane and cylindrical Couette-Poiseuille flow. For circular Couette flow the acceleration is centripetal. Therefore, when solving for the flow using the Navier-Stokes equations, the inertial term drops out of the momentum balance in the direction of the flow.

3.1.2 The driving force

The flow could be drive by one of the following three agencies:

1. **Pressure gradient:** A gradient of pressure is said to have been set and driving the flow when an external agency, e.g., a pump, increases the pressure at the inlet of the pipe or channel, or decreases it far downstream. In this case, the flow is driven by the pressure decreasing along the length of the channel in the direction of the flow. The resulting flow is called a Poiseuille flow, after the French physicist and physiologist Jean Léonard Marie Poiseuille, who used this type of analysis to study blood flow through capillaries. The general type of question that arises for these types of flow are about the pressure gradient needed to drive a certain volumetric flow rate through the domain.

2. **Gravity:** Similar to pressure gradient, the flow could be driven by an external volumetric force, such as gravity.

3. **Motion of the walls:** An external agency could push on the walls to move along the direction of the flow, thereby dragging the fluid next to them. The general question that comes up in such cases is the shear stress needed to move the walls at prescribed speeds. This type of flow is called Couette flow, after the French physicist Maurice Marie Alfred Couette.

Note that in circular Couette flow either a pressure gradient or a constant force of gravity cannot drive the flow. This is so because of periodicity in the direction of the flow. Pressure cannot continually decrease along the flow because after an angle of 2π we return back to the original point, where we expect to find the same pressure as we started. Therefore, circular Couette flow may only be driven by motion of the walls.

All three methods of driving the flow can be used in instruments that measure the dynamic viscosity of the fluid.

3.2 Select solutions of Navier-Stokes

Here we present some examples of solutions of Navier-Stokes equations for steady one-dimensional flows.

3.2.1 Plane Poiseuille flow

This case corresponds to the pressure-driven flow in a channel. The walls of the channel are stationary in the lab frame in this case.

Given the flow profile $\boldsymbol{u} = u(y)\boldsymbol{e}_x$, the equation of continuity is trivially satisfied and the two-dimensional Navier-Stokes equations in Cartesian coordinates simplify to

$$\frac{\partial p}{\partial x} = \mu \frac{\partial^2 u}{\partial y^2}, \tag{3.1a}$$

$$\frac{\partial p}{\partial y} = 0. \tag{3.1b}$$

According to (3.1b), the pressure does not vary with y, and hence we can assert that $p(x)$ alone. Then in (3.1a), the left-hand side depends on x alone, whereas the right-hand side on y alone, implying that both sides must be a constant. Physically, because the flow does not change with x, we do not expect the pressure gradient to vary with x either. Using the constancy of $\partial p/\partial x$, (3.1b) may be readily integrated to yield

$$u(y) = \frac{1}{2\mu}\frac{\partial p}{\partial x} y^2 + Ay + B, \tag{3.2}$$

where A and B are constants of integration. They are determined by the no-slip boundary conditions on the two walls

$$u\left(y=\pm\frac{H}{2}\right)=0, \tag{3.3}$$

which yields

$$u(y)=-\frac{1}{2\mu}\frac{\partial p}{\partial x}\left(\frac{H^2}{4}-y^2\right). \tag{3.4}$$

The flow rate per unit width Q/w is

$$\frac{Q}{w}=-\int_{-H/2}^{H/2}u(y)\;dy=-\frac{1}{2\mu}\frac{\partial p}{\partial x}\left(\frac{H^2y}{4}-\frac{y^3}{3}\right)_{-H/2}^{H/2}=-\frac{H^3}{12\mu}\frac{\partial p}{\partial x}. \tag{3.5}$$

This is the relation between the pressure-gradient and the flow rate that is practically sought. Experimental measurements of H, $\partial p/\partial x$ and Q/w may be used to determine the dynamic viscosity μ.

Also worth noting is the result that the average fluid speed $Q/(wH)$ is 2/3 of the maximum fluid speed $u(y=0)$.

3.2.2 Hagen-Poiseuille

This is the flow driven by a pressure gradient inside a pipe. In this case, the inside cylinder shown in Figure 3.1(b) is absent the fluid fills all the space inside the outer cylinder. Also, the outer cylinder is stationary.

Continuity is trivially satisfied by the velocity profile given by $\boldsymbol{u}=w(r)\boldsymbol{e}_z$. The radial and axial components of the Navier-Stokes equations, (2.70a) and (2.70c), respectively, simplify to

$$\frac{\partial p}{\partial r}=0, \tag{3.6}$$

$$\frac{\partial p}{\partial z}=\frac{\mu}{r}\frac{\partial}{\partial r}\left(r\frac{\partial w}{\partial r}\right). \tag{3.7}$$

Similar to the case of plane Poiseuille flow, here the pressure depends on z alone and must decrease at a rate independent of z, i.e. $\partial p/\partial z$ is a constant. The z-momentum equation may then be integrated ot yield

$$r\frac{\partial w}{\partial r}=\frac{r^2}{2\mu}\frac{\partial p}{\partial z}+A,\qquad w(r)=\frac{r^2}{4\mu}\frac{\partial p}{\partial z}+A\ln r+B, \tag{3.8}$$

where A and B are constants of integration.

At this stage, it appears that we only have one boundary condition $w(r=R)=0$ corresponding to no-slip on the outer wall at $r=R$. It is commonly (and in my opinion incorrectly) stated that a second condition is that the velocity cannot blow up on the axis $r=0$ and hence $A=0$. Then, one uses no-slip on $r=R$ to obtain

$$w(r)=-\frac{1}{4\mu}\frac{\partial p}{\partial z}(R^2-r^2). \tag{3.9}$$

A much more convincing argument for setting $A=0$ is as follows. Consider the surface of an imaginary cylinder at radius r. The axial force exerted by the fluid inside of the cylinder on the fluid outside is

$$\tilde{F}_z=2\pi r\sigma_{rz}=2\pi r\mu\frac{\partial w}{\partial r}=2\pi\left(\frac{r^2}{2\mu}\frac{\partial p}{\partial z}+A\right), \tag{3.10}$$

where we have used (3.8) in the final step. Now consider the limit of r approaching zero, i.e. the imaginary cylinder shrinking towards the axis. In the limit $\tilde{F}_z$ approaches $2\pi A$. Based on this, it seems that A is related to the influence of an imaginary agency on the axis of the cylinder to exert a shear force on the fluid. Since we do not have any such agency present in the problem under consideration, $\tilde{F}_z$ must vanish, and so must A.

The relation between pressure gradient and the volumetric flow rate through the pipe Q may also be found easily as

$$Q=\int_0^R 2\pi r w(r)\;dr=-\frac{2\pi}{4\mu}\frac{\partial p}{\partial z}\int_0^R r(R^2-r^2)\;dr=-\frac{2\pi}{4\mu}\frac{\partial p}{\partial z}\left(\frac{R^2r^2}{2}-\frac{r^4}{4}\right)_0^R=-\frac{\pi R^4}{8\mu}\frac{\partial p}{\partial z}. \tag{3.11}$$

Also worth noting is the result that the average fluid speed $Q/(\pi R^2)$ is 1/2 of the maximum fluid speed $w(r=0)$.

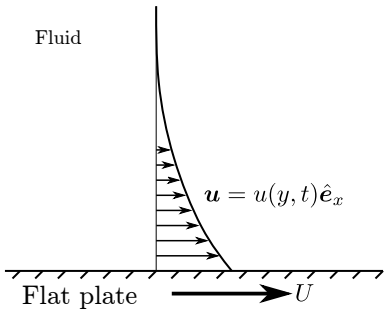


Figure 3.2: Schematic setup of the Stokes first problem.

3.3 Unsteady flows

If condition of steadiness is relaxed, then the governing equations simplify to a partial differential equation for the flow profile that depends only on one spatial coordinate and time. Here we present one such example. The same three geometrical categories and the three kinds of driving forces can be used to classify this type of flow. We will not repeat them here.

Because of the partial nature of the differential equation, the solution procedure depends on the particular problem and can be a little involved. It is illustrated for the case of Stokes first problem below.

3.3.1 Stokes first problem

An infinite flat plate borders a semi-infinite layer of fluid of density ρ and viscosity μ, see Figure 3.2. The fluid and the plate are initially stationary. At $t = 0$, the plate impulsively starts to translate parallel to itself with speed U. This drags the fluid adjacent to the plate.

In this case, the fluid velocity may be assumed to be one-dimensional as $\boldsymbol{u} = u(y,t)\boldsymbol{e}_x$. With this assumption, the velocity profile satisfies continuity exactly. The governing Navier-Stokes equations simplify to

$$\rho\frac{\partial u}{\partial t} = -\frac{\partial p}{\partial x} + \mu\frac{\partial^2 u}{\partial y^2}, \tag{3.12}$$

$$0 = -\frac{\partial p}{\partial y}. \tag{3.13}$$

The y-momentum balance (3.13) implies p does not vary with y. Combining this with the condition far away from the plate, where the fluid is stationary and therefore the pressure is a constant, implies that the pressure is constant everywhere. This implies that the pressure gradient in (3.12) vanishes.

The governing equations and the boundary conditions now become

$$\frac{\partial u}{\partial t} = \nu\frac{\partial^2 u}{\partial y^2}, \tag{3.14a}$$

$$u(y = 0, t) = U, \tag{3.14b}$$

$$u(y \to \infty, t) = 0, \tag{3.14c}$$

$$u(y, t = 0) = 0. \tag{3.14d}$$

We will look for self-similar solutions to this problem. A self-similar solution is one that can be rescaled back to a single profile by suitably rescaling the dependent and independent variables. A scaling argument, similar to those presented in Chapter 4, can be made to determine the right scaling. But for now, we will use the ansatz that the solution is of the form

$$u(y,t) = u(\xi) = U f(\xi), \tag{3.15}$$

where

$$\xi = \frac{y}{t^\alpha}. \tag{3.16}$$

We first need to find which values of α satisfy (3.14) (if any), by substituting the ansatz into it. Using the chain rule, we obtain

$$-\frac{\alpha}{t}\xi f'(\xi) = \frac{\nu}{t^{2\alpha}} f''(\xi). \tag{3.17}$$

Rearranging we obtain

$$-\alpha t^{2\alpha-1} = \nu \frac{f''(\xi)}{\xi f'(\xi)}. \tag{3.18}$$

The left hand side only depends on t, while the right hand side only depends on ξ. This is only possible if both sides are equal to a constant, which is only possible if $\alpha = 1/2$. Therefore, the equation for $f(\xi)$ becomes

$$f''(\xi) + \frac{\xi}{2\nu} f'(\xi) = 0. \tag{3.19}$$

We also need to check that the ansatz satisfies the initial and boundary conditions. This is the case, as

$$f(\xi) = 1, \qquad \text{at } \xi = 0 \qquad \text{(equivalent to } y = 0\text{)}, \tag{3.20a}$$
$$f(\xi) \to 0, \qquad \text{as } \xi \to +\infty \qquad \text{(equivalent to } y \to \infty \text{ and } t \to 0\text{)}. \tag{3.20b}$$

Solving (3.19) yields

$$f(\xi) = A + B\text{erf}\left(\frac{\xi}{\sqrt{2\nu}}\right), \tag{3.21}$$

where $\text{erf}(x) = \frac{2}{\pi}\int_0^x e^{-s^2} ds$ is the error function. The constants A and B are determined by the boundary conditions (3.20). The final solution is

$$f(\xi) = 1 - \text{erf}\left(\frac{\xi}{\sqrt{2\nu}}\right) = \text{erfc}\left(\frac{\xi}{\sqrt{2\nu}}\right), \tag{3.22}$$

where $\text{erfc}(x) = 1 - \text{erf}(x)$ is the complementary error function.

Going back to the original variables, the solution is

$$u(y,t) = U\text{erfc}\left(\frac{y}{\sqrt{2\nu t}}\right). \tag{3.23}$$

Interested students can try out the following problems of this kind:

1. the flow in the annular gap between two co-axial cylinders (see Figure 3.1c) driven by the rotation of the outer or the inner cylinder,
2. the flow driven simultaneously by pressure gradient, gravity and the motion of walls in the planar or cylindrical geometry (see Figure 3.1ab),
3. the analogue of Stokes first problem, but where the wall velocity $U = \alpha t$, i.e. the wall accelerates with a constant acceleration α,
4. Stokes second problem, where the wall velocity oscillates sinusoidally for perpetuity.

Chapter 4

Dimensional Analysis

Dimensional analysis applies the principles of manipulating dimensions of physical quantities for reducing the complexity of a problem. It is best illustrated through examples.



Figure 4.1: Schematic of a two-dimensional flow past a cylinder.

4.1 Reducing parameters in a set of equations

A process called "non-dimensionalisation" is used to reduce the number of parameters the solution of a set of equations depend on. Here, we non-dimensionalise the Navier-Stokes equations for flow past an infinitely-long cylinder.

Consider a fluid of density ρ and viscosity μ flowing with speed U along the x-axis past an infinitely long cylinder of radius R, as shown in Figure 4.1. Let us write the Navier-Stokes equations and the boundary conditions.

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0, \tag{4.1a}$$

$$\rho\left(\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y}\right) = -\frac{\partial p}{\partial x} + \mu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right), \tag{4.1b}$$

$$\rho\left(\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y}\right) = -\frac{\partial p}{\partial y} + \mu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right), \tag{4.1c}$$

$$\boldsymbol{u}(\boldsymbol{x} \to \infty, t) = U\boldsymbol{e}_x, \tag{4.1d}$$

$$\boldsymbol{u}(|\boldsymbol{x}| = R, t) = \boldsymbol{0}. \tag{4.1e}$$

We would like to determine the drag, D, per unit length on the cylinder

$$D = \int_{\partial\Omega} \boldsymbol{e}_x \cdot \boldsymbol{T} \cdot \boldsymbol{n}\, \mathrm{d}A, \quad \text{where} \quad T_{ij} = -p\delta_{ij} + \mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right). \tag{4.2}$$

First notice the parameters of this problem: ρ, μ, U and R. We use three of these parameters (R, U and ρ) to "non-dimensionalise" all the quantities in the problem, i.e., we rescale

$$t = \frac{R}{U}\tilde{t}, \quad \boldsymbol{x} = R\tilde{\boldsymbol{x}}, \quad \boldsymbol{u} = U\tilde{\boldsymbol{u}}, \quad p = \rho U^2\tilde{p}, \quad \text{and} \quad \boldsymbol{T} = \rho U^2\tilde{\boldsymbol{T}}. \tag{4.3}$$

Note that $\tilde{x}$, $\tilde{\boldsymbol{u}}$, $\tilde{p}$ and $\tilde{\boldsymbol{T}}$ are all dimensionless.

This transformation converts (4.1) to

$$\frac{\partial \tilde{u}}{\partial \tilde{x}} + \frac{\partial \tilde{v}}{\partial \tilde{y}} = 0, \tag{4.4a}$$

$$\left(\frac{\partial \tilde{u}}{\partial \tilde{t}} + \tilde{u}\frac{\partial \tilde{u}}{\partial \tilde{x}} + \tilde{v}\frac{\partial \tilde{u}}{\partial \tilde{y}}\right) = -\frac{\partial \tilde{p}}{\partial \tilde{x}} + \frac{1}{\mathrm{Re}}\left(\frac{\partial^2 \tilde{u}}{\partial \tilde{x}^2} + \frac{\partial^2 \tilde{u}}{\partial \tilde{y}^2}\right), \tag{4.4b}$$

$$\left(\frac{\partial \tilde{v}}{\partial \tilde{t}} + \tilde{u}\frac{\partial \tilde{v}}{\partial \tilde{x}} + \tilde{v}\frac{\partial \tilde{v}}{\partial \tilde{y}}\right) = -\frac{\partial \tilde{p}}{\partial \tilde{y}} + \frac{1}{\mathrm{Re}}\left(\frac{\partial^2 \tilde{v}}{\partial \tilde{x}^2} + \frac{\partial^2 \tilde{v}}{\partial \tilde{y}^2}\right), \tag{4.4c}$$

$$\tilde{\boldsymbol{u}}(\tilde{\boldsymbol{x}} \to \infty, \tilde{t}) = \boldsymbol{e}_x, \tag{4.4d}$$

$$\tilde{\boldsymbol{u}}(|\tilde{\boldsymbol{x}}| = 1, t) = \boldsymbol{0}, \tag{4.4e}$$

where $\mathrm{Re} = \dfrac{\rho U R}{\mu}$ is called the Reynolds number after Osborne Reynolds. The Reynolds number is the ratio of the strengths of inertial effects to viscous effects in the fluid.

The rescaled version of drag from (4.2) is

$$D = \rho U^2 R \int_{\partial\Omega} \boldsymbol{e}_x \cdot \boldsymbol{T} \cdot \boldsymbol{n} \, \mathrm{d}A, \quad \text{where} \quad \tilde{T}_{ij} = -\tilde{p}\delta_{ij} + \frac{1}{\mathrm{Re}}\left(\frac{\partial \tilde{u}_i}{\partial \tilde{x}_j} + \frac{\partial \tilde{u}_j}{\partial \tilde{x}_i}\right). \tag{4.5}$$

Note that the dimensionless equations (4.4) are equivalent to the dimensional one (4.1) with $\rho = 1$, $U = 1$, $R = 1$ and $\mu = \mathrm{Re}^{-1}$. This means the flows for different sets of parameters are simply rescaled versions of each other if the Re for them is identical. Using this, the drag on these cylinders will also be related. Or, rewriting (4.5) as

$$C_D = \frac{D}{\frac{1}{2}\rho U^2 R} = 2\int_{\partial\Omega} \boldsymbol{e}_x \cdot \boldsymbol{T} \cdot \boldsymbol{n} \, \mathrm{d}A, \tag{4.6}$$

implies that

$$C_D = F(\mathrm{Re}) \text{ alone.} \tag{4.7}$$

Here C_D is called the drag coefficient, and the factor of $1/2$ in the denominator is for historical reasons. The specific form of C_D depends on the shape of the immersed body. In the case of a cylinder, C_D is as shown in Figure 4.2. Note the drop in C_D around $\mathrm{Re} = 5 \cdot 10^5$. This is called the "drag crisis" and it is caused by the transition from laminar to turbulent flow.

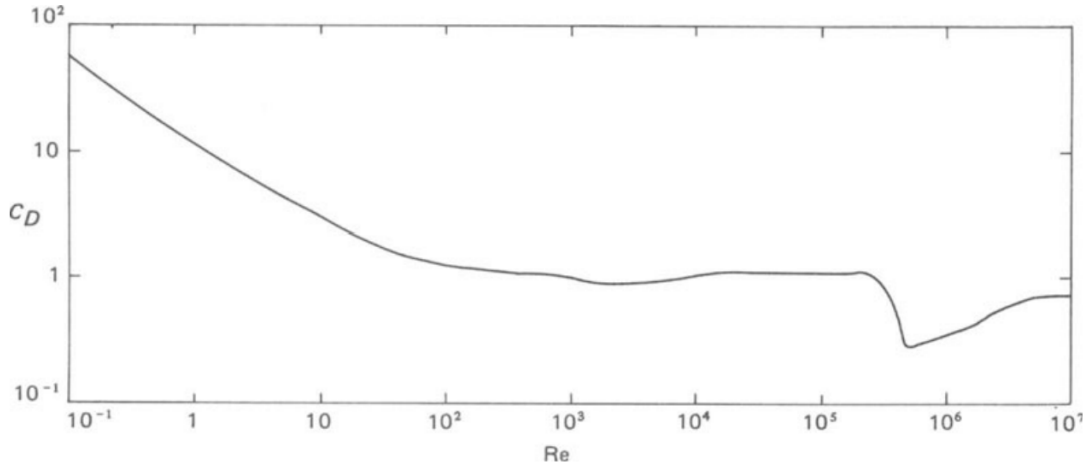


Figure 4.2: Drag coefficient for flow past a cylinder as a function of the Reynolds number (reproduced from D.J. Tritton, Physical Fluid Dynamics, Oxford Science Publs.).

4.2 Deducing dependence on parameters

Equation 4.7 may be derived without reference to partial differential equations, simply by using the arbitrariness of the system of units. Clearly, the drag on the cylinder depends on R, U, μ and ρ, i.e.,

$$D = \frac{1}{2} f(\rho, \mu, R, U), \tag{4.8}$$

where the factor of $1/2$ is included to conform to convention that will be introduced later. In a different set of units, where length L, time T and mass M transforms as

$$\tilde{L} = \alpha L, \qquad \tilde{T} = \beta T, \qquad \tilde{M} = \gamma M, \tag{4.9}$$

where the symbols with the tilde refer to transformed units. Under this transformation, the parameters transform as

$$\tilde{\rho} = \frac{\tilde{M}}{\tilde{M}^3} = \frac{\gamma}{\alpha^3}\frac{M}{L^3} = \frac{\gamma}{\alpha^3}\rho, \tag{4.10}$$

$$\tilde{\mu} = \frac{\tilde{M}}{\tilde{L}\tilde{T}} = \frac{\gamma}{\alpha\beta}\frac{M}{LT} = \frac{\gamma}{\alpha\beta}\mu, \tag{4.11}$$

$$\tilde{R} = \tilde{L} = \alpha L = \alpha R, \tag{4.12}$$

$$\tilde{U} = \frac{\tilde{L}}{\tilde{T}} = \frac{\alpha}{\beta}\frac{L}{T} = \frac{\alpha}{\beta}U, \tag{4.13}$$

$$\tilde{D} = \frac{\tilde{M}}{\tilde{T}^2} = \frac{\gamma}{\beta^2}\frac{M}{T^2} = \frac{\gamma}{\beta^2}D. \tag{4.14}$$

By the arbitrariness of system of units, (4.8) also holds in the transformed system, i.e.,

$$\tilde{D} = \frac{1}{2} f(\tilde{\rho}, \tilde{\mu}, \tilde{R}, \tilde{U}). \tag{4.15}$$

Transforming it back implies

$$\frac{\gamma}{\beta^2}D = \frac{1}{2} f\left(\frac{\gamma}{\alpha^3}\rho,\ \frac{\gamma}{\alpha\beta}\mu,\ \alpha R,\ \frac{\alpha}{\beta}U\right). \tag{4.16}$$

Because the scaling factors α, β and γ could be chosen at will, we choose

$$\alpha = 1/R, \qquad \beta = \frac{U}{R}, \qquad \gamma = \frac{1}{\rho R^3}. \tag{4.17}$$

Substituting in (4.15) gives

$$\frac{D}{\frac{1}{2}\rho U^2 R} = f\left(1, \frac{\mu}{\rho U R}, 1, 1\right) = F(\mathrm{Re}), \tag{4.18}$$

where the inserted factor of $1/2$ turns the left hand side to the drag coefficient as in (4.7).

First, let us derive (4.7) using a short-cut. Notice that there are two ways to construct the dimensions of D from those of ρ, μ, R and U: (i) $\rho U^2 R$, and (ii) μU. The former neglects viscosity and the latter inertia. Since both have dimensions of D, their ratio is dimensionless; in fact

$$\frac{\rho U^2 R}{\mu U} = \mathrm{Re}. \tag{4.19}$$

Because there are 3 basic dimensions of mass, length and time that make up the dimensions of 4 parameters, there is only 4-3=1 dimensionless parameter that could be constructed from the 4 parameters.

We can go a little further and derive more insight into the functional form of $F(\mathrm{Re})$. Suppose that Re is large, so inertia dominates and viscous effects are negligible. Under these circumstances, μ is eliminated as a parameter, so

$$D = f_I(\rho, U, R) = \frac{1}{2} C_D \rho U^2 R. \tag{4.20}$$

In the opposite extreme, if the fluid is extremely viscous, Re is small, the density of the fluid (which quantifies inertia) could be neglected and

$$D = f_V(\mu, U) = C_V \mu U, \tag{4.21}$$

where C_V is a dimensionless constant. The relation could be rearranged to determine the functional form of $F(\mathrm{Re})$ in the limit of small or large Re.

$$F(Re) = C_D = \text{constant} \qquad \ldots \text{ Re large,} \tag{4.22}$$

$$= \frac{C_V \mu U}{\frac{1}{2}\rho U^2 R} = \frac{2C_V}{\mathrm{Re}} \qquad \ldots \text{ Re small.} \tag{4.23}$$

In this way, dimensional analysis can assist in reducing the number of parameters in the physical relation between variables describing a phenomenon.

Chapter 5

Hydrostatics and Bernoulli equation

In this chapter, we will treat two closely related topics in fluid mechanics – hydrostatics and the Bernoulli equation. The reader is presumably acquainted with the concepts in hydrostatics; here we include it for completeness and to illustrate the proofs underlying some celebrated principles. The Bernoulli equation is one of the most important results in fluid dynamics because, unlike the Navier-Stokes equations, it yields answers to questions inspired by applications.

5.1 Hydrostatics

Hydrostatics is the study of the mechanics of static fluids. In this case $\boldsymbol{u} = 0$ and the Navier-Stokes equations reduce to

$$\nabla p = \rho \boldsymbol{g}, \tag{5.1}$$

which may be readily integrated to yield

$$p = p_0 + \rho \boldsymbol{g} \cdot (\boldsymbol{x} - \boldsymbol{x}_0), \tag{5.2}$$

where p_0 is the pressure at a reference point $\boldsymbol{x}_0$. The general result is that pressure increases with depth at a rate that balances the increase of weight of the fluid per unit area.

5.1.1 Archimedes principle

A cornerstone of hydrostatics is force exerted by a static fluid on immersed (or floating) bodies given by the Archimedes principle. We can now derive this principle for a body occupying a volume Ω as follows. The net hydrostatic force on the body, $\boldsymbol{F}_h$, is due to hydrostatic pressure

$$\boldsymbol{F}_h = \int_{\partial\Omega} -p\boldsymbol{n} \, \mathrm{d}A = \int_{\partial\Omega} (p_0 + \rho \boldsymbol{g} \cdot (\boldsymbol{x} - \boldsymbol{x}_0)) \, \boldsymbol{n} \, \mathrm{d}A, \tag{5.3}$$

where $\boldsymbol{n}$ is the unit normal pointing out of the body. This expression can be simplified using the divergence theorem as

$$\boldsymbol{F}_h = \int_{\Omega} \rho \boldsymbol{g} \, \mathrm{d}V. \tag{5.4}$$

In other words, the net force on the immersed body is equal to the net volumetric force acting on the fluid displaced by the body. This is the statement of Archimedes principle.

5.2 Bernoulli equation

The Bernoulli equation is a celebrated result in fluid dynamics, which could be considered as a generalization of hydrostatics. There are two different conditions for incompressible flow under which two different versions

of Bernoulli equation apply. Here we present those two. Analogues of these equations for compressible flow are also possible, which we do not treat.

Underlying both of these versions is an expression of incompressible momentum balance in differential form that uses the identity

$$\boldsymbol{u} \times \boldsymbol{\omega} = \boldsymbol{u} \times (\nabla \times \boldsymbol{u}) = \nabla \left(\frac{1}{2} |\boldsymbol{u}|^2 \right) - \boldsymbol{u} \cdot \nabla \boldsymbol{u}. \tag{5.5}$$

Using (5.5) to re-write the $\boldsymbol{u} \cdot \nabla \boldsymbol{u}$ term in the momentum balance yields

$$\left(\rho \frac{\partial \boldsymbol{u}}{\partial t} + \nabla \left(\frac{1}{2} \rho |\boldsymbol{u}|^2 + p - \rho \boldsymbol{g} \cdot \boldsymbol{x} \right) \right) = \mu \nabla^2 \boldsymbol{u} + \rho \boldsymbol{u} \times \boldsymbol{\omega}. \tag{5.6}$$

5.2.1 Steady incompressible inviscid flow

For a steady inviscid flow, (5.6) simplifies to

$$\nabla \tilde{B} = \rho \boldsymbol{u} \times \boldsymbol{\omega}, \qquad \text{where} \quad \tilde{B} = \frac{1}{2} \rho |\boldsymbol{u}|^2 + p - \rho \boldsymbol{g} \cdot \boldsymbol{x} \tag{5.7}$$

The derivative of $\tilde{B}$ in the direction of either $\boldsymbol{u}$ or $\boldsymbol{\omega}$ is zero, i.e.,

$$\boldsymbol{u} \cdot \nabla \tilde{B} = \rho \boldsymbol{u} \cdot (\boldsymbol{u} \times \boldsymbol{\omega}) = 0 \text{ and } \boldsymbol{\omega} \cdot \nabla \tilde{B} = \rho \boldsymbol{\omega} \cdot (\boldsymbol{u} \times \boldsymbol{\omega}) = 0. \tag{5.8}$$

Therefore, $\tilde{B}$ is constant along a curve that is everywhere tangent to the velocity (i.e. along a streamline) or to the vorticity (i.e. along a curve known as the vortex line).

5.2.2 Unsteady incompressible potential flow

Potential flow implies that the velocity is the gradient of a scalar function, $\phi(\boldsymbol{x}, t)$, called the vector potential, i.e., $\boldsymbol{u} = \nabla \phi$, everywhere in the domain of interest. Due to incompressibility, the velocity potential satisfies the Laplace equation

$$\nabla \cdot \boldsymbol{u} = 0 \quad \Longrightarrow \quad \nabla \cdot (\nabla \phi) = \nabla^2 \phi = 0. \tag{5.9}$$

Furthermore, owing to the identity that the curl of a gradient vanishes,

$$\boldsymbol{\omega} = \nabla \times \boldsymbol{u} = \nabla \times \nabla \phi = 0. \tag{5.10}$$

The two terms on the right-hand side of (5.6) vanish because $\nabla^2 \boldsymbol{u} = \nabla^2 (\nabla \phi) = \nabla (\nabla^2 \phi) = \boldsymbol{0}$, and $\boldsymbol{\omega} = 0$. The unsteady term on the left hand side can also be included inside the gradient to yield

$$\nabla \hat{B} = \boldsymbol{0}, \qquad \text{where} \qquad \hat{B} = \rho \frac{\partial \phi}{\partial t} + \frac{1}{2} \rho |\boldsymbol{u}|^2 + p - \rho \boldsymbol{g} \cdot \boldsymbol{x}. \tag{5.11}$$

Because the gradient of $\hat{B}$ vanishes everywhere, $\hat{B}$ does not vary with the spatial coordinates, but may depend on time. Note that there is no condition on the viscosity of the fluid in this case.

5.3 Applications and examples

The main use of Bernoulli equation is the determine pressure once the form of velocity is known. In the discussion surrounding the Navier-Stokes equations (see §2.12.2), the difficulty of determining pressure for an incompressible flow was identified. In general, for incompressible fluids, pressure is a variable that is determined implicitly through the condition that the fluid density is not altered by the flow. This difficulty is overcome by the Bernoulli equation when one of the conditions for its validity applies. Also note that the two forms of Bernoulli equation reduce to hydrostatics for the case $\boldsymbol{u} \equiv \boldsymbol{0}$. Here we will present examples of the application of Bernoulli equation.

In many cases, the application of Bernoulli in the steady inviscid case does not require an explicit specification of the complete velocity profile. Instead, the velocity is known at some specific location, and use is made of it to derive relations of practical use.

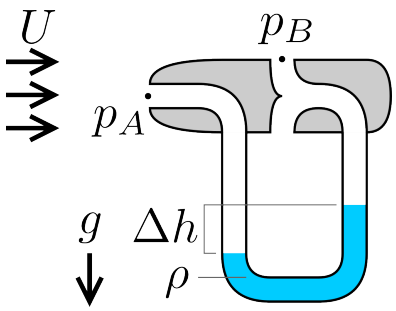


Figure 5.1: Depiction of a Pitot tube. There is an outer flow (of air) incoming at speed U. The liquid in the tube is at rest.

5.3.1 The Pitot tube

The Pitot tube is a device that allows to measure the fluid velocity and it is widely used to determine aircraft velocity. As shown in Figure 5.1, the device consists of a curved tube, one end of which points into the flow while the other is perpendicular to it. We will assume that the fluid on the outside is air, with density ρ_{a}. The tube is filled by a liquid with density ρ, and the air flow causes a height difference in the liquid. We can use the Bernoulli equation to understand how it works and relate the height difference across both ends of the fluid with the flow velocity.

Consider the fluid velocity at the two ends of the tube. At the inlet (A), the air cannot go anywhere (the path is blocked by the liquid in the tube) and therefore the velocity must be zero. This is know as a stagnation point. At the other end, the air velocity is the same as the far-field velocity, so it is equal to inlet velocity u. We assume the flow is irrotational, so we can use the Bernoulli equation for a potential flow. As the flow is steady, this yields

$$p_A = p_B + \frac{1}{2}\rho_{\mathrm{a}} U^2. \tag{5.12}$$

We now need to relate p_A and p_B. We assuming the density of air is negligible, compared with the density of the fluid in the tube, therefore we can just look at the pressure at the two surfaces of the fluid. There is no motion so we can use the hydrostatic equation to find

$$p_A - p_B = \rho g \Delta h. \tag{5.13}$$

Combining both equations we get

$$p_A - p_B = \frac{1}{2}\rho_{\mathrm{a}} U^2 = \rho g \Delta h, \tag{5.14}$$

and, therefore,

$$U = \sqrt{\frac{2\rho g \Delta h}{\rho_{\mathrm{a}}}}. \tag{5.15}$$

5.3.2 Flow through an opening in a tank

Consider water flowing out of a tank through a opening of area A, which is small relative to the area of the tank viewed from above, as shown in Figure 5.2, thereby draining the tank. The question is to determine the rate at which the tank empties. We will see that the rate of emptying will depend on the level of water, H, above the level of the opening.

Here note that the flow is not steady. This is because as the water-level falls, the driving force H decreases, thereby reducing the flow with time. But if the water-level doesn't fall too fast, for example, the opening being too small to drain the water, then a steady approximation might be worthwhile to make. Nor is water inviscid, but we imagine considering the limit of the fluid being inviscid to examine what the consequence of that assumption would be. If the consequence of these approximations agrees with observations, then we will deduce that the assumptions are useful. So let's get on with it.

We are not given any details of the velocity field. So we will assume the velocity to be uniform, with a value U, across the area of the opening. Now let's trace back a streamline from some point near the opening to another point where we might be able to approximate the velocity and pressure. Note that, by the definition

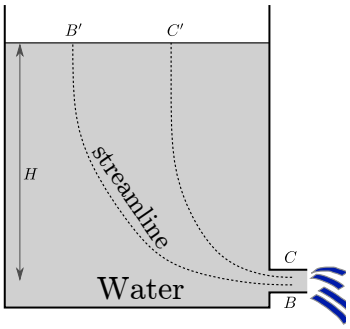


Figure 5.2: Schematic setup for flow out of an opening in a tank.

of a streamline, it depicts the instantaneous chain of displacement of fluid particles. Since the fluid exiting the opening is ultimately being displaced by the falling water-level, every streamline that starts in the opening must terminate at the free surface. Thus we consider a streamline BB' or CC' for our consideration.

On such a streamline, Bernoulli equation for steady inviscid flow states

$$\tilde{B} = \frac{1}{2}\rho|\boldsymbol{u}|^2 + p - \rho\boldsymbol{g}\cdot\boldsymbol{x} = \text{constant}, \tag{5.16}$$

and, therefore,

$$p_B + \frac{1}{2}\rho U^2 - \rho\boldsymbol{g}\cdot\boldsymbol{x}_B = p_{B'} + \frac{1}{2}\rho U_{B'}^2 - \rho\boldsymbol{g}\cdot\boldsymbol{x}_{B'}. \tag{5.17}$$

Here we use our approximation $U_{B'} \ll U$, so the term $\rho U_{B'}^2/2$ on the right-hand side is negligible. We also use that both p_B and $p_{B'}$ are close to atmospheric values (because both the free surface and the opening are exposed to open air), so they cancel each other from the two sides of this equation. Then using $\boldsymbol{g}\cdot(\boldsymbol{x}_B - \boldsymbol{x}_{B'}) = gH$ and solving for U yields

$$U = \sqrt{2gH}. \tag{5.18}$$

Water exits the opening at the same speed as an object freely falling under gravitational acceleration g from a height H.

5.3.3 A water wave

The Cartesian components of velocity (u, v, w) in a semi-infinite layer of fluid $(y \le 0)$ of density ρ are given by

$$u = k\sin(\omega t)\cos(kx)e^{-|k|y}, \qquad v = -|k|\sin(\omega t)\sin(kx)e^{-|k|y}, \qquad w = 0,$$

where ω and k are constants and t is time. Gravity or any other volumetric force is absent. This velocity field closely approximates the flow under a sinusoidal train of water waves. The question is to deduce the pressure on the surface $y = 0$.

This is unsteady flow. It will be extremely useful if we could apply the unsteady version of Bernoulli equation here. Let us first examine if the conditions for it are satisfied.

1. **Is the flow incompressible?**

$$\nabla\cdot\boldsymbol{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = \sin(\omega t)e^{-|k|y}\left(-k^2\sin(kx) + |k|^2\sin(kx) + 0\right) = 0. \tag{5.19}$$

   Yes!

2. **Is the flow irrotational?** The velocity is two-dimensional, so we only need to verify the z-component of vorticity

$$\omega_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = \sin(\omega t)e^{-|k|y}\left(-k|k|\cos(kx) - (-k|k|)\cos(kx)\right). \tag{5.20}$$

Yes! So the velocity may be written as the gradient of a potential ϕ.

$$\frac{\partial \phi}{\partial x} = u = k \sin(\omega t) \cos(kx) e^{-|k|y}, \quad \frac{\partial \phi}{\partial y} = v = -|k| \sin(\omega t) \sin(kx) e^{-|k|y}, \tag{5.21}$$

which yields

$$\phi = \sin(\omega t) \sin(kx) e^{-|k|y}. \tag{5.22}$$

According to Bernoulli (in the absence of gravity),

$$p = -\rho \frac{\partial \phi}{\partial t} - \frac{1}{2} \rho |\boldsymbol{u}|^2 = -\rho \omega \cos(\omega t) \sin(kx) e^{-|k|y} - \frac{1}{2} \rho k^2 \sin^2(\omega t) e^{-2|k|y}. \tag{5.23}$$

On $y = 0$, the pressure is

$$p|_{y=0} = -\rho \omega \cos(\omega t) \sin(kx) - \frac{1}{2} \rho k^2 \sin^2(\omega t). \tag{5.24}$$

A useful homework is to deduce the pressure using the Navier-Stokes equations and verify that it agrees with the one obtained using Bernoulli.

Chapter 6

Potential flow

The case where the fluid velocity in given by

$$\boldsymbol{u} = \nabla\phi, \qquad \nabla^2\phi = 0, \tag{6.1}$$

is called potential flow. Velocity profiles of this type arise in a number of situations in practice, but most importantly this theory allows fluid dynamicists to construct flow profiles that automatically satisfy conservation of mass and momentum. Here we treat this subject.

6.1 Conditions for and implications of potential flow

The conditions under which potential flow is realized are those of incompressibility and irrotationality, i.e.,

$$\nabla \cdot \boldsymbol{u} = 0 \qquad \text{and} \qquad \nabla \times \boldsymbol{u} = \boldsymbol{0}, \text{ respectively.} \tag{6.2}$$

A necessary and sufficient condition for irrotationality is

$$\nabla \times \boldsymbol{u} = 0 \qquad \Longleftrightarrow \qquad \boldsymbol{u} = \nabla\phi, \tag{6.3}$$

where ϕ is a scalar field known as the velocity potential. Substituting this form of $\boldsymbol{u}$ into the incompressibility conditions yields Laplace equation $\nabla^2\phi = 0$ for ϕ. In this way, mass is conserved.

We have seen in (5.11), that momentum is conserved if the pressure satisfies the Bernoulli equation, i.e.,

$$\rho\frac{\partial\phi}{\partial t} + \frac{1}{2}\rho|\boldsymbol{u}|^2 + p - \rho\boldsymbol{g}\cdot\boldsymbol{x} = \text{constant in } \boldsymbol{x}. \tag{6.4}$$

In this manner, potential flow satisfies mass and momentum conservation for a viscous fluid.

To construct a potential flow, one solves the Laplace equation (6.1) for ϕ subject to appropriate boundary conditions. Given the nature of Laplace equation, on the boundaries only one of the two components of velocity may be imposed, but not both. On the boundary between the fluid and a solid object, a condition of no-normal flow is imposed, i.e.,

$$\nabla\phi \cdot \boldsymbol{n} = 0, \qquad \text{on solid walls.} \tag{6.5}$$

Therefore, in general, the no-slip condition cannot be satisfied by potential flow past solid objects and potential flow generally slips past the walls. Regardless, potential flow is incredibly successful in providing insight into situations where solid bodies are either not present or could be abstracted away.

6.2 Incompressible and irrotational flow in two-dimensions

While the general concept of potential flow applies in three dimensions, the application of complex variables makes potential flow much more powerful in two dimensions. Essentially, the use of complex variables trivializes

the solution of Laplace equation. The only remaining complexity that remains is the satisfaction of the boundary conditions. In the rest of this chapter, we will focus on two-dimensional incompressible irrotational flow.

Irrotationality in two dimension implies for the components of velocity $\boldsymbol{u} = (u, v)$

$$\nabla \times \boldsymbol{u} = \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \boldsymbol{e}_z = \boldsymbol{0} \qquad \Longleftrightarrow \qquad u = \frac{\partial \phi}{\partial x}, \quad v = \frac{\partial \phi}{\partial y}. \tag{6.6}$$

Similarly, incompressibility implies

$$\nabla \cdot \boldsymbol{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \qquad \Longleftrightarrow \qquad u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}, \tag{6.7}$$

where ψ is a field called stream function.

This type of reciprocal relation between the derivatives of two scalar functions, in our case ϕ and ψ, are called the Cauchy-Riemann conditions:

$$\frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y}, \qquad \frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x}. \tag{6.8}$$

Owing to this relation, both ϕ and ψ satisfy Laplace equation, i.e., $\nabla^2 \phi = \nabla^2 \psi = 0$. Furthermore, ϕ and ψ are the real and imaginary part of a holomorphic function w of the complex variable z, i.e.

$$w(z) = \phi(x, y) + i\psi(x, y), \qquad \text{where} \qquad z = x + iy. \tag{6.9}$$

Thus, the strategy for generating potential flows is to construct holomorphic functions $w(z)$ called complex potentials that satisfy the necessary boundary condition. The fluid velocity is given by the relation

$$u - iv = \frac{dw}{dz}. \tag{6.10}$$

Here are some geometric relations between ϕ, ψ and streamlines.

6.2.1 Stream function and streamlines

Contours of constant stream function are streamlines of the flow. To see this, consider the directional derivative of the stream function in the direction of the streamline, i.e. in the direction of velocity

$$\nabla \psi \cdot \boldsymbol{u} = \frac{\partial \psi}{\partial x} u + \frac{\partial \psi}{\partial y} v = -vu + uv = 0. \tag{6.11}$$

Thus, irrespective of location in the flow, ψ does not vary in the direction of streamlines. Therefore, ψ is a constant along a streamline. In other words, streamlines are contours of constant stream function.

6.2.2 Stream function and velocity potential

Contours of constant stream function and those of constant velocity potential meet orthogonally. To see this, consider the dot product of $\nabla \phi$ and $\nabla \psi$, the normals of the contours at a given point:

$$\nabla \phi \cdot \nabla \psi = \frac{\partial \psi}{\partial x} \frac{\partial \phi}{\partial x} + \frac{\partial \psi}{\partial y} \frac{\partial \phi}{\partial y} = 0. \tag{6.12}$$

Since the normals are orthogonal, the tangents must be too, and therefore the curves must be too.

6.2.3 Polar coordinates

In some cases we might find more convenient to work in polar coordinates. The definition of the velocity potential is conceptually the same, but we need to be careful with the coordinate system. We now define $\boldsymbol{u} = (u, v)$ to be the velocities in the radial and angular directions, respectively. The velocity potential ϕ satisfies $\boldsymbol{u} = \nabla \phi$ which, in polar coordinates, is

$$u = \frac{\partial \phi}{\partial r}, \qquad v = \frac{1}{r} \frac{\partial \phi}{\partial \theta}. \tag{6.13}$$

The contour lines of the stream function must still be perpendicular to those of the velocity potential, therefore they are defined as

$$u = \frac{1}{r}\frac{\partial \psi}{\partial \theta}, \qquad v = -\frac{\partial \psi}{\partial r}. \tag{6.14}$$

6.3 Elementary complex potentials

Some basic elements that are used to construct the complex potentials.

6.3.1 Uniform flow

Consider the complex potential

$$w(z) = Ue^{-i\alpha}z \qquad \Longrightarrow \qquad \phi = U(x\cos\alpha + y\sin\alpha), \quad \psi = U(y\cos\alpha - x\sin\alpha). \tag{6.15}$$

The flow velocity corresponding to this potential is

$$u = U\cos\alpha, \qquad v = U\sin\alpha, \tag{6.16}$$

i.e., a uniform flow at an angle α to the x-axis. An example of this flow is shown in Figure 6.1(a).

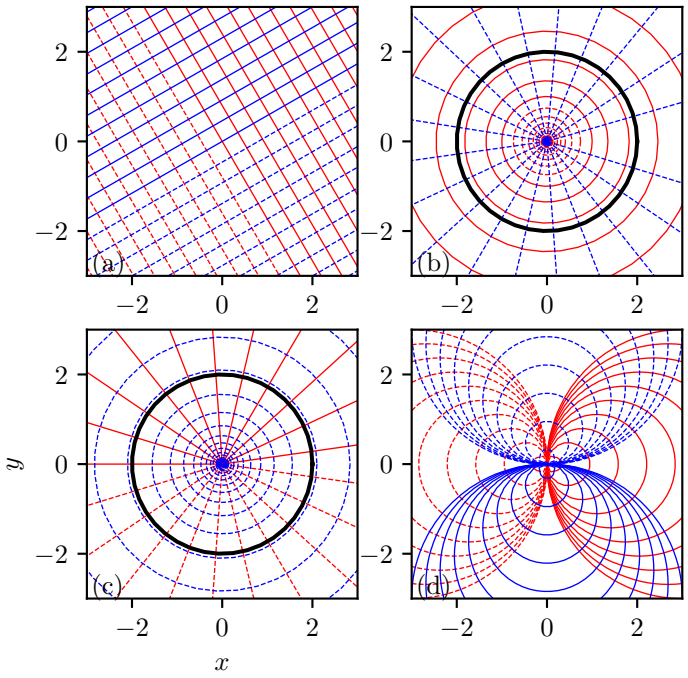


Figure 6.1: Visualization of Elementary flows. Streamlines are in blue and equipotential lines are in red. (a) Uniform flow at an angle of 30° to the x-axis. Potential $w(z) = ze^{i\pi/6}$. (b) Flow due to a point source. Potential $w(z) = \log z$. (c) Flow due to a point vortex. Potential $w(z) = i\log z$. (d) Flow due to a point dipole. Potential $w(z) = 1/z$. The black circle in panels (b) and (c) depicts the contour used for illustrating the concentrated nature of the point source and vortex.

6.3.2 Point source

Consider the complex potential

$$w(z) = \frac{Q}{2\pi}\log z \qquad \Longrightarrow \qquad \phi = \frac{Q}{2\pi}\log r, \quad \psi = \frac{Q\theta}{2\pi}, \tag{6.17}$$

where $z = re^{i\theta}$ is the polar representation of the complex variable z. The flow velocity corresponding to this potential is

$$\boldsymbol{u} = \frac{Q}{2\pi r}\boldsymbol{e}_r, \tag{6.18}$$

i.e., a radially outward flow decaying inversely proportional with distance. An example of this flow is shown in Figure 6.1(b). For what follows, let $\boldsymbol{u}\cdot\boldsymbol{e}_r = u_r$ and $\boldsymbol{u}\cdot\boldsymbol{e}_\theta = u_\theta$.

To understand why the flow corresponding to this potential is termed as the point source, consider the divergence and curl of the velocity field:

$$\nabla\cdot\boldsymbol{u} = \frac{1}{r}\left(\frac{\partial(ru_r)}{\partial r} + \frac{\partial u_\theta}{\partial\theta}\right) = 0, \tag{6.19}$$

$$\nabla\times\boldsymbol{u} = \frac{1}{r}\left(\frac{\partial(ru_\theta)}{\partial r} - \frac{\partial u_r}{\partial\theta}\right)\boldsymbol{e}_z = \boldsymbol{0} \tag{6.20}$$

for $r > 0$, but this calculation says nothing about $r = 0$. To deduce that, consider the line integrals along the circle of radius r

$$\int_C \boldsymbol{u}\cdot\boldsymbol{n}\ \mathrm{d}s = \int_0^{2\pi}\frac{Q}{2\pi r}\boldsymbol{e}_r\cdot\boldsymbol{e}_r\ r d\theta = Q, \tag{6.21}$$

$$\int_C \boldsymbol{u}\cdot\boldsymbol{t}\ \mathrm{d}s = \int_0^{2\pi}\frac{Q}{2\pi r}\boldsymbol{e}_r\cdot\boldsymbol{e}_\theta\ r d\theta = 0, \tag{6.22}$$

where $\boldsymbol{n}$ and $\boldsymbol{t}$ are the unit normal and tangential vectors to the curve C. Given that, due to the divergence and Stokes theorems, respectively,

$$\int_C \boldsymbol{u}\cdot\boldsymbol{n}\ \mathrm{d}s = \int_A \nabla\cdot\boldsymbol{u}\ \mathrm{d}A, \quad \text{and} \tag{6.23}$$

$$\int_C \boldsymbol{u}\cdot\boldsymbol{t}\ \mathrm{d}s = \int_A (\nabla\times\boldsymbol{u})\cdot\boldsymbol{e}_z\ \mathrm{d}A, \tag{6.24}$$

the inescapable conclusion is that for this flow $\nabla\cdot\boldsymbol{u} = Q\delta(\boldsymbol{x})$, where δ is the Dirac-delta function and $\nabla\times\boldsymbol{u} = \boldsymbol{0}$. Thus, this flow is irrotational everywhere, but incompressible only for $r > 0$. Volume is not conserved at $r = 0$, but instead is generated there. Hence the terminology to describe this flow as a "point source".

6.3.3 Point vortex

Consider the complex potential

$$w(z) = -\frac{i\Gamma}{2\pi}\log z \qquad \Longrightarrow \qquad \phi = \frac{\Gamma\theta}{2\pi}, \quad \psi = -\frac{\Gamma}{2\pi}\log r. \tag{6.25}$$

The flow velocity corresponding to this potential is

$$\boldsymbol{u} = \frac{\Gamma}{2\pi r}\boldsymbol{e}_\theta, \tag{6.26}$$

i.e., an azimuthal flow decaying inversely proportional with distance. An example of this flow is shown in Figure 6.1(c).

We again examine $\nabla\cdot\boldsymbol{u}$ and $\nabla\times\boldsymbol{u}$ to understand the terminology associated with this flow. Just as for the point source,

$$\nabla\cdot\boldsymbol{u} = \frac{1}{r}\left(\frac{\partial(ru_r)}{\partial r} + \frac{\partial u_\theta}{\partial\theta}\right) = 0, \tag{6.27}$$

$$\nabla\times\boldsymbol{u} = \frac{1}{r}\left(\frac{\partial(ru_\theta)}{\partial r} - \frac{\partial u_r}{\partial\theta}\right)\boldsymbol{e}_z = \boldsymbol{0} \tag{6.28}$$

for $r > 0$, but which says nothing about $r = 0$. The line integrals along the circle of radius r

$$\int_C \boldsymbol{u}\cdot\boldsymbol{n}\ \mathrm{d}s = \int_0^{2\pi}\frac{\Gamma}{2\pi r}\boldsymbol{e}_\theta\cdot\boldsymbol{e}_r\ r d\theta = 0, \tag{6.29}$$

$$\int_C \boldsymbol{u}\cdot\boldsymbol{t}\ \mathrm{d}s = \int_0^{2\pi}\frac{\Gamma}{2\pi r}\boldsymbol{e}_\theta\cdot\boldsymbol{e}_\theta\ r d\theta = \Gamma, \tag{6.30}$$

which, in view of (6.23-6.24) lead to the inescapable conclusion that the flow is incompressible everywhere but irrotational only for $r > 0$. The vorticity is, in fact, $\nabla\times\boldsymbol{u} = \Gamma\delta(\boldsymbol{x})\boldsymbol{e}_z$. Hence the flow due to this potential is termed as the "point vortex".

6.3.4 Point dipole

The flow constructed by the linear superposition of a point source and an equal point sink (i.e., a source with negative strength) located vanishingly small distances apart constitutes a point dipole. The complex potential is constructed as

$$w(z) = \lim_{\epsilon \to 0} \frac{Q}{2\pi} \left(\log(z+\epsilon) - \log(z-\epsilon) \right). \tag{6.31}$$

Of course, in the limit $\epsilon \to 0$, the potential would vanish, unless the source-sink pair also strengthened in the process. In particular, let $Q = D/(2\epsilon)$, so that the source-sink pair strength is inversely proportional to the distance between them. Here D is the strength of the point dipole. With this assumption,

$$w(z) = \lim_{\epsilon \to 0} \frac{D}{2\pi} \left(\frac{\log(z+\epsilon) - \log(z-\epsilon)}{2\epsilon} \right) = \frac{D}{2\pi z}. \tag{6.32}$$

The flow due to the point dipole is depicted in Figure 6.1(d). The streamlines and equipotential lines are both sets of circles that are tangential to the x and y axes, respectively, at the origin.

6.3.5 Higher multipoles

Note that, due to its construction, the potential due to a point dipole is the derivative of the potential due to a point source. This can be extended to higher orders called "multipoles". For example, a point quadrupole is a point dipole of point dipoles with a complex potential proportional to $1/z^2$. And octupole is a point dipole of quadrupoles with potential $1/z^3$. And so on.

In this manner, the general complex potential with the far-field approaching uniform flow may be written as

$$w(z) = Uz + \frac{Q - i\Gamma}{2\pi} \log z + \frac{a_1}{z} + \frac{a_2}{z^2} + \frac{a_3}{z^3} + \dots. \tag{6.33}$$

Here, the coefficients U, Q, Γ, a_1, a_2, etc, may be functions of time.

6.4 Potential flow past immersed bodies

The coefficients Q, Γ, a_1, a_2, ..., may be chosen to construct the complex potential for flow past bodies. Here are some examples.

6.4.1 Rankine half-body

The linear superposition of a uniform flow and a point source may be used to represent flow past a semi-infinite body, called the Rankine half-body. The complex potential is

$$w(z) = Uz + \frac{Q}{2\pi} \log z. \tag{6.34}$$

The flow streamlines and equipotential lines are shown in Figure 6.2(a). Any region of the plane on one side of a streamline may be replaced by a solid body and the complex potential in the remaining region describes potential flow past this solid body. For example, consider the streamline given by $\psi = Q/2$, where

$$\psi = \mathrm{Im}(w) = Ur\sin\theta + \frac{Q\theta}{2\pi} \quad \text{so the streamline is} \quad r = \frac{Q(\pi - \theta)}{2\pi U \sin\theta}. \tag{6.35}$$

The streamline is shown in Figure 6.2(b) and the area inside the streamline, shaded blue in the figure, is the Rankine half-body. The complex potential $w(z)$ in (6.34) describes the flow past this Rankine half-body.

6.4.2 Rankine oval

The linear superposition of a uniform flow, a point source and a point sink may be used to represent flow past a finite body, called the Rankine oval. The complex potential is

$$w(z) = Uz + \frac{Q}{2\pi} \left(\log(z-1) - \log(z+1) \right). \tag{6.36}$$

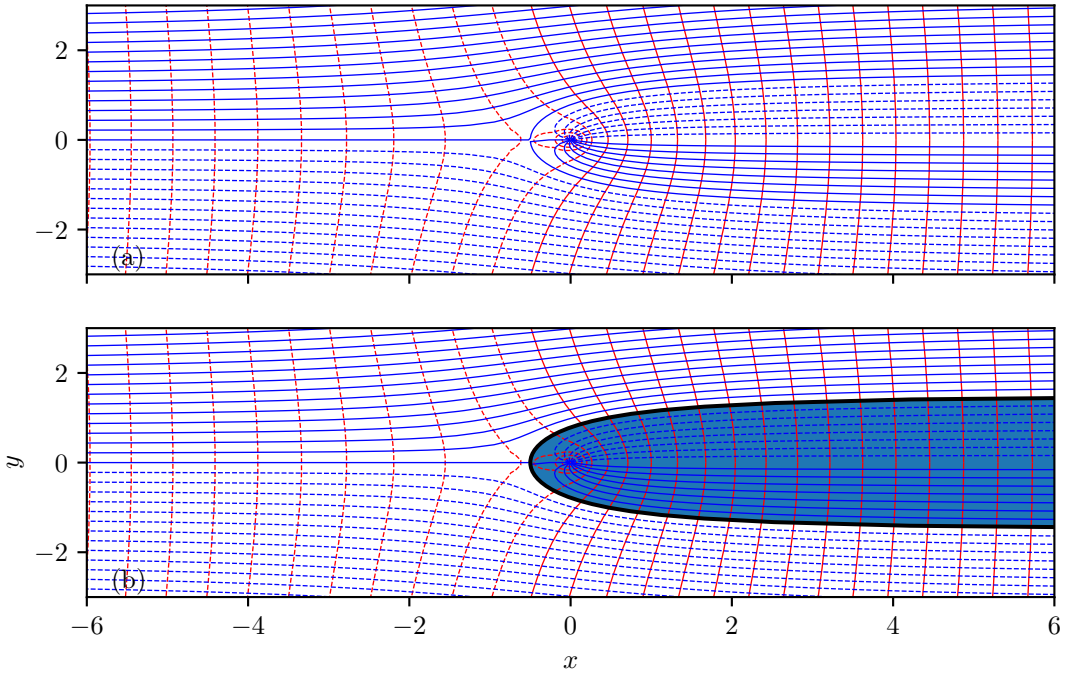


Figure 6.2: Flow around a Rankine half-body. (a) The flow streamlines (blue) and equipotential lines (red) for the potential given by (6.34) for $U = 1$ and $Q = \pi$. (b) Same as (a) but with the streamline $\psi = Q/2$ drawn in solid black and the area inside this streamline shaded in blue. The shaded region is the Rankine half-body.

The flow streamlines and equipotential lines are shown in Figure 6.3(a). Consider the streamline given by $\psi = 0$, where

$$\psi = \mathrm{Im}(w) = Ur\sin\theta + \frac{Q(\theta_1 - \theta_2)}{2\pi} \quad \text{so the streamline is given by} \quad Ur\sin\theta + \frac{Q(\theta_1 - \theta_2)}{2\pi} = 0, \tag{6.37}$$

where $\tan\theta_1 = r\sin\theta/(r\cos\theta + 1)$ and $\tan\theta_2 = r\sin\theta/(r\cos\theta - 1)$. The streamline is shown in Figure 6.3(b) and the area inside the streamline, shaded blue in the figure, is the Rankine oval. The complex potential $w(z)$ in (6.36) describes the flow past this Rankine oval.

6.4.3 Circle

The Rankine oval approaches a circle as the source and the sink approach each other to form a dipole. The complex potential in this case is

$$w(z) = U\left(z + \frac{a^2}{z}\right). \tag{6.38}$$

The flow streamlines and equipotential lines are shown in Figure 6.4(a). Consider the streamline given by $\psi = 0$, where

$$\psi = \mathrm{Im}(w) = U\sin\theta\left(r - \frac{a^2}{r}\right), \quad \text{so that } \psi = 0 \text{ on } \quad r = a. \tag{6.39}$$

The streamline is shown in Figure 6.4(b) and the area inside the streamline, shaded blue in the figure, is a circle of radius a. The complex potential $w(z)$ in (6.38) describes the flow past this circle.

An arbitrary point vortex centred at the center of the circle may be added to the mix without violating the boundary condition on the circle. The most general potential flow around a circle is given by

$$w(z) = U\left(z + \frac{a^2}{z}\right) - \frac{i\Gamma}{2\pi}\log z. \tag{6.40}$$

Here Γ is the circulation around the circle, perhaps due to the spinning of the circle. The flow due to this potential is shown in Figure 6.4(c).

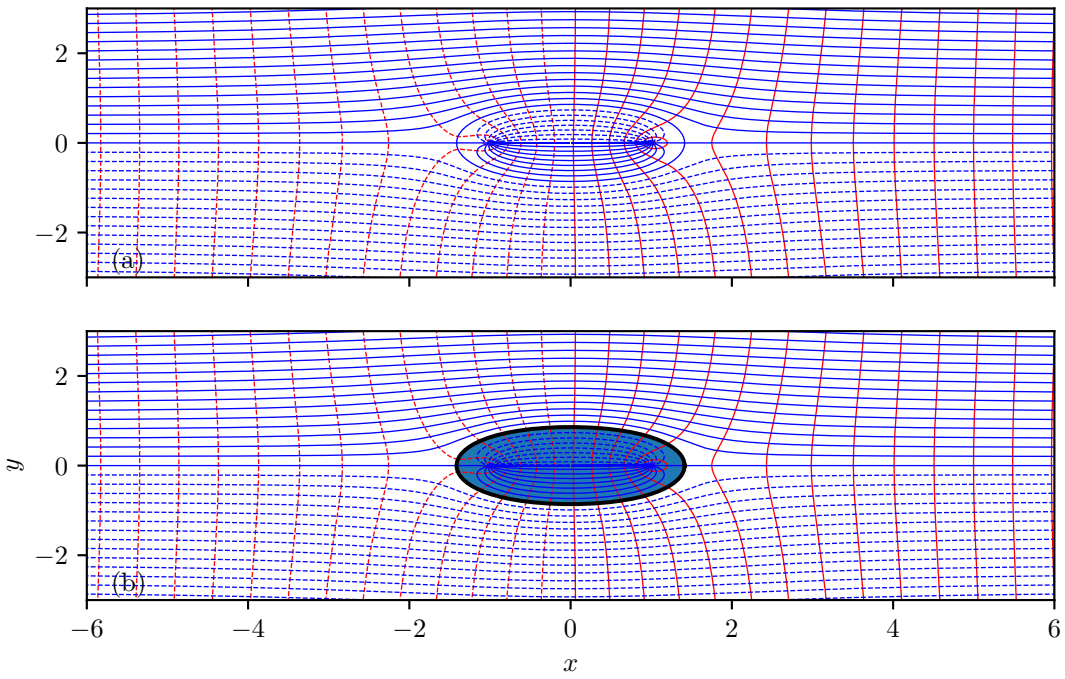


Figure 6.3: Flow around a Rankine oval. (a) The flow streamlines (blue) and equipotential lines (red) for the potential given by (6.36) for $U = 1$ and $Q = \pi$. (b) Same as (a) but with the streamline $\psi = 0$ drawn in solid black and the area inside this streamline shaded in blue. The shaded region is the Rankine half-body.

6.5 Force on a immersed body

The force on the body due to the potential flow (6.33) may be calculated by integrating the force due to the fluid pressure. The force on the body per-unit-length-into-the-page is

$$\boldsymbol{F} = \int_C -p\boldsymbol{n}\ \mathrm{d}s, \tag{6.41}$$

where C is the bounding closed streamline defining the body. The fluid pressure is determined by Bernoulli equations as

$$p = -\rho\left(\frac{\partial\phi}{\partial t} + \frac{1}{2}|\boldsymbol{u}|^2\right) + \rho\boldsymbol{g}\cdot\boldsymbol{x}, \tag{6.42}$$

The contribution to the force due to gravity is the force of buoyancy from Archimedes principle. Here we calculate the contribution from the remaining two terms.

6.5.1 Force in a steady potential flow

In steady potential flow, only the term $\rho|\boldsymbol{u}|^2/2$ contributes to the net unbalanced force beyond buoyancy. In complex variables, let the components of the force $\boldsymbol{F} = F_x\boldsymbol{e}_x + F_y\boldsymbol{e}_y$ be written as $F_x + iF_y$. If the tangent to C makes an angle θ to the x-axis and runs counter-clockwise around the body, then $\boldsymbol{u} = |\boldsymbol{u}|e^{i\theta}$ because the flow is tangential to the streamline. Also, because the components of velocity are $w'(z) = u - iv = |\boldsymbol{u}|e^{-i\theta}$, $|\boldsymbol{u}|^2 = w'(z)^2e^{2i\theta}$. The product of the unit normal and the infinitesimal line element is $-ie^{i\theta}\ \mathrm{d}s$. It is then useful to consider the complex conjugate of the force, which is given by

$$F_x - iF_y = \int_C \frac{1}{2}\rho w'(z)^2e^{2i\theta}\ (ie^{-i\theta}\ \mathrm{d}s) = \frac{i\rho}{2}\int_C w'(z)^2\ dz, \tag{6.43}$$

where the complex integration infinitesimal $dz = e^{i\theta}\ \mathrm{d}s$.

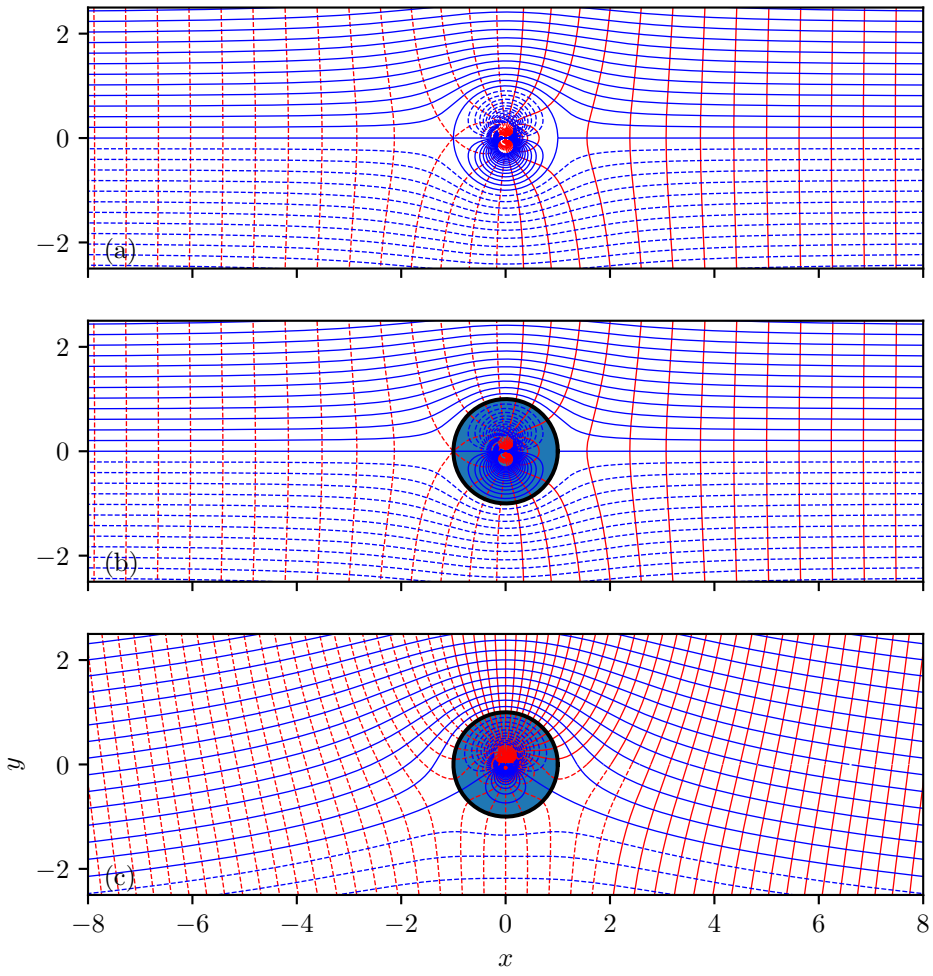


Figure 6.4: Flow around a Rankine oval. (a) The flow streamlines (blue) and equipotential lines (red) for the potential given by (6.38) for $U = 1$ and $a = 1$. (b) Same as (a) but with the streamline $\psi = 0$ drawn in solid black and the area inside this streamline shaded in blue. The shaded region is a circle. (c) The flow around a spinning circle corresponding to the complex potential given by (6.40) for $U = 1$, $a = 1$, $\Gamma = 2\pi$.

Using this expression for force (per-unit-length), and substituting (6.33) in (6.43) yields

$$F_x - iF_y = -\rho U(Q - i\Gamma), \tag{6.44}$$

where Cauchy's residue theorem is employed for calculating the contour integral.

Law of conservation of mass demands that for a body immersed in the flow $Q = 0$, and therefore,

$$F_x = 0 \qquad \text{and} \qquad F_y = -\rho U\Gamma. \tag{6.45}$$

The drag on an object is the force it experiences in the direction of the flow, while the lift is the force perpendicular to the flow. According to (6.45), the magnitude of the lift on an immersed body due to the potential flow is $|\rho U\Gamma|$, which is the statement of the Kutta-Joukowsky (also spelled Joukowski, Zhukowski or Zhukovsky) theorem in aerodynamics. The lift on the spinning circle, due to this theorem, is $|\rho U\Gamma|$.

The fact that a body immersed in a potential flow does not experience any drag is the subject of the D'Alembert paradox. Neither the Rankine oval, nor the circle, nor body of any other shape experiences any drag in potential flow. D'Alembert paradox exist because potential flow does not satisfy the no-slip condition. The friction of a viscous fluid with the solid boundary generates vorticity, which is transported by the fluid to

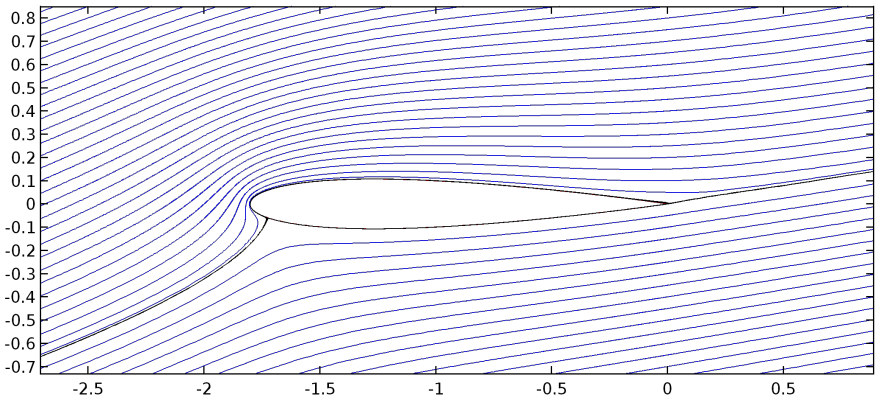


Figure 6.5: Flow around an airfoil at an angle of attack computed using the commercial software COMSOL. The streamlines are shown in blue, and the black curve is the stagnation streamline, part of which makes the airfoil shape. The angle of attack for this case is 14°.

regions away from the immersed body. In this manner, application of the no-slip boundary condition on viscous fluids invalidates the assumptions underlying potential flow, especially that the flow be irrotational. In fact, the shedding of vorticity into the flow is the origin of fluid dynamical drag (this we state without proof at this point).

6.5.2 Added mass from the unsteady term

Consider the contribution of the term $\rho\dfrac{\partial\phi}{\partial t}$. The resulting force is

$$\int_C \rho\frac{\partial\phi}{\partial t}\boldsymbol{n}\ \mathrm{d}s. \tag{6.46}$$

Let us examine this expression for the case of flow around a circle in which the speed of the flow U can now change in time, so that $w(z)$ is given by (6.38). This corresponds to

$$\phi = \mathrm{Re}(w) = U\cos\theta\left(r+\frac{a^2}{r}\right), \qquad \text{which on } r=a \text{ is} \qquad \phi = 2Ua\cos\theta. \tag{6.47}$$

The integral for force yields

$$\boldsymbol{F} = \int_0^{2\pi} 2\rho\frac{dU}{dt}a\cos\theta(\cos\theta\boldsymbol{e}_x+\sin\theta\boldsymbol{e}_y)a\ d\theta = \rho\pi a^2\frac{dU}{dt}\boldsymbol{e}_x. \tag{6.48}$$

This result may be interpreted in the following way. Any acceleration of the circle implies an acceleration of the fluid around it. This acceleration needs an additional force proportional to the acceleration. The coefficient of proportionality between this additional force and the acceleration is the "added mass" the fluid presents to the motion of the circle. In this case, the added mass in $\rho\pi a^2$.

6.6 Flow around an airfoil

The lift generated by an airfoil is a central result in aerodynamics, which makes use of potential flow. Some salient features of this flow are visualized in Figure 6.5. The stagnation streamline is incident on the lower surface of the airfoil, which develops the stagnation pressure under the airfoil. The streamlines curve around the leading edge of the airfoil, generating a low-pressure area above the airfoil. Finally and crucially, note that the stagnation streamline separates from the plate at the trailing edge. These are the features that will guide us in constructing the appropriate potential flow.

Here we will present one calculation, that of the lift on the prototypical airfoil – a flat plate, which highlights this central result. The underlying potential flow is constructed in this case using a technique known as conformal mapping (the same technique can be used to construct flow around airfoils of other shapes).

A conformal map is a map between two complex variables, which we take to be z and ζ. Here $z = x + iy$ corresponds to the physical spatial variable, in which the flat plate exists, and ζ is the mapped variable, in which the flat plate is mapped to a circle. The particular map we use is called the Joukowsky map:

$$z = f(\zeta) = (1+\epsilon)\zeta + (1-\epsilon)\frac{a^2}{\zeta}. \tag{6.49}$$

A circle in ζ-plane, $\zeta = ae^{i\theta}$ is mapped to

$$z = a(1+\epsilon)e^{i\theta} + a(1-\epsilon)e^{-i\theta} = 2a\cos\theta + 2ia\epsilon\sin\theta, \tag{6.50}$$

which is an ellipse of major axis $2a$ along the x-axis and minor axis $2a\epsilon$ along the y-axis. In the limit $\epsilon \to 0$, the ellipse approaches a flat plate $-2a \le x \le 2a$. The point $\zeta = a$ is mapped to the trailing edge of the plate $z = 2a$. The chord of the flat plate is $c = 4a$.

While it is possible to solve for the inverse map $\zeta = f^{-1}(z)$ in closed form, the idea is to not need the explicit expression for the inverse. However, we do need the inverse map to be analytic so that the complex potential

$$W(z) = w\left(f^{-1}(z)\right) \tag{6.51}$$

automatically deforms the streamlines and converts the circular streamline in the ζ coordinates to be a flat plate in the z coordinates, whilst preserving the analyticity of the complex potential.

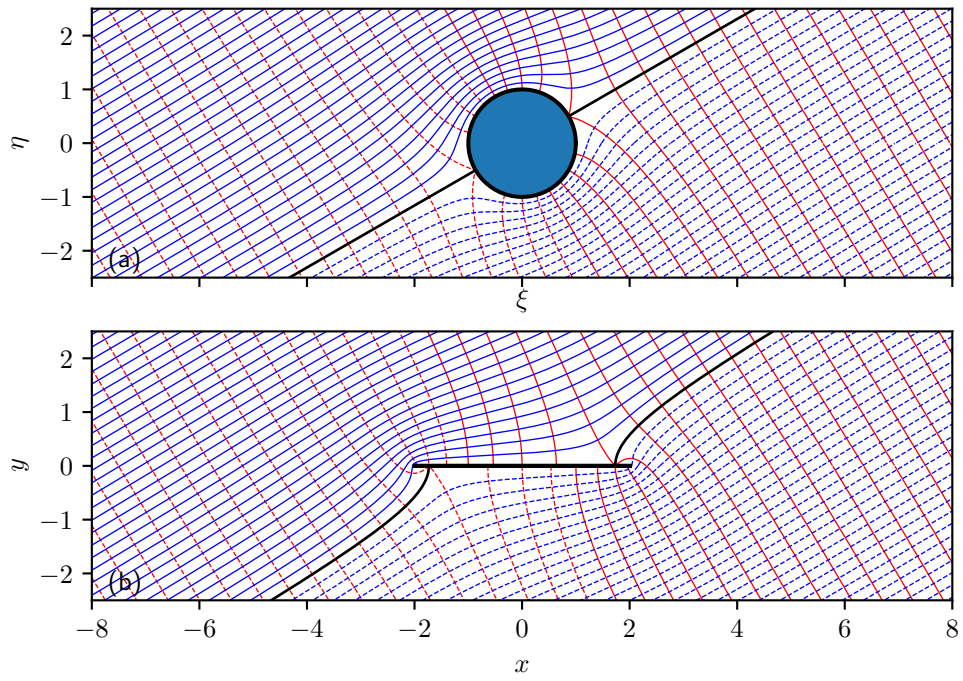


Figure 6.6: Flow around a flat plate at an angle of 30° to the flow and $\Gamma = 0$. (a) The flow in the $\zeta = \xi + i\eta$ coordinates. Streamlines are in blue and equipotential lines in red. Black curve shows the stagnation streamline. (b) The same flow in the $z = x + iy$ coordinates.

We start with the complex potential (6.40) for flow around a circle, but now expressed in ζ and with an angle of attack α, i.e.,

$$w(\zeta) = U\left(\zeta e^{-i\alpha} + \frac{a^2}{\zeta e^{-i\alpha}}\right) - \frac{i\Gamma}{2\pi}\log\zeta. \tag{6.52}$$

As $\zeta \to \infty$, the largest term in $w(\zeta)$ is $U\zeta e^{-i\alpha}$, corresponding with uniform flow at an angle α to the flat plate. The complex velocity in the ζ plane is

$$w'(\zeta) = U\left(e^{-i\alpha} - \frac{a^2}{\zeta^2}e^{i\alpha}\right) - \frac{i\Gamma}{2\pi\zeta}. \tag{6.53}$$

The velocity in the z-coordinates is

$$\frac{dW}{dz} = \frac{w'(\zeta)}{z'(\zeta)} = \frac{U\left(e^{-i\alpha} - \dfrac{a^2}{\zeta^2}e^{i\alpha}\right) - \dfrac{i\Gamma}{2\pi\zeta}}{1 - \dfrac{a^2}{\zeta^2}}. \tag{6.54}$$

As $\zeta \to \infty$, dW/dz approaches U.

Consider the case $\Gamma = 0$. An example of the flow corresponding to this case is shown in Figure 6.6. Note that the stagnation point in the ζ-plane is at $\zeta = \pm ae^{i\alpha}$. This point maps in the z-plane to $z = -2a\cos\alpha$ below the flat plate, and $z = 2a\cos\alpha$ above it. Note that in this flow, the stagnation streamline does not separate from the trailing edge $z = 2a$.

The stagnation point may be moved by judiciously choosing the value of Γ. The condition that the trailing edge be the rear stagnation point is called the *Kutta condition*. We impose the Kutta condition by ensuring that the stagnation point in the ζ-coordinates is at $\zeta = a$. Substituting the velocity $w'(\zeta) = 0$ at $\zeta = a$ yields

$$\Gamma = -4\pi U a \sin\alpha \tag{6.55}$$

The streamlines and equipotential lines corresponding to the value of Γ that enforces the Kutta condition, shown in Figure 6.7, illustrate this condition.

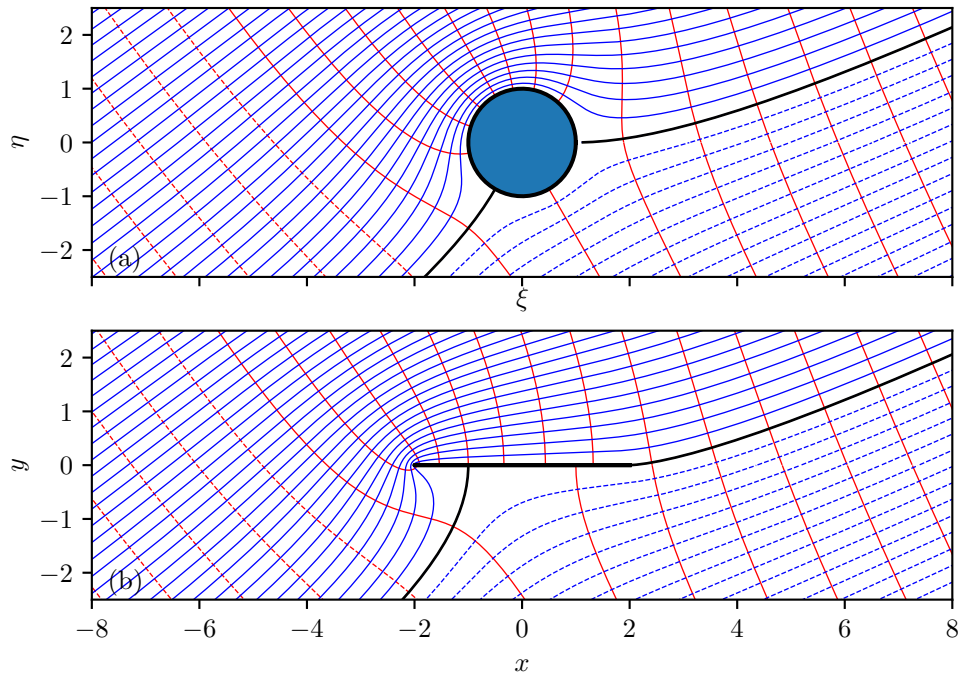


Figure 6.7: Flow around a flat plate at an angle of 30° to the flow and Γ given by (6.55). (a) The flow in the $\zeta = \xi + i\eta$ coordinates. Streamlines are in blue and equipotential lines in red. Black curve shows the stagnation streamline. (b) The same flow in the $z = x + iy$ coordinates.

The lift F_y on the flat-plate is given by

$$F_y = -\rho U\Gamma = 4\pi\rho U^2 a\sin\alpha. \tag{6.56}$$

This result can be converted to a dimensionless form in terms of the coefficient of lift, C_L, defined as

$$C_L = \frac{F_y}{\frac{1}{2}\rho U^2 c}. \tag{6.57}$$

Using the result (6.56), the lift coefficient depends on the angle of attack as

$$C_L = \frac{4\pi\rho U^2 a\sin\alpha}{\frac{1}{2}\rho U^2(4a)} = 2\pi\sin\alpha. \tag{6.58}$$

Figure 6.8 shows that the results of (6.58) compare well with the lift coefficient computed using the commercial software COMSOL for small α. The agreement worsens around $\alpha \approx 10^\circ$ due to a phenomenon called stall, in which the stagnation streamline also separates from the leading edge. A careful treatment of the influence of viscosity is needed to gain a deeper understanding of stall .

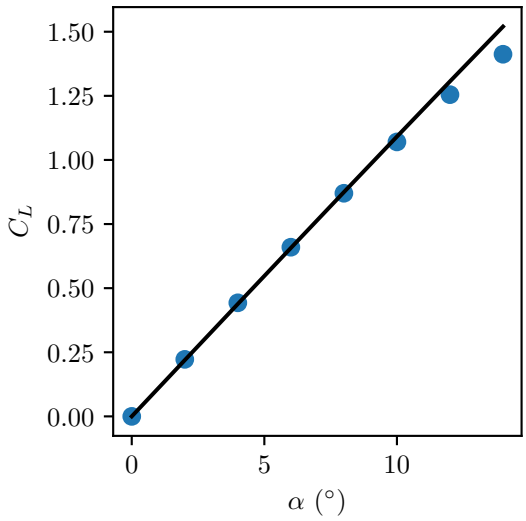


Figure 6.8: Lift coefficient C_L as a function of angle of attack α. Blue circles show the lift calculated from computations in COMSOL. Black line shows results of (6.58).

Chapter 7

Boundary layers

Boundary layers are regions of space adjacent to boundaries of fluid domains where affects of viscosity may not be ignored no matter how small the coefficient of viscosity may be. In particular, although the viscosity is small, the no-slip condition forces large velocity gradients near the wall, making viscous stresses non-negligible there. The characteristics of and the dynamics within boundary layers are of interest especially when the coefficient of viscosity is vanishlingly small. The example of the boundary layer next to a flat plate aligned with the flow, the topic of this chapter, illustrates the essential features.

7.1 Flow next to a flat plate

Consider a uniform flow of speed U along the x-axis being incident on a flat plate of infinitesimal thickness and length L aligned with the flow, as schematically shown in Figure 7.1. The plate is oriented perpendicular to the plane of paper. The fluid has viscosity μ and density ρ. We ignore gravity in this example. The potential flow $\boldsymbol{u} = U\boldsymbol{e}_x$ with a uniform pressure satisfies mass and momentum conservation, and the no-penetration condition on the flat plate, but not the no-slip condition. For an inviscid fluid, this is a perfectly valid flow, but not for a viscous fluid, no matter how small the viscosity coefficient. In this manner, this example offers the opportunity to study how the imposition of the no-slip condition modifies the simplest of potential flows in the presence of the slightest of viscosity. Hence, we choose a vanishingly small coefficient of viscosity μ (more precisely, we will have a large Reynolds number).

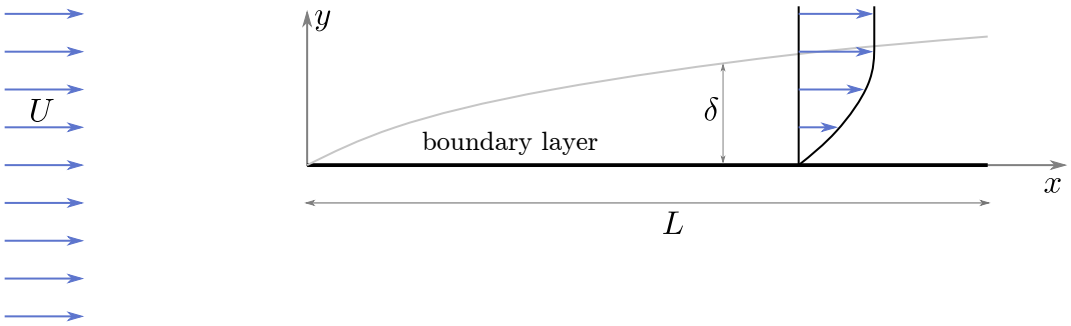


Figure 7.1: Schematic of a flow past a flat plate aligned with the flow. Here the thickness of the boundary layer is denoted as δ.

7.1.1 Qualitative description of the flow

Let us first develop a picture of the dynamics of this flow in our minds. The approach we take is to make various assumptions about the flow now and verify them later. This flow is driven by the inertia of the fluid in the freestream. The fluid particles away from the plate start out with speed U and will continue to do so unless an unbalanced force makes them decelerate. The action of the no-slip boundary condition and the shear

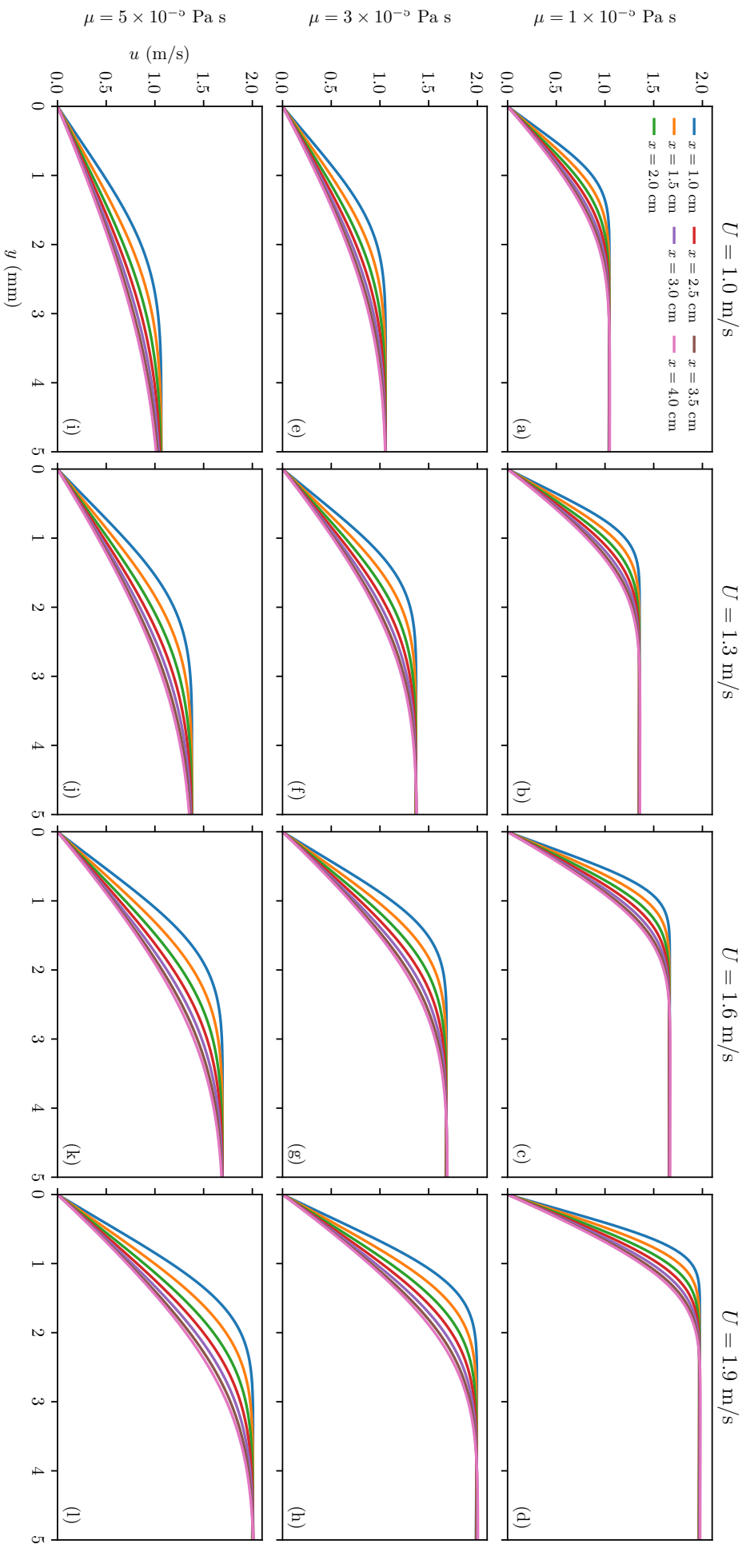


Figure 7.2: Profiles of u versus y taken at 7 distances from the leading edge of the plate. (a-l) Columns correspond to constant U listed at the top of each columns and rows to constant μ listed to the left.

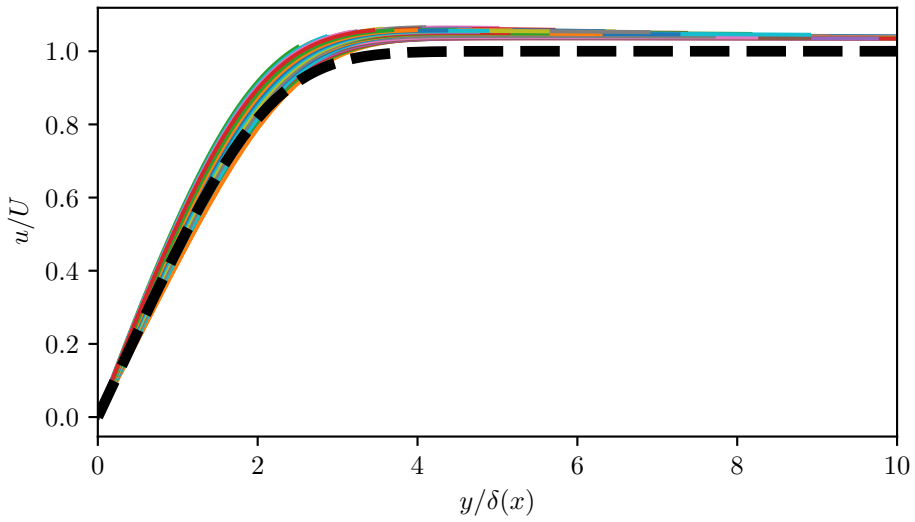


Figure 7.3: Collapse of the profiles of u versus y when rescaled. Each of the curves in Figure 7.2 is plotted as the coloured curves in this figure. Here δ is from (7.18). The dashed black curve corresponds to the solution of (7.24-7.25).

stress that develops in the fluid is expected to slow down the fluid in the vicinity of the flat plate. A typical profile u versus y is shown in Figure 7.1. The boundary layer, the region where the flow speed appreciably falls blow the freestream value U, is expected to be of thickness δ that is also vanishingly small with the coefficient of viscosity. Outside this layer, we expect the flow to be approximately uniform, and certainly described by potential flow. A natural consequence of the slowing down of the flow in the boundary layer is that the fluid upstream that is incident just above the plate now needs to be deflected away from the plate. Therefore, a y components of velocity also naturally appears in the flow. However, because of the thinness of the boundary layer, the amount of deflection is weak and the y component is expected to be smaller in magnitude than the x component.

A computational solution of Navier-Stokes equations is used to assist with the qualitative description of the flow. The commercial finite elements software COMSOL is used to obtain the solution. The particular case considered is for $L = 5$ cm, $\rho = 1$ kg/m^3 and several values for μ and U (listed in Figure 7.2. The plate in this case has a finite thickness equal to 1/64 the length. The computational box is chosen to be 10 cm $\times$ 10 cm.

To obtain some insight into the thickness of the boundary layer, Figure 7.2 shows the computed profiles of velocity for different values of U and μ. It is evident that increasing U decreases δ, while increasing μ increases it. The growth of the boundary layer with x, the distance form the leading edge, is also evident. These are the features we seek to explain in the subsequent analysis.

A scaling estimate for δ may be made in the following way. A fluid particle starting from the leading edge of the plate traveling with speed U takes time $t = x/U$ to traverse the distance x. Perpendicular to the direction of travel, viscosity influences a distance $\delta = \sqrt{\nu t}$ in time t , where $\nu = \mu/\rho$ is the kinematic viscosity. In this case, this implies $\delta = \sqrt{\nu x/U}$. Miraculously, replotting a rescaled version of the data shown in Figure 7.2 collapses each one of those profiles of u versus y very close to a universal curves. Figure 7.3 demonstrates this collapse where y is rescaled using $\delta(x)$ and u with U. To explain this miraculous collapse, to determine the universal shape the velocity collapses to, and to better understand the underlying fluid dynamics, we proceed with certain simplifying assumptions about the flow.

It can be seen from this estimate that the boundary layer becomes vanishingly thin as ν approaches zero. The slope of the growing boundary layer thickness is proportional to δ/x, and therefore, we expect the y component of velocity to be of the magnitude $U\delta/x$. Subsequent analysis proceeds under the assumption that δ is much smaller than L in the following analysis to derive the so-called Prandtl boundary-layer equations.

Equation	**Terms and their magnitudes**				
Mass conservation (7.1a)	$\frac{\partial u}{\partial x} \sim \frac{U}{L}$	$\frac{\partial v}{\partial y} \sim \frac{V}{\delta}$			
x-momentum conservation (7.1b)	$\rho u\frac{\partial u}{\partial x} \sim \frac{\rho U^2}{L}$	$\rho v\frac{\partial u}{\partial y} \sim \frac{\rho UV}{\delta}$	$\frac{\partial p}{\partial x} \sim \frac{P}{L}$	$\mu\frac{\partial^2 u}{\partial x^2} \sim \frac{\mu U}{L^2}$	$\mu\frac{\partial^2 u}{\partial y^2} \sim \frac{\mu U}{\delta^2}$
y-momentum conservation (7.1c)	$\rho u\frac{\partial v}{\partial x} \sim \frac{\rho UV}{L}$	$\rho v\frac{\partial v}{\partial y} \sim \frac{\rho V^2}{\delta}$	$\frac{\partial p}{\partial y} \sim \frac{P}{\delta}$	$\mu\frac{\partial^2 v}{\partial x^2} \sim \frac{\mu V}{L^2}$	$\mu\frac{\partial^2 v}{\partial y^2} \sim \frac{\mu V}{\delta^2}$
(magnitudes of (7.1c) simplified)	$\frac{\rho U^2}{L}\frac{1}{\sqrt{\mathrm{Re}}}$	$\frac{\rho U^2}{L}\frac{1}{\sqrt{\mathrm{Re}}}$	$\frac{\rho U^2}{L}\sqrt{\mathrm{Re}}$	$\frac{\rho U^2}{L}\frac{1}{\sqrt{\mathrm{Re}^3}}$	$\frac{\rho U^2}{L}\frac{1}{\sqrt{\mathrm{Re}}}$

Table 7.1: Scaling estimate of the magnitude of terms in (7.1).

7.1.2 Derivation of the Prandtl boundary layer equations

In the limit of small viscosity, and especially due to the boundary layer being thinner than the length of the plate, the governing Navier-Stokes equations simplify to the simpler Prandtl boundary layer equations. To derive them, let us start with the steady two-dimensional Navier-Stokes equations

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0, \tag{7.1a}$$

$$\rho\left(u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y}\right) = -\frac{\partial p}{\partial x} + \mu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right), \tag{7.1b}$$

$$\rho\left(u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y}\right) = -\frac{\partial p}{\partial y} + \mu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right). \tag{7.1c}$$

Noting the reflection symmetry of the flow around $y = 0$, we only solve for u, v and p for $y >= 0$. These equations are to be solved with the boundary conditions

$$u(x \geq 0, y = 0) = 0, \tag{7.2a}$$

$$v(x \geq 0, y = 0) = 0, \tag{7.2b}$$

$$u(x, y \to \infty) = U. \tag{7.2c}$$

Scaling estimate of the size of terms in Navier-Stokes equations

The first step is to develop a scaling estimate for each of the terms in these equations. Here our objective is to identify how each term in (7.1) scales with the parameters of the problem, viz., ρ, μ, U and L, and then balance the magnitudes of the dominant terms.

Each dependent and independent variable is assigned a constant magnitude in terms of the parameters governing the problem. For example, the magnitude of u is taken to be U, expressed as $u \sim U$. Similarly, the scale for x is taken to be L, i.e. $x \sim L$. We do not yet know the magnitudes for v, y and p, so we define symbols V, δ and P, respectively, to represent them. The magnitudes of partial derivatives are estimated using ratios of the variables in the derivative, e.g., $\frac{\partial u}{\partial x} \sim \frac{U}{L}$. Our task is to express these undefined symbols in terms of the known parameters using dominant balances.

As an illustration, consider the mass conservation equation (7.1a). The magnitude of the two terms in this equation are presented in the second row of Table 7.1. Balancing the magnitudes of the two terms yields the magnitude for v as

$$\frac{U}{L} = \frac{V}{\delta} \qquad \Longrightarrow \qquad V = \frac{U\delta}{L}. \tag{7.3}$$

Note that this is consistent with our estimates in the simple picture in §7.1.1.

Table 7.1 also shows the magnitude estimates for the momentum conservation equations in (7.1). Using (7.3), it is evident that in (7.1b)

$$\frac{\rho U^2}{L} = \frac{\rho UV}{\delta}, \tag{7.4}$$

implying that both the terms $\rho u\dfrac{\partial u}{\partial x}$ and $\rho v\dfrac{\partial u}{\partial y}$ are of equal magnitude. Furthermore, given that the pressure field is established to enforce incompressibility, its magnitude is set up to balance the strongest term in the equations. This implies that the $\rho u\dfrac{\partial u}{\partial x}$ is of a magnitude comparable to $\dfrac{\partial p}{\partial x}$, or

$$\frac{P}{L} = \frac{\rho U^2}{L} \qquad \Longrightarrow \qquad P = \rho U^2. \tag{7.5}$$

Between the two viscous terms, $\mu\dfrac{\partial^2 u}{\partial x^2}$ and $\mu\dfrac{\partial^2 u}{\partial y^2}$, which scale as $\mu U/L^2$ and $\mu U/\delta^2$, respectively, because $\delta \ll L$ it is evident that, $\mu\dfrac{\partial^2 u}{\partial x^2} \ll \mu\dfrac{\partial^2 u}{\partial y^2}$. Thus, the term $\mu\dfrac{\partial^2 u}{\partial x^2}$ is perhaps negligible compared to at least one other term in (7.1b). Neglecting the $\mu\dfrac{\partial^2 u}{\partial y^2}$ also will render the dominant balance equivalent to that of an inviscid fluid, which is contrary to our original intention of studying the influence of viscosity. Therefore, balancing the dominant of the two viscous terms $\mu\dfrac{\partial^2 u}{\partial y^2}$ with the inertial terms yields

$$\frac{\rho U^2}{L} = \frac{\mu U}{\delta^2} \qquad \Longrightarrow \qquad \delta = \sqrt{\frac{\nu L}{U}}. \tag{7.6}$$

This expression is eerily similar to one derived in §7.1.1, except that x is replaced by L. We can now verify our assumption

$$\frac{\delta}{L} \ll 1 \qquad \Longleftrightarrow \qquad \sqrt{\frac{\nu}{UL}} = \mathrm{Re}^{-1/2} \ll 1 \qquad \Longleftrightarrow \qquad \mathrm{Re} \gg 1, \tag{7.7}$$

where we defne the Reynolds number $\mathrm{Re} = UL/\nu$.

We now turn our attention to (7.1c). Substituting V from (7.3) and P from (7.5) into (7.1c) yields the magnitude of every term in (7.1c). In light of the assumption $\delta \ll L$ yields the most dominant term in that equation to be $\dfrac{\partial p}{\partial y}$, which scales as $\dfrac{\rho U^2}{L} \times \sqrt{\mathrm{Re}}$, while all the other terms are negligible in comparison. See Table 7.1 for details.

With these estimates, (7.1) simplify to

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0, \tag{7.8a}$$

$$\rho\left(u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y}\right) = -\frac{\partial p}{\partial x} + \mu\frac{\partial^2 u}{\partial y^2}, \tag{7.8b}$$

$$0 = -\frac{\partial p}{\partial y}, \tag{7.8c}$$

which are subject to boundary conditions given by (7.2). The set of equations (7.8) are known as the Prandtl boundary-layer equations after Ludwig Prandtl, who first identified the presence of boundary layers in high Reynolds number flow of viscous fluids, and the approximation leading to them is called the Prandtl boundary-layer approximation.

For flow past a flat plate, the flow is uniform outside the boundary layer, and consequently the pressure is uniform there. Combined with (7.8c), this implies that the pressure is uniform everywhere and $\dfrac{\partial p}{\partial x} = 0$. Thus, for the case of flow past a flat plate, (7.8) simplify further to

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0, \tag{7.9a}$$

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = \nu\frac{\partial^2 u}{\partial y^2}. \tag{7.9b}$$

7.1.3 Self-similar flow past a semi-infinite flat plate

In a region closer to the leading edge of the plate, say a distance x, which is much larger than ν/U but much smaller than the plate length, L, a self-similar structure emerges for the flow. We expose this structure in this section.

Scale invariance

The scale invariance of the geometry of the flow past a long plate, approximated as semi-infinite, is evident. Upon zooming out, the plate appears semi-infinite and the uniform flow remains uniform. The governing dynamics in the boundary layer are also scale invariant. To see this, rescale the variables in (7.9) and (7.2) as

$$u = \alpha\tilde{u}, \quad v = \beta\tilde{v}, \quad x = \gamma\tilde{x} \quad \text{and} \quad y = \delta\tilde{y}, \tag{7.10}$$

where α, β, γ and δ are positive transformation factors. The rescaled equations (7.9) and (7.2) are

$$\frac{\alpha}{\gamma}\frac{\partial\tilde{u}}{\partial\tilde{x}} + \frac{\beta}{\delta}\frac{\partial\tilde{v}}{\partial\tilde{y}} = 0, \tag{7.11a}$$

$$\frac{\alpha^2}{\gamma}\tilde{u}\frac{\partial\tilde{u}}{\partial\tilde{x}} + \frac{\alpha\beta}{\delta}\tilde{v}\frac{\partial\tilde{u}}{\partial\tilde{y}} = \frac{\alpha}{\delta^2}\nu\frac{\partial^2\tilde{u}}{\partial\tilde{y}^2}, \tag{7.11b}$$

and

$$\alpha\tilde{u}(\alpha\tilde{x} \geq 0, \delta\tilde{y} = 0) = 0, \qquad \Longrightarrow \qquad \tilde{u}(\tilde{x} \geq 0, \tilde{y} = 0) = 0 \tag{7.12a}$$

$$\beta\tilde{v}(\alpha\tilde{x} \geq 0, \delta\tilde{y} = 0) = 0, \qquad \Longrightarrow \qquad \tilde{v}(\tilde{x} \geq 0, \tilde{y} = 0) = 0 \tag{7.12b}$$

$$\alpha\tilde{u}(\alpha\tilde{x}, \delta\tilde{y} \to \infty) = U. \qquad \Longrightarrow \qquad \tilde{u}(\tilde{x}, \tilde{y} \to \infty) = \frac{U}{\alpha} \tag{7.12c}$$

Equation (7.11) and (7.12) are identical to (7.9) and (7.2) if

$$\frac{\beta}{\delta} = \frac{\alpha}{\gamma}, \quad \frac{\alpha^2}{\gamma} = \frac{\alpha}{\delta^2}, \quad \text{and} \quad \alpha = 1. \tag{7.13}$$

These relations between the scaling factors are satisfied if $\alpha = 1$, $\delta = \sqrt{\gamma}$ and $\beta = 1/\sqrt{\gamma}$, irrespective of the value of γ. This demonstrates the invariance of the governing equations, and thereby the flow, to the (anisotropic) scaling of the independent and dependent variables.

The scale invariance implies that the flow velocity at points related by the transformation is also related. In particular, If the functional form of $u(x, y)$ is given by $u(x, y) = F(x, y)$, then an identical form describes the velocity in the transformed coordinates, i.e., $\tilde{u}(\tilde{x}, \tilde{y}) = F(\tilde{x}, \tilde{y})$. Given the relation between u and $\tilde{u}$ from (7.10),

$$F(x, y) = \alpha F(\tilde{x}, \tilde{y}) = \alpha F\left(\frac{x}{\gamma}, \frac{y}{\delta}\right). \tag{7.14}$$

Expressing α and δ in terms of γ and F in terms of u yields

$$u(x, y) = u\left(\frac{x}{\gamma}, \frac{y}{\sqrt{\gamma}}\right). \tag{7.15}$$

Choosing $\gamma = x$ reduces this relation to a form useful for our interpretation as

$$u(x, y) = u\left(1, \frac{y}{\sqrt{x}}\right). \tag{7.16}$$

This equation states that the profile of u versus y at any location x is related to that at $x = 1$ (arbitrary units) by a simple rescaling of y by $\sqrt{x}$. Similarly,

$$v(x, y) = \frac{1}{\sqrt{x}}v\left(1, \frac{y}{\sqrt{x}}\right). \tag{7.17}$$

In this way, we expect the velocity profile to not be independently dependent on x and y, but merely on the combination $y/\sqrt{x}$ alone.

Determination of the Blasius self-similar profile

Once the possibility of self-similarity is established, the following approach can be used to simplify the governing equations. First, a precursory method incorporates the dependence of the dimensional parameters into the scaling invariance. At this stage, we simply replace derivatives in (7.9) with ratios, and replace the dependent and independent variables with the scaling estimates in terms of known problem parameters, very similar to scaling estimates in §7.1.2. In this analysis, armed by the knowledge of scale invariance, we recognize that x itself represents the scale for the independent variable x, and $\delta(x)$ for y. The scales for u and v are U and $U\delta(x)/x$, respectively, in view of (7.9a). The dominant balance between terms in (7.9b) is

$$\frac{U^2}{x} = 2\nu\frac{U}{\delta(x)^2} \qquad \Longrightarrow \qquad \delta(x) = \sqrt{\frac{2\nu x}{U}}, \tag{7.18}$$

in agreement with the intuition presented in §7.1.1, except for the factor of 2. (The explanation for this factor being an algebraic convenience will appear soon.) This approach also explains the appearance of L in the expression for δ in (7.6) instead of x as in (7.18). Because of the scale-invariance, the plate length L is not the appropriate scale for x, but x itself is. Apart from this difference, the balance of terms that led to (7.18) and (7.6) are identical. Equation (7.18) now incorporates the dimensional dependence on the parameters ν and U.

We proceed to determine the self-similar flow profile by incorporating the results (7.18) and (7.16) by asserting that

$$u = Uf'(\xi), \quad \text{where} \quad \xi = \frac{y}{\delta(x)}. \tag{7.19}$$

Here the variables ξ and $f(\xi)$ are dimensionless and $'$ denotes differentiation with respect to the argument. There exist multiple equivalent formulations, which represent the same flow, from which we choose one.

To facilitate the subsequent analysis, note that

$$\frac{d\delta}{dx} = \sqrt{\frac{\nu}{2Ux}} = \frac{\delta}{2x} \quad \text{so} \quad \frac{\partial\xi}{\partial x} = -\frac{y}{\delta^2}\delta'(x) = -\frac{\xi}{2x} \quad \text{and} \quad \frac{\partial\xi}{\partial y} = \frac{1}{\delta}. \tag{7.20}$$

Using (7.20),

$$\frac{\partial u}{\partial x} = Uf''(\xi)\frac{\partial\xi}{\partial x} = -\frac{U}{2x}\xi f''(\xi), \tag{7.21a}$$

$$\frac{\partial u}{\partial y} = Uf''(\xi)\frac{\partial\xi}{\partial y} = \frac{U}{\delta}f''(\xi), \tag{7.21b}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{U}{\delta^2}f'''(\xi), \tag{7.21c}$$

$$\frac{\partial v}{\partial y} = \frac{1}{\delta}\frac{\partial v}{\partial\xi}. \tag{7.21d}$$

Substitution into (7.9a) yields

$$\frac{\partial v}{\partial\xi} = U\frac{\delta}{2x}\xi f''(\xi), \qquad \Longrightarrow \qquad v = U\frac{\delta}{2x}\left(\xi f'(\xi) - f(\xi)\right). \tag{7.22}$$

Substituting (7.19) and (7.22) in (7.9b) yields

$$\frac{U^2}{2x}\left(-\xi f'(\xi)f''(\xi) + \left(\xi f'(\xi) - f(\xi)\right)f''(\xi)\right) = \nu\frac{U}{\delta^2}f'''(\xi). \tag{7.23}$$

This equation reveals the utility of the dominant balance using scaling. Note that in (7.18) we balanced the terms U^2/x and $\nu U/\delta^2$, which are also the dimensional pre-factors of the terms in (7.23). The additional factor of 2 in (7.18) is simply to account for the factor of 2 that arises in (7.23) – this factor is not necessary for deriving the self-similar flow profile. The essential reason underlying the utility of the method to derive (7.18) is that, according to the chain rule applied in (7.20) and (7.21), differentiation with respect to x yields a dimensional factor of $1/x$, and with respect to y yields $1/\delta$. These were precisely the rules we used to derive (7.18).

In view of (7.18), equation (7.23) simplifies to

$$f'''(\xi) + f(\xi)f''(\xi) = 0. \tag{7.24}$$

The boundary conditions (7.2) simplify to

$$f'(\xi = 0) = 0, \tag{7.25a}$$
$$f(\xi = 0) = 0, \tag{7.25b}$$
$$f'(\xi \to \infty) = 1. \tag{7.25c}$$

Equations (7.24-7.25) are called the Blasius equations for the self-similar profile. They constitute an ordinary differential boundary value problem, which cannot be solved in closed form analytically but can be easily solved numerically. The solution is shown in Figure 7.3 as the dashed curve, which agrees very well with the numerically computed profiles.

7.1.4 Drag on the plate

With the complete flow field described analytically, we now return to our original motivation of estimating the drag on the plate because of viscous shear stress. The uniform potential flow violates the no-slip boundary condition and, in the process, also fails to account for the drag on the plate. The boundary layer approximation retains the no-slip condition, and thus might be able to explain the drag on the plate.

The drag on the upper surface of the plate is calculated by integrating the tangential component of the shear stress. The components of the total (i.e., Cauchy) stress tensor are

$$\boldsymbol{T} = -p \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \mu \begin{bmatrix} 2\dfrac{\partial u}{\partial x} & \dfrac{\partial u}{\partial y} + \dfrac{\partial v}{\partial x} \\ \dfrac{\partial u}{\partial y} + \dfrac{\partial v}{\partial x} & 2\dfrac{\partial v}{\partial y} \end{bmatrix}. \tag{7.26}$$

The tangential component of the traction on the plate is

$$\boldsymbol{e}_x \cdot \boldsymbol{T} \cdot \boldsymbol{e}_y = \mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \approx \mu \frac{\partial u}{\partial y}. \tag{7.27}$$

Note that the component of stress due to pressure does not exert a tangential force on the plate. And using scaling estimates developed in §7.1.2, amongst the terms in (7.27), $\dfrac{\partial u}{\partial y}$ scales as U/δ and dominates over $\dfrac{\partial v}{\partial x}$, which scales as $V/L = U\delta/L^2$. Therefore, to the level of approximation we have retained, the term $\dfrac{\partial v}{\partial x}$ may be neglected.

The total force, D, per unit length out of the plane of the paper now may be estimated using (7.21b) as

$$D = \int_0^L \mu \frac{\partial u}{\partial y}|_{y=0} \, dx = \int_0^L \frac{\mu U}{\delta(x)} f''(\xi = 0) \, dx = \sqrt{\frac{\mu^2 U^3}{2\nu}} f''(\xi) \int_0^L \frac{dx}{\sqrt{x}} = \sqrt{2\mu\rho U^3 L} f''(0). \tag{7.28}$$

The value of $f''(0)$ from the solution of (7.24-7.25) is 0.4696. Substituting this value yields an expression for the drag on a flat plate align with a flow at high Reynolds number as

$$D = 0.664\sqrt{\rho\mu U^3 L}. \tag{7.29}$$

A comparison of the results of (7.29) with that of the complete solution computed using COMSOL is shown in Table 7.2. The simple expression in (7.29) agrees well with the results of the full computational solution. Reasons for the remaining slight disagreement are the finite thickness of the plate in the COMSOL simulation and the approximations made in the boundary layer theory.

According to (7.29), longer plates experience greater drag but drag does not grow proportional to the plate length. It is so because the boundary layer is thinner near the leading edge of the plate, where the plate experiences most drag. The drag is super-linear in the freestream fluid speed, growing as $U^{3/2}$, which is because the boundary layer thickness itself decreases with U thereby enhancing the shear stress on the plate. Also, in the limit of vanishing viscosity, the drag decreases to zero proportional to $\sqrt{\mu}$. The physical explanation for why the drag might vanish with vanishing fluid density is left for the astute student to discover.

μ (Pa s) / U (m/s)	$\mu = 10^{-5}$	3×10^{-5}	5×10^{-5}	μ (Pa s) / U (m/s)	$\mu = 10^{-5}$	3×10^{-5}	5×10^{-5}
$U = 1.0$	4.13	7.85	10.6	$U = 1.0$	4.70	8.13	10.5
$U = 1.3$	5.99	11.4	15.4	$U = 1.3$	6.96	12.1	15.6
$U = 1.6$	8.04	15.3	20.6	$U = 1.6$	9.50	16.5	21.2
$U = 1.9$	10.3	19.5	26.3	$U = 1.9$	12.3	21.3	27.5

Table 7.2: Drag on the plate computed using COMSOL (left) compared with the result of the boundary layer approximation (7.29) (right).